CONDITIONAL LOCAL TYPE I NAVIER–STOKES REDUCTION: CONCENTRATION, SEREGIN LIMITS, AND LIOUVILLE-CLASS CLOSURES

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Abstract. The concentration, compactness, and Liouville-class reductions are proved for the conditional local Type I Navier–Stokes reduction. These reductions consist of the local Caffarelli–Kohn–Nirenberg concentration theorem and Type I/Type II dichotomy, the admissible Type I Seregin blow-up extraction, the decomposition of normalized centered ancient limits into specified Liouville classes and a residual class, the endpoint exclusion of the uniformly L^3-tight class, and the structured ancient-solution exclusions covered by classical and perturbative Liouville theorems. The residual class is kept as the explicit residual-class hypothesis Hypothesis 16.1.

Contents

Introduction and Structure of the Argument 4
Part I. Local Concentration and the Type I/Type II Dichotomy 5
Overview 5
1. Introduction 5
2. Suitable Weak Solutions and Singular Points 5
3. Parabolic Rescaling and Scale-Invariant Quantities 6
4. ε-Regularity and the Vanishing Case 8
5. Positive Local Concentration 9
6. No-Escape for Finite-Energy Type I Singularities 11
7. The Type I/Type II Blow-Up Rate Dichotomy 15
8. Local Pointwise Type I Reduction to Seregin Extraction 16
Part II. Type I Blow-up Limits and the Residual Hypothesis 20
Overview 20
9. Introduction and main results 20
9.1. Type I blow-up limits 20
9.2. Normalized Seregin limits 20
9.3. Liouville alternatives and the remaining case 21
9.4. Main theorems 21
9.5. Dependencies and conditional hypothesis 22
9.6. Organization 22
10. Suitable solutions, scale-invariant quantities, and local concentration 22
11. The Type I blow-up sequence and compactness 26
11.1. The rescaled sequence 26
11.2. Inherited Type I bounds 26
11.3. Local energy, pressure, and time-derivative estimates 26
11.4. Compactness away from the terminal time 29

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11.5. The ancient suitable weak limit 30
12. Persistence of concentration and nontriviality 31
12.1. Persistence of concentration away from the terminal time 31
12.2. Scale selection 31
12.3. Nonzero ancient limits 32
13. Centered variables and normalized Seregin limits 32
13.1. The centered self-similar equation 32
13.2. Boundedness, smoothness, and local energy 33
13.3. Normalized Seregin ancient limits 34
14. Liouville classes and the remaining class 35
15. Liouville hypotheses and class exclusions 36
16. The remaining-case criterion 39
17. Elementary symmetries of the centered equation 39
Part III. Uniformly Tight Ancient Dynamics 41
Overview 41
18. Introduction and main results 41
18.1. The uniformly tight ancient class 41
18.2. Minimal elements and compact hulls 41
18.3. Invariant measures and covariance 42
18.4. Endpoint L^3 closure of the tight class 42
18.5. Main results 43
18.6. Relation to the Type I reduction 44
18.7. Organization 44
19. Centered ancient solutions and the local smooth topology 44
19.1. The centered equation 44
19.2. The Ornstein–Uhlenbeck–Stokes semigroup 44
19.3. Riesz transform convention 45
19.4. Bounded mild ancient solutions 45
19.5. Time translation 45
19.6. The locally smooth topology 46
20. Smoothing, pressure reconstruction, and compactness 46
20.1. Local smoothing 46
20.2. Pressure reconstruction 47
20.3. Stability under locally smooth limits 48
20.4. Compactness of bounded families 49
21. Uniform L^3-tightness and stationary L^3 rigidity 49
21.1. Uniform L^3-tightness 49
21.2. Closedness under locally smooth convergence 49
21.3. Stationary time-translate limits 50
21.4. Stationary self-similar rigidity 50
22. Normalized collections and minimal ancient solutions 51
22.1. Closed normalized collections 51
22.2. Time-centering 51
22.3. Stability of local nonvanishing 51
22.4. Lower semicontinuity of size 51
22.5. Existence of a minimal element 52
22.6. The uniformly tight normalized class 52
23. Compact trajectory hulls 53
23.1. Trajectory hulls 53

23.2. Compactness of bounded hulls 53
23.3. Compact invariant sets 54
23.4. Minimal compact invariant subsets 54
23.5. The zero trajectory 54
24. Invariant measures and the covariance obstruction 55
24.1. Krylov–Bogolyubov averages 55
24.2. Support on minimal hulls 55
24.3. Stationary statistical identity 55
24.4. Barycenter and covariance 56
25. Endpoint L^3 closure of the uniformly tight class 57
25.1. Physical pullback 57
25.2. Critical L^3 invariance 58
25.3. Mildness after finite terminal-time restriction 59
25.4. Endpoint L^3 ancient Liouville theorem 59
26. Tight-Class Liouville Criterion and Type I Reduction 60
26.1. Normalized tight collections 60
26.2. Use in the Type I reduction 60
Part IV. Structured Ancient Solution Exclusions 62
Overview 62
27. Introduction 62
27.1. Ancient solutions from Type I blow-up 62
27.2. Axisymmetric Liouville hypotheses 62
27.3. Lyapunov hypotheses 63
27.4. Main results 63
27.5. Relation to the preceding parts 64
27.6. Summary of hypotheses 64
27.7. Organization 64
28. Ancient Type I limits and self-similar variables 64
28.1. Type I ancient solutions 64
28.2. Nonzero Type I blow-up limits 64
28.3. Backward self-similar variables 64
28.4. Preservation of axisymmetry and swirl variables 65
28.5. Time shifts away from the singular time 66
28.6. PDE hypotheses 66
29. Axisymmetric notation and cited Liouville theorems 66
29.1. Cylindrical variables and the swirl scalar 66
29.2. Constant solutions and Galilean reduction 67
29.3. Cited Liouville theorems 67
30. Unforced axisymmetric Liouville consequences 68
30.1. No swirl 68
30.2. Pointwise scale-invariant control 68
30.3. Finite L^p control of Γ 68
30.4. Periodic bounded swirl 69
30.5. Proof of the axisymmetric Liouville theorem 69
31. Perturbative weighted-vorticity Lyapunov rigidity 69
31.1. The rescaled vorticity equation 69
31.2. Gallay–Wayne small-data decay 69
31.3. The Lyapunov functional 70
31.4. Ancient rigidity in the small weighted ball 70

32. Summary of PDE consequences 71
32.1. Zero conclusions 71
32.2. Consequences for Type I blow-up arguments 71
32.3. Summary table 71
References 72

Introduction and Structure of the Argument

The conditional local Type I reduction reduces a local pointwise Type I singularity to a normalized, nonzero, bounded centered ancient solution and excludes the non-residual Liouville classes obtained from standard local compactness and explicit Liouville theorems. The residual class is not excluded unconditionally; its emptiness is isolated as the residual-class hypothesis Hypothesis 16.1.

The reduction has four components. First, the local concentration theorem shows that a singular point carries positive Caffarelli–Kohn–Nirenberg critical mass at arbitrarily small scales and gives the local Type I/local Type II dichotomy. It also identifies the endpoint weak-Serrin estimate used to obtain terminal velocity no-escape for local pointwise Type I rescalings. Second, the Type I blow-up analysis gives the admissible Seregin extraction: an admissible Type I extraction produces a smooth bounded ancient solution of the centered self-similar equation with positive compact local L^3 velocity mass. It then defines the Liouville classes and the residual class.

Third, the uniformly tight analysis excludes nonzero uniformly L^3-tight normalized Seregin limits. Boundedness plus uniform L^3-tightness gives a backward sequence of finite critical norm after physical pullback, so the Albritton–Barker endpoint ancient Liouville theorem applies. Fourth, the structured part gives the axisymmetric and perturbative weighted-vorticity exclusions used in the Liouville-class analysis. These results turn the corresponding structured hypotheses into the zero ancient solution, contradicting the local nonvanishing obtained from the Seregin extraction.

Consequently, a normalized Seregin limit generated by an admissible Type I extraction must either belong to one of the excluded Liouville classes or to the residual class. The Type I contradiction is therefore conditional on the residual-class hypothesis Hypothesis 16.1.

Part I. Local Concentration and the Type I/Type II Dichotomy

Overview. This part proves a local analytic reduction for possible finite-time singularities of the three-dimensional Navier–Stokes equations. For a suitable weak solution, every singular point at time T has arbitrarily small backward parabolic cylinders on which both the standard mean-subtracted Caffarelli–Kohn–Nirenberg scale-invariant quantity $C + D$ and the scale-invariant velocity quantity C are bounded below by positive constants. If these local scale-invariant quantities tend to zero at each point of the singular set, the solution is regular at time T by the ε-regularity theorem. Along any such positive concentration sequence, the solution either satisfies a local Type I scale-invariant bound or belongs to the local Type II alternative. For local pointwise Type I terminal singularities of suitable weak solutions, the endpoint weak-Serrin argument gives the velocity no-escape condition used in the Type I blow-up analysis.

1. Introduction

Let (u, p) be a suitable weak solution of the three-dimensional incompressible Navier–Stokes equations

$$\partial_t u + u \cdot \nabla u + \nabla p = \Delta u, \qquad \nabla \cdot u = 0.$$

The viscosity is normalized to one. The weak solution theory goes back to Leray [5]. Suitable weak solutions and their partial regularity theory were introduced and developed by Scheffer [7] and by Caffarelli, Kohn, and Nirenberg [2]; see also Lin [6]. The local reduction below connects a possible singularity at time T with the Type I/Type II blow-up rate dichotomy.

Scale-invariant regularity and blow-up criteria provide the standard background for this formulation; see Serrin [9], Giga [4], Escauriaza–Seregin–Šverak [3], and Seregin [8]. The principal regularity theorem is the Caffarelli–Kohn–Nirenberg ε-regularity theorem.

The proof has four components. First, the local scale-invariant quantities C and D are finite for suitable weak solutions and invariant under the usual parabolic scaling. Second, the Caffarelli–Kohn–Nirenberg ε-regularity theorem implies that vanishing of $C + D$ at a candidate singular point gives local regularity, while the velocity ε-regularity criterion gives positive concentration in the scale-invariant $L^3_{x,t}$ quantity. Third, a finite-time singularity therefore produces sequences of shrinking cylinders on which $C + D$, and separately C, are bounded below by positive constants. Finally, once such a sequence is fixed, it either satisfies a Type I scale-invariant bound or belongs to the local Type II alternative.

The concentration and Type I/Type II dichotomy proofs are local in spacetime. Apart from the standard Caffarelli–Kohn–Nirenberg ε-regularity theorem, all quantities and alternatives used in that local analysis are defined in this part. The local Type I reduction section shows how the endpoint weak-Serrin estimate upgrades the local pointwise Type I envelope to terminal velocity no-escape through the local endpoint weak-Serrin Type I estimate.

2. Suitable Weak Solutions and Singular Points

Definition 2.1 (Suitable weak solution)**.** Let $I \subset \mathbb{R}$ be an open interval. A pair (u, p) is a suitable weak solution on $\mathbb{R}^3 \times I$ if

$$u \in L^\infty_t L^2_{x,\mathrm{loc}} \cap L^2_t H^1_{x,\mathrm{loc}}, \qquad p \in L^{3/2}_{\mathrm{loc}}, \qquad \nabla \cdot u = 0,$$

the Navier–Stokes equations hold in the sense of distributions, and the local energy inequality holds: for almost every $t_1 < t_2$ in I and every nonnegative $\phi \in C_c^\infty(\mathbb{R}^3 \times I)$,

$$\begin{aligned}&\int_{\mathbb{R}^3} \frac{|u(x,t_2)|^2}{2}\phi(x,t_2)\,dx + \int_{t_1}^{t_2}\int_{\mathbb{R}^3} |\nabla u|^2 \phi\,dx\,dt \\ &\le \int_{\mathbb{R}^3} \frac{|u(x,t_1)|^2}{2}\phi(x,t_1)\,dx + \int_{t_1}^{t_2}\int_{\mathbb{R}^3} \frac{|u|^2}{2}(\partial_t\phi + \Delta\phi)\,dx\,dt \\ &\quad + \int_{t_1}^{t_2}\int_{\mathbb{R}^3} \left(\frac{|u|^2}{2} + p\right) u\cdot\nabla\phi\,dx\,dt.\end{aligned}$$

The pressure is understood modulo functions of time.

For $z_0 = (x_0, t_0)$ and $r > 0$, write

$$Q_r(z_0) = B_r(x_0) \times (t_0 - r^2, t_0).$$

Also write $Q_r(x_0, t_0)$ for $Q_r((x_0, t_0))$.

Definition 2.2 (Singular set at a fixed time)**.** For a time T, define

$$\Sigma(T) = \{x \in \mathbb{R}^3 : \ u \text{ is not locally bounded in any } Q_r(x, T)\}.$$

Equivalently, $x \notin \Sigma(T)$ if there are $r > 0$ and $M < \infty$ such that $\|u\|_{L^\infty(Q_r(x,T))} \le M$.

If a finite-time singularity occurs at time T, then $\Sigma(T) \neq \emptyset$ in this local sense.

3. Parabolic Rescaling and Scale-Invariant Quantities

Fix $z_0 = (x_0, T)$. For $r > 0$, define the parabolic rescaling

$$u^{(r)}(y,s) = r\,u(x_0 + ry, T + r^2 s), \qquad p^{(r)}(y,s) = r^2 p(x_0 + ry, T + r^2 s).$$

The rescaled variables are defined on the set of (y, s) for which $(x_0 + ry, T + r^2 s)$ belongs to the original spacetime domain.

Proposition 3.1 (Invariance under parabolic rescaling)**.** *If (u, p) is a suitable weak solution near (x_0, T), then $(u^{(r)}, p^{(r)})$ is a suitable weak solution on the rescaled domain. The local energy inequality is preserved, and the pressure remains defined modulo functions of the rescaled time.*

Proof. The distributional equations and the divergence condition follow from the change of variables $x = x_0 + ry$, $t = T + r^2 s$. The local energy inequality is verified to fix the normalization.

Let $\psi \in C_c^\infty$ be nonnegative and compactly supported in the rescaled domain. Define

$$\phi(x,t) = \psi\left(\frac{x - x_0}{r}, \frac{t - T}{r^2}\right).$$

If $y = (x - x_0)/r$ and $s = (t - T)/r^2$, then

$$\partial_t\phi = r^{-2}(\partial_s\psi)(y,s), \qquad \nabla_x\phi = r^{-1}(\nabla_y\psi)(y,s), \qquad \Delta_x\phi = r^{-2}(\Delta_y\psi)(y,s).$$

For almost every $s_1 < s_2$, the corresponding times $T + r^2 s_1 < T + r^2 s_2$ are admissible in the original local energy inequality. Apply that inequality for (u, p) to ϕ on $[T + r^2 s_1, T + r^2 s_2]$. After the change of variables $x = x_0 + ry$, $t = T + r^2 s$, using

$$u(x_0 + ry, T + r^2 s) = r^{-1}u^{(r)}(y,s), \qquad p(x_0 + ry, T + r^2 s) = r^{-2}p^{(r)}(y,s),$$

each term contains the same common factor r. Dividing by this factor gives the rescaled inequality

$$\begin{aligned}&\int_{\mathbb{R}^3} \frac{|u^{(r)}(y,s_2)|^2}{2}\psi(y,s_2)\,dy + \int_{s_1}^{s_2}\int_{\mathbb{R}^3} |\nabla_y u^{(r)}|^2\psi\,dy\,ds \\ &\le \int_{\mathbb{R}^3} \frac{|u^{(r)}(y,s_1)|^2}{2}\psi(y,s_1)\,dy + \int_{s_1}^{s_2}\int_{\mathbb{R}^3} \frac{|u^{(r)}|^2}{2}(\partial_s\psi + \Delta_y\psi)\,dy\,ds \\ &\quad + \int_{s_1}^{s_2}\int_{\mathbb{R}^3} \left(\frac{|u^{(r)}|^2}{2} + p^{(r)}\right) u^{(r)}\cdot\nabla_y\psi\,dy\,ds.\end{aligned}$$

Thus $(u^{(r)}, p^{(r)})$ is suitable on the rescaled domain. The pressure transforms according to the Navier–Stokes scaling, and addition of a function of t becomes addition of a function of s. □

Definition 3.2 (Scale-invariant local quantities)**.** For $z_0 = (x_0, t_0)$ and $r > 0$, set

$$C(z_0, r) = r^{-2}\int_{Q_r(z_0)} |u|^3\,dx\,dt$$

and

$$D(z_0, r) = r^{-2}\int_{t_0-r^2}^{t_0}\int_{B_r(x_0)} |p(x,t) - (p)_{B_r(x_0)}(t)|^{3/2}\,dx\,dt.$$

Here $(p)_{B_r(x_0)}(t)$ denotes the spatial mean of $p(\cdot, t)$ on $B_r(x_0)$.

The subtraction of the spatial mean removes the pressure ambiguity in the form used by the Caffarelli–Kohn–Nirenberg criterion.

Proposition 3.3 (Local finiteness and scaling)**.** *Let (u, p) be a suitable weak solution near $z_0 = (x_0, T)$. Then $C(z_0, r)$ and $D(z_0, r)$ are finite for every sufficiently small r. Moreover, for every fixed $R < \infty$,*

$$\begin{aligned}&\int_{-R^2}^{0}\int_{B_R} \left(|u^{(r)}|^3 + |p^{(r)} - (p^{(r)})_{B_R}(s)|^{3/2}\right) dy\,ds \\ &\qquad = R^2\big(C(z_0, rR) + D(z_0, rR)\big).\end{aligned}$$

Proof. Local finiteness of the velocity term follows from

$$u \in L^\infty_t L^2_{x,\mathrm{loc}} \cap L^2_t H^1_{x,\mathrm{loc}} \subset L^3_{\mathrm{loc}}$$

on finite cylinders. Indeed, if $B \Subset \mathbb{R}^3$ and $(a, b) \Subset I$, Sobolev and interpolation give

$$\int_a^b \|u(t)\|^3_{L^3(B)}\,dt \le C\left(\operatorname*{ess\,sup}_{a<t<b}\|u(t)\|_{L^2(B)}\right)^{3/2}\left(\int_a^b \|u(t)\|^2_{H^1(B)}\,dt\right)^{3/4}(b-a)^{1/4}.$$

Thus $u \in L^3_{\mathrm{loc}}$. The pressure term is finite because $p \in L^{3/2}_{\mathrm{loc}}$, and subtracting a spatial mean over the same ball preserves local $L^{3/2}$ integrability. More explicitly, for a ball B,

$$|p - (p)_B(t)|^{3/2} \le C\left(|p|^{3/2} + |(p)_B(t)|^{3/2}\right),$$

while Jensen's inequality gives

$$|(p)_B(t)|^{3/2} \le \frac{1}{|B|}\int_B |p(x,t)|^{3/2}\,dx.$$

After integration over B and over time, the mean-subtracted pressure term is controlled by the local $L^{3/2}$ norm of p.

For the scaling identity, use $x = x_0 + ry$, $t = T + r^2 s$. The velocity term transforms as

$$\int_{-R^2}^{0}\int_{B_R} |u^{(r)}|^3\,dy\,ds = r^{-2}\int_{T-(rR)^2}^{T}\int_{B_{rR}(x_0)} |u|^3\,dx\,dt.$$

The pressure means transform according to

$$(p^{(r)})_{B_R}(s) = r^2 (p)_{B_{rR}(x_0)}(T + r^2 s),$$

because

$$\frac{1}{|B_R|}\int_{B_R} r^2 p(x_0 + ry, T + r^2 s)\,dy = \frac{r^2}{|B_{rR}|}\int_{B_{rR}(x_0)} p(x, T + r^2 s)\,dx.$$

Therefore

$$\begin{aligned}\int_{-R^2}^{0}\int_{B_R} &|p^{(r)} - (p^{(r)})_{B_R}(s)|^{3/2}\,dy\,ds\\ &= r^{-2}\int_{T-(rR)^2}^{T}\int_{B_{rR}(x_0)} |p - (p)_{B_{rR}(x_0)}(t)|^{3/2}\,dx\,dt.\end{aligned}$$

Adding the velocity and pressure identities gives the displayed formula. □

4. ε-Regularity and the Vanishing Case

Theorem 4.1 (Caffarelli–Kohn–Nirenberg ε-regularity)**.** *There exists a universal constant $\varepsilon_0 > 0$ with the following property. Let (u, p) be a suitable weak solution in $Q_r(z_0)$. If*

$$C(z_0, r) + D(z_0, r) < \varepsilon_0,$$

then u is locally bounded in a smaller cylinder, for instance in $Q_{r/2}(z_0)$. In particular, if $z_0 = (x_0, T)$, then $x_0 \notin \Sigma(T)$.

Proof. This is the local ε-regularity theorem of Caffarelli–Kohn–Nirenberg [2]; see also Lin [6]. It is stated here with the spatial mean subtraction used in D. Equivalent representatives of the pressure differ by functions of time, and the spatial mean subtraction on $B_r(x_0)$ removes this ambiguity. □

Theorem 4.2 (Velocity ε-regularity)**.** *There exists a universal constant $\varepsilon_v > 0$ with the following property. Let (u, p) be a suitable weak solution in $Q_r(z_0)$. If*

$$C(z_0, r) < \varepsilon_v,$$

then u is locally bounded in $Q_{r/2}(z_0)$. In particular, if $z_0 = (x_0, T)$, then $x_0 \notin \Sigma(T)$.

Proof. This is the standard $L^3_{x,t}$ form of the Caffarelli–Kohn–Nirenberg ε-regularity criterion. It follows from the standard local pressure decay lemma together with Theorem 4.1; equivalently, it is the $L^3_{x,t}$-smallness criterion proved in the Caffarelli–Kohn–Nirenberg theory and in Lin's direct proof [2, 6]. The statement is scale invariant because C is invariant under the parabolic Navier–Stokes scaling. □

Definition 4.3 (Vanishing local scale-invariant quantities at time T)**.** We say that the local scale-invariant quantities vanish at time T if, for every $x_0 \in \Sigma(T)$,

$$\limsup_{r\downarrow 0}\big(C((x_0, T), r) + D((x_0, T), r)\big) = 0.$$

This condition is vacuous when $\Sigma(T) = \emptyset$.

Proposition 4.4 (Equivalent rescaled formulation)**.** *The preceding condition is equivalent to the following statement: for every $x_0 \in \Sigma(T)$ and every fixed $R < \infty$,*

$$\lim_{r\downarrow 0} \int_{-R^2}^{0} \int_{B_R} \left(|u^{(r)}|^3 + |p^{(r)} - (p^{(r)})_{B_R}(s)|^{3/2} \right) dy\, ds = 0.$$

Proof. By Proposition 3.3, the rescaled integral over $B_R \times (-R^2, 0)$ equals

$$R^2\big(C((x_0,T), rR) + D((x_0,T), rR)\big).$$

Thus vanishing of $C + D$ implies vanishing on every fixed rescaled cylinder. Conversely, taking $R = 1$ gives the original pointwise condition. □

Theorem 4.5 (Regularity under vanishing local quantities)**.** *Let (u,p) be a suitable weak solution on $\mathbb{R}^3 \times (T-\delta, T)$ for some $\delta > 0$. If the local scale-invariant quantities vanish at time T, then $\Sigma(T) = \emptyset$. Thus no finite-time singularity occurs at time T in the no-concentration case.*

Proof. Suppose, toward a contradiction, that $x_0 \in \Sigma(T)$. By the vanishing hypothesis, choose $r > 0$ such that

$$C((x_0,T), r) + D((x_0,T), r) < \varepsilon_0.$$

Theorem 4.1 implies that u is bounded in $Q_{r/2}(x_0,T)$, hence $x_0 \notin \Sigma(T)$. This contradicts the choice of x_0. Hence $\Sigma(T) = \emptyset$. □

Corollary 4.6 (Vanishing local quantities exclude a terminal singularity)**.** *Let (u,p) be suitable near (x_0,T). If*

$$\limsup_{r\downarrow 0}\big(C((x_0,T), r) + D((x_0,T), r)\big) = 0,$$

then u is bounded in some backward cylinder $Q_\rho(x_0,T)$, and hence $x_0 \notin \Sigma(T)$. Equivalently, membership in $\Sigma(T)$ requires that $C+D$ have a positive lower bound along a sequence of shrinking backward parabolic cylinders.

Proof. The hypothesis gives a scale at which $C+D$ is below the universal threshold in Theorem 4.1. The ε-regularity theorem gives local boundedness in a backward cylinder ending at T, so $x_0 \notin \Sigma(T)$. □

5. Positive Local Concentration

Lemma 5.1 (Failure of vanishing gives a positive concentration sequence)**.** *Suppose the local scale-invariant quantities do not vanish at time T. Then there exist $x_0 \in \Sigma(T)$, a constant $\eta > 0$, and radii $r_n \downarrow 0$ such that*

$$C((x_0,T), r_n) + D((x_0,T), r_n) \geq \eta \qquad \textit{for all } n.$$

Equivalently, the rescaled sequence around (x_0,T) has a positive lower bound for the corresponding local scale-invariant quantities on a fixed bounded cylinder.

Proof. Negating the definition of vanishing gives a point $x_0 \in \Sigma(T)$ with positive limsup of

$$C((x_0,T), r) + D((x_0,T), r)$$

as $r \downarrow 0$. Choose $\eta > 0$ below that limsup and choose a sequence $r_n \downarrow 0$ along which the displayed lower bound holds. The rescaled formulation follows from Proposition 3.3. □

Definition 5.2 (Positive local concentration sequence)**.** Let $x_0 \in \Sigma(T)$. A sequence $r_n \downarrow 0$ is a positive local concentration sequence at (x_0,T) if there is $\eta > 0$ such that

$$C((x_0,T), r_n) + D((x_0,T), r_n) \geq \eta \qquad \text{for all } n.$$

The corresponding parabolic rescalings are those defined by

$$u_n(y,s) = r_n u(x_0 + r_n y, T + r_n^2 s), \qquad p_n(y,s) = r_n^2 p(x_0 + r_n y, T + r_n^2 s).$$

Proposition 5.3 (Persistence of positive concentration)**.** *Let (u_n, p_n) be suitable weak solutions on a fixed cylinder $B_R \times (-R^2, 0)$. Suppose that*

$$\int_{-R^2}^{0} \int_{B_R} \left(|u_n|^3 + |p_n - (p_n)_{B_R}(s)|^{3/2} \right) dy\, ds \geq \eta > 0$$

for all n. Suppose further that, after passing to a subsequence,

$$u_n \to U \quad \text{in } L^3(B_R \times (-R^2, 0))$$

and

$$p_n - (p_n)_{B_R}(s) \to \Pi \quad \text{in } L^{3/2}(B_R \times (-R^2, 0)).$$

Then

$$\int_{-R^2}^{0} \int_{B_R} \left(|U|^3 + |\Pi|^{3/2} \right) dy\, ds \geq \eta.$$

In particular, the limiting pair retains the same positive lower bound for the mean-subtracted local scale-invariant quantity. If $\Pi = P - (P)_{B_R}(s)$, this is the corresponding Caffarelli–Kohn–Nirenberg quantity of (U, P) on $B_R \times (-R^2, 0)$.

Proof. Strong convergence in L^3 gives

$$\int_{-R^2}^{0} \int_{B_R} |u_n|^3\, dy\, ds \longrightarrow \int_{-R^2}^{0} \int_{B_R} |U|^3\, dy\, ds.$$

Strong convergence in $L^{3/2}$ gives the analogous convergence of the mean-subtracted pressure integrals. Passing to the limit in the lower bound yields the asserted inequality. □

Remark 5.4. Proposition 5.3 is an auxiliary compactness observation, independent of the proof of the dichotomy below.

Proposition 5.5 (Local alternative)**.** *Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T - \delta, T)$ for some $\delta > 0$. Exactly one of the following alternatives holds.*

(i) *For every $x_0 \in \Sigma(T)$,*

$$\limsup_{r \downarrow 0} \big(C((x_0, T), r) + D((x_0, T), r) \big) = 0.$$

(ii) *There exist $x_0 \in \Sigma(T)$, $\eta > 0$, and $r_n \downarrow 0$ such that*

$$C((x_0, T), r_n) + D((x_0, T), r_n) \geq \eta \qquad \text{for all } n.$$

Proof. The first alternative is a universal assertion over the singular set, and the second is its failure expressed by a sequence. If the first alternative fails, then for some $x_0 \in \Sigma(T)$ the corresponding limsup is positive; choosing a positive number below this limsup and a sequence along which the lower bound holds gives the second alternative. Conversely, the second alternative contradicts the first. Hence precisely one alternative holds. □

Theorem 5.6 (Finite-time singularity yields positive local scale-invariant concentration)**.** *Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T - \delta, T)$. If $\Sigma(T) \neq \emptyset$, then there are $x_0 \in \Sigma(T)$, $\eta > 0$, and $r_n \downarrow 0$ such that*

$$C((x_0, T), r_n) + D((x_0, T), r_n) \geq \eta \qquad \text{for all } n.$$

Proof. If the first alternative in Proposition 5.5 held, then Theorem 4.5 would imply $\Sigma(T) = \emptyset$, contrary to the hypothesis. Therefore the second alternative in Proposition 5.5 holds. □

Corollary 5.7 (Singular points carry positive velocity concentration)**.** *Let (u,p) be a suitable weak solution on $\mathbb{R}^3 \times (T-\delta,T)$. If $x_0 \in \Sigma(T)$, then there are $\eta_v > 0$ and $r_n \downarrow 0$ such that*

$$C((x_0,T),r_n) \geq \eta_v \qquad \text{for all } n.$$

Proof. If the conclusion failed, then

$$\limsup_{r\downarrow 0} C((x_0,T),r) = 0.$$

Choose $r > 0$ so small that

$$C((x_0,T),r) < \varepsilon_v,$$

where ε_v is the constant in Theorem 4.2. The velocity ε-regularity theorem gives boundedness of u in $Q_{r/2}(x_0,T)$, contradicting $x_0 \in \Sigma(T)$. Hence the limsup is positive; taking any η_v below it and passing to a sequence with $C((x_0,T),r_n) \geq \eta_v$ gives the result. ∎

6. No-Escape for Finite-Energy Type I Singularities

The preceding concentration result gives the positive velocity concentration sequence used in the Type I blow-up analysis. In the Seregin extraction theorem Theorem 9.1, the additional compactness hypothesis is a no-escape condition excluding the possibility that all of the selected scale-invariant velocity mass collapses into the final rescaled time slice. The next theorem shows that this no-escape condition is forced by the finite-energy Type I hypotheses.

For this section we use, in addition to C and D, the standard local kinetic-energy and dissipation quantities

$$A(z_0,r) = r^{-1} \operatorname*{ess\,sup}_{t_0-r^2<t<t_0} \int_{B_r(x_0)} |u(x,t)|^2\,dx,$$

and

$$E(z_0,r) = r^{-1} \int_{t_0-r^2}^{t_0} \int_{B_r(x_0)} |\nabla u(x,t)|^2\,dx\,dt.$$

These quantities are invariant under the Navier–Stokes scaling.

Definition 6.1 (Finite-energy Type I terminal singularity)**.** Let (u,p) be a suitable weak solution on $\mathbb{R}^3 \times (0,T)$. We say that $z_* = (x_*,T)$ is a finite-energy Type I terminal singularity if $x_* \in \Sigma(T)$,

$$u \in L^\infty(0,T;L^2(\mathbb{R}^3)) \cap L^2(0,T;\dot H^1(\mathbb{R}^3)),$$

and there are $T_0 < T$ and $M < \infty$ such that

$$\|u(t)\|_{L^\infty(\mathbb{R}^3)} \leq \frac{M}{\sqrt{T-t}}, \qquad T_0 < t < T. \tag{6.1}$$

The main estimate is a uniformly local energy propagation bound. It is stated at unit scale because the application is obtained by parabolic rescaling from (x_*,T). The estimate is a local differential inequality: the unknown is the uniformly-local kinetic energy, and the Type I envelope turns the nonlinear pressure contribution into an integrable linear coefficient. No bound below depends on the size of the global L^2 energy or the global dissipation; finite energy is used to place the pressure in the standard whole-space Leray gauge and to provide fixed-scale terminal L^3 absolute continuity for each fixed rescaling.

The standard Leray pressure gauge is used throughout this section. At almost every terminal time on which the Type I bound holds, $u(t) \in L^2(\mathbb{R}^3) \cap L^\infty(\mathbb{R}^3)$, hence $u_i(t)u_j(t) \in L^{3/2}(\mathbb{R}^3)$. The whole-space pressure

$$p^L(t) = \mathcal{R}_i\mathcal{R}_j(u_i(t)u_j(t))$$

is then defined by the Calderon–Zygmund theory. The distributional pressure equation gives $-\Delta p = \partial_i\partial_j(u_iu_j) = -\Delta p^L$, and the whole-space de Rham argument applied to the momentum equation shows that $p - p^L$ is a function of time. Thus replacing p by p^L is only a pressure-gauge change and does not alter suitability. This gauge choice uses the finite-energy setting to define the representative; the uniform constants below, in particular B_{uloc} and B_A, do not contain $\|u\|_{L^\infty(0,T;L^2(\mathbb{R}^3))}$ or $\int_0^T \|\nabla u(t)\|^2_{L^2(\mathbb{R}^3)}\,dt$.

Lemma 6.2 (Uniformly-local energy propagation under a Type I envelope)**.** *Let (v,π) be a suitable weak solution on $\mathbb{R}^3 \times (-4,0)$ with $v(s) \in L^2(\mathbb{R}^3)$ for a.e. s, and take π to be the whole-space Leray pressure representative, modulo functions of time. Assume that*

$$|v(x,s)| \le M(-s)^{-1/2}, \qquad \text{for a.e. } (x,s) \in \mathbb{R}^3 \times (-4,0), \tag{6.2}$$

and that for some admissible initial time $s_0 \in (-4,-3)$,

$$\alpha_0 := \sup_{a\in\mathbb{R}^3} \int_{B_2(a)} |v(x,s_0)|^2\,dx \le K. \tag{6.3}$$

Then there is a constant $B_{\mathrm{uloc}} = B_{\mathrm{uloc}}(M,K) < \infty$ such that

$$\operatorname*{ess\,sup}_{-1<s<0} \sup_{a\in\mathbb{R}^3} \int_{B_1(a)} |v(x,s)|^2\,dx + \sup_{a\in\mathbb{R}^3} \int_{-1}^0 \int_{B_1(a)} |\nabla v|^2\,dx\,ds \le B_{\mathrm{uloc}}. \tag{6.4}$$

Proof. The estimate supplies the terminal no-escape bound for Theorem 6.5. Let

$$\alpha(s) = \sup_{a\in\mathbb{R}^3} \int_{B_1(a)} |v(x,s)|^2\,dx.$$

By (6.2),

$$\alpha(s)^{1/2} \le CM(-s)^{-1/2}, \qquad -4 < s < 0. \tag{6.5}$$

Fix $a \in \mathbb{R}^3$ and choose $\phi_a \in C_c^\infty(B_2(a))$ with $0 \le \phi_a \le 1$, $\phi_a \equiv 1$ on $B_1(a)$, and $|\nabla\phi_a| + |\Delta\phi_a| \le C$, uniformly in a. Applying the local energy inequality on (s_0,t), using the standard monotone approximation of temporal cutoffs and with the pressure gauge chosen below, gives for a.e. $t \in (s_0,0)$

$$\begin{aligned}
&\int |v(t)|^2\phi_a\,dx + 2\int_{s_0}^t \int |\nabla v|^2\phi_a\,dx\,ds \\
&\quad\le \int |v(s_0)|^2\phi_a\,dx + C\int_{s_0}^t \int_{B_2(a)} |v|^2\,dx\,ds + C\int_{s_0}^t \int_{B_2(a)} |v|^3\,dx\,ds \\
&\qquad + C\int_{s_0}^t \int_{B_2(a)} |\pi - c_a(s)|\,|v|\,dx\,ds.
\end{aligned} \tag{6.6}$$

Here $c_a(s)$ is an arbitrary function of time; adding or subtracting such a function does not change the local energy inequality, since $\nabla\cdot v = 0$ and ϕ_a is compactly supported.

The velocity terms are controlled by the uniformly local kinetic energy. Since $B_2(a)$ is covered by finitely many unit balls,

$$\int_{B_2(a)} |v|^2\,dx \le C\alpha(s),$$

and by (6.2),

$$\int_{B_2(a)} |v|^3\,dx \le M(-s)^{-1/2} \int_{B_2(a)} |v|^2\,dx \le CM(-s)^{-1/2}\alpha(s). \tag{6.7}$$

It remains to estimate the pressure term uniformly in a.

Use the whole-space pressure representative $\pi = \mathcal{R}_i\mathcal{R}_j(v_iv_j)$, modulo functions of time. Let $\chi_a \in C_c^\infty(B_8(a))$, $\chi_a \equiv 1$ on $B_4(a)$, and write inside $B_4(a)$

$$\pi = q_a + h_a, \qquad q_a = \mathcal{R}_i\mathcal{R}_j(\chi_a v_i v_j).$$

Then h_a is harmonic in $B_4(a)$. Calderon–Zygmund boundedness [10] and (6.2) give, for a.e. s,

$$\begin{aligned}\|q_a(s)\|_{L^2(B_4(a))} &\le C\|\chi_a v(s)\otimes v(s)\|_{L^2(\mathbb{R}^3)}\\ &\le CM(-s)^{-1/2}\|v(s)\|_{L^2(B_8(a))}\\ &\le CM(-s)^{-1/2}\alpha(s)^{1/2},\end{aligned}$$

where the last step again uses a finite covering by unit balls. Hence

$$\int_{B_2(a)} |q_a|\,|v|\,dx \le CM(-s)^{-1/2}\alpha(s). \tag{6.8}$$

For the harmonic part, set $c_a(s) = h_a(a,s)$. The kernel estimate

$$|K_{ij}(x-z) - K_{ij}(a-z)| \le C|x-a|\,|z-a|^{-4}, \qquad x\in B_2(a), \quad |z-a|>4,$$

where K_{ij} is the kernel of $\mathcal{R}_i\mathcal{R}_j$, gives

$$|h_a(x,s) - c_a(s)| \le C\sum_{m=2}^{\infty} 2^{-4m}\int_{2^m<|z-a|<2^{m+1}} |v(z,s)|^2\,dz.$$

The annulus $\{2^m < |z-a| < 2^{m+1}\}$ is covered by at most $C2^{3m}$ unit balls. Therefore

$$\|h_a(s) - c_a(s)\|_{L^\infty(B_2(a))} \le C\sum_{m=2}^{\infty} 2^{-m}\alpha(s) \le C\alpha(s). \tag{6.9}$$

Consequently,

$$\begin{aligned}\int_{B_2(a)} |h_a - c_a|\,|v|\,dx &\le C\alpha(s)\int_{B_2(a)} |v|\,dx\\ &\le C\alpha(s)^{3/2}\\ &\le CM(-s)^{-1/2}\alpha(s),\end{aligned} \tag{6.10}$$

where the last inequality is precisely the Type I absorbing estimate (6.5). Combining (6.8) and (6.10),

$$\int_{B_2(a)} |\pi - c_a(s)|\,|v|\,dx \le CM(-s)^{-1/2}\alpha(s). \tag{6.11}$$

Insert (6.7) and (6.11) into (6.6), then take the supremum over a. Since $(-s)^{-1/2} \in L^1(-4,0)$, Gronwall's inequality yields

$$\operatorname*{ess\,sup}_{s_0<s<0} \alpha(s) \le CK\exp\left(C\int_{s_0}^0 \bigl(1 + M(-s)^{-1/2}\bigr)\,ds\right) \le B_1(M,K).$$

Returning to (6.6), using this bound, and then letting $t \uparrow 0$ through admissible times gives the corresponding uniformly-local dissipation estimate on $(-1,0)$. This proves (6.4). □

Remark 6.3 (Local energy estimate as an absorbing bound). The preceding proof can be read as a three-term estimate for the local quantity $\alpha(s)$. The diffusion and transport terms are linear in α, with coefficient $1+M(-s)^{-1/2}$. The local pressure term is also linear after Calderon–Zygmund and the Type I envelope. The remaining nonlinear term is the harmonic pressure flux, which is $O(\alpha^{3/2})$; the pointwise Type I envelope implies $\alpha^{1/2} \le CM(-s)^{-1/2}$, reducing it to the same integrable linear coefficient. Thus large values of α cannot persist: the resulting differential inequality bounds α in terms of constants depending only on M and the initial uniformly local energy level.

Proposition 6.4 (Terminal kinetic tightness)**.** *Let (u, p, z_*), $z_* = (x_*, T)$, be a finite-energy Type I terminal singularity in the sense of Definition 6.1. Then there exist $r_0 > 0$ and $B_A < \infty$, with B_A depending only on the Type I constant M, such that*

$$A(z_*, r) \le B_A, \qquad 0 < r < r_0. \tag{6.12}$$

Moreover,

$$r^{-1} \int_{T-r^2}^{T} \int_{B_r(x_*)} |\nabla u|^2 \, dx \, dt \le B_A, \qquad 0 < r < r_0. \tag{6.13}$$

Proof. Choose $r_0 > 0$ so small that $T - 4r^2 > T_0$ for $0 < r < r_0$. For such r, use the whole-space Leray pressure gauge fixed above and define the rescaled solution

$$v^{(r)}(y, s) = r u(x_* + ry, T + r^2 s), \qquad \pi^{(r)}(y, s) = r^2 p(x_* + ry, T + r^2 s),$$

on $\mathbb{R}^3 \times (-4, 0)$. The Type I bound (6.1) gives

$$|v^{(r)}(y, s)| \le M(-s)^{-1/2}, \qquad -4 < s < 0.$$

For any admissible $s_0 \in (-4, -3)$ at which the Type I bound holds, and any $a \in \mathbb{R}^3$,

$$\begin{aligned} \int_{B_2(a)} |v^{(r)}(y, s_0)|^2 \, dy &= r^{-1} \int_{B_{2r}(x_* + ra)} |u(x, T + r^2 s_0)|^2 \, dx \\ &\le r^{-1} |B_{2r}| \frac{M^2}{r^2 |s_0|} \le C M^2. \end{aligned}$$

Such times form a full-measure subset of $(-4, -3)$. The preceding estimate is uniform in this choice, so Lemma 6.2 applies with $K = CM^2$, uniformly in r. Hence

$$\operatorname*{ess\,sup}_{-1<s<0} \int_{B_1} |v^{(r)}(y, s)|^2 \, dy + \int_{-1}^{0} \int_{B_1} |\nabla v^{(r)}|^2 \, dy \, ds \le B_A.$$

Undoing the parabolic change of variables gives precisely (6.12) and (6.13). The constant B_A depends only on M; no global energy norm of u has been used in the estimate. ☐

Theorem 6.5 (Finite-energy velocity no-escape)**.** *Let (u, p, z_*), $z_* = (x_*, T)$, be a finite-energy Type I terminal singularity. Let $\lambda_k \downarrow 0$ be any sequence of scales, and set*

$$u_k(x, t) = \lambda_k u(x_* + \lambda_k x, T + \lambda_k^2 t).$$

Then

$$\lim_{\sigma \downarrow 0} \sup_k \int_{-\sigma}^{0} \int_{B_1} |u_k(x, t)|^3 \, dx \, dt = 0. \tag{6.14}$$

Proof. By Proposition 6.4, after discarding at most finitely many indices,

$$\operatorname*{ess\,sup}_{-1<t<0} \int_{B_1} |u_k(x, t)|^2 \, dx \le B_A.$$

The Type I bound gives, for the same tail of the sequence,

$$\|u_k(t)\|_{L^\infty(\mathbb{R}^3)} \le M(-t)^{-1/2}, \qquad -1 < t < 0.$$

Therefore, for $0 < \sigma < 1$,

$$\int_{-\sigma}^{0}\int_{B_1}|u_k|^3\,dx\,dt \le \int_{-\sigma}^{0}\|u_k(t)\|_{L^\infty(B_1)}\int_{B_1}|u_k(x,t)|^2\,dx\,dt$$
$$\le MB_A\int_{-\sigma}^{0}(-t)^{-1/2}\,dt = 2MB_A\sigma^{1/2}.$$

This bound is uniform over the tail. For each of the finitely many discarded indices k, choose $\sigma_k > 0$ so that $T+\lambda_k^2 t > T_0$ for $-\sigma_k < t < 0$. On this terminal subinterval the Type I envelope and the finite-energy assumption imply

$$\int_{-\sigma_k}^{0}\int_{B_1}|u_k|^3\,dx\,dt \le \int_{-\sigma_k}^{0}\|u_k(t)\|_{L^\infty(B_1)}\int_{B_1}|u_k(x,t)|^2\,dx\,dt < \infty.$$

Indeed, on $(-\sigma_k, 0)$,

$$\|u_k(t)\|_{L^\infty(B_1)} \le M(-t)^{-1/2}, \qquad \int_{B_1}|u_k(x,t)|^2\,dx \le \lambda_k^{-1}\|u(T+\lambda_k^2 t)\|_{L^2(\mathbb{R}^3)}^2,$$

and the right-hand side is finite for this fixed k by finite energy. Thus each discarded index has absolute continuity of its L^3-integral at the terminal slice. Taking the maximum over this finite set gives the same limit for the full supremum over k. This proves (6.14). □

Corollary 6.6 (Admissibility of finite-energy Type I concentration sequences)**.** *Let (u, p, z_*), $z_* = (x_*, T)$, be a finite-energy Type I terminal singularity. Then there exist $\eta_v > 0$ and $\lambda_k \downarrow 0$ such that*

$$C(z_*, \lambda_k) \ge \eta_v \qquad \text{for every } k,$$

and the corresponding rescaled velocities satisfy the no-escape condition (6.14). Hence the concentration and no-escape admissibility condition used in the Type I blow-up-limit theorem Theorem 9.1 follows for finite-energy Type I singularities.

Proof. Since $x_* \in \Sigma(T)$, Corollary 5.7 gives $\eta_v > 0$ and $\lambda_k \downarrow 0$ with $C(z_*, \lambda_k) \ge \eta_v$ for all k. The no-escape condition for this same sequence follows from Theorem 6.5. □

7. The Type I/Type II Blow-Up Rate Dichotomy

Let $x_0 \in \Sigma(T)$, $\eta > 0$, and $r_n \downarrow 0$ be given by Theorem 5.6. Define

$$u_n(y,s) = r_n u(x_0 + r_n y, T + r_n^2 s), \qquad p_n(y,s) = r_n^2 p(x_0 + r_n y, T + r_n^2 s).$$

Then (u_n, p_n) are suitable weak solutions on expanding backward cylinders, and by Proposition 3.3 they retain a positive lower bound for the mean-subtracted local scale-invariant quantity on a fixed bounded cylinder.

Definition 7.1 (Type I concentration sequence)**.** Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T-\delta, T)$, and let $r_n \downarrow 0$ be a positive local concentration sequence at (x_0, T). The sequence is called Type I if the concentration point (x_0, T) satisfies a local Type I scale-invariant bound: there are $\rho > 0$ and $M < \infty$, with $\rho^2 < \delta$, such that

$$\operatorname*{ess\,sup}_{T-\rho^2<t<T}\sqrt{T-t}\,\|u(t)\|_{L^\infty(B_\rho(x_0))} \le M.$$

This is the Type I alternative.

Definition 7.2 (Local Type II alternative)**.** Let (u,p) be a suitable weak solution on $\mathbb{R}^3\times(T-\delta,T)$, and let $r_n \downarrow 0$ be a positive local concentration sequence at (x_0,T). The sequence belongs to the local Type II alternative if the concentration point (x_0,T) satisfies no local Type I scale-invariant bound. Equivalently, for every $\rho>0$ with $\rho^2<\delta$,

$$\operatorname*{ess\,sup}_{T-\rho^2<t<T} \sqrt{T-t}\,\|u(t)\|_{L^\infty(B_\rho(x_0))} = \infty.$$

Theorem 7.3 (Local blow-up rate dichotomy)**.** *Let (u,p) be a suitable weak solution on $\mathbb{R}^3\times(T-\delta,T)$, and suppose $\Sigma(T)\neq\emptyset$. Then there exist $x_0\in\Sigma(T)$, $\eta>0$, and $r_n\downarrow 0$ such that*

$$C((x_0,T),r_n) + D((x_0,T),r_n) \geq \eta \qquad \textit{for all } n.$$

The sequence r_n is a positive local concentration sequence. The corresponding concentration regime satisfies either the local Type I bound or the local Type II alternative. Consequently, once the vanishing case has been excluded by ε-regularity, every finite-time singular case belongs to precisely one of the two blow-up rate alternatives.

Proof. If the local scale-invariant quantities vanished at every point of $\Sigma(T)$, then Theorem 4.5 would give $\Sigma(T)=\emptyset$, contrary to the hypothesis. Therefore Theorem 5.6 gives $x_0\in\Sigma(T)$, $\eta>0$, and $r_n\downarrow 0$ with the stated lower bound. The rescaled sequence has a positive local scale-invariant lower bound by Proposition 3.3. By definition, either the Type I bound in Definition 7.1 holds, or the local Type II alternative in Definition 7.2 holds. □

Remark 7.4. This theorem is local. Its conclusion is the existence of a positive mean-subtracted scale-invariant concentration sequence and the dichotomy between a local Type I bound and failure of every such bound at the concentration point.

8. Local Pointwise Type I Reduction to Seregin Extraction

The following consequence of the weak-Serrin endpoint argument is used. The local pointwise Type I alternative in Definition 7.1 is strong enough, after rescaling at the singular point, to give compactness on every compact negative-time cylinder. The endpoint weak-Serrin implication reads: for suitable weak solutions the local pointwise Type I envelope belongs to $L^{2,\infty}_t L^\infty_x$, hence the Albritton–Barker weak Serrin local Type I estimate gives the scale-invariant A-bound on smaller cylinders and therefore terminal velocity no-escape. Thus positive local Type I concentration sequences are admissible for the local Seregin extraction in Lemma 8.5.

Throughout this section $z_*=(x_*,T)$, and

$$Q_1(0,0)=B_1(0)\times(-1,0).$$

Definition 8.1 (Admissible local Type I concentration sequence)**.** For a local pointwise Type I point $z_*=(x_*,T)$, a sequence $r_n\downarrow 0$ is an admissible local Type I concentration sequence if the rescaled velocities

$$u_n(x,t) = r_n u(x_*+r_n x, T+r_n^2 t)$$

satisfy, for some $\eta_v>0$,

$$\int_{Q_1(0,0)} |u_n|^3\,dx\,dt \geq \eta_v \qquad \text{for every } n,$$

and

$$\lim_{\sigma\downarrow 0}\sup_n \int_{-\sigma}^0\int_{B_1} |u_n|^3\,dx\,dt = 0.$$

Proposition 8.2 (Local pointwise Type I rescalings have terminal compactness). *Let (u,p) be a suitable weak solution on $\mathbb{R}^3 \times (0,T)$ satisfying the finite-energy bounds*

$$u \in L^\infty(0,T;L^2(\mathbb{R}^3)) \cap L^2(0,T;\dot H^1(\mathbb{R}^3)).$$

Let $z_* = (x_*,T)$ be a singular point, and assume that z_* satisfies the local pointwise Type I bound: there exist $\rho > 0$ and $M < \infty$ such that

$$\operatorname*{ess\,sup}_{T-\rho^2<t<T} \sqrt{T-t}\,\|u(t)\|_{L^\infty(B_\rho(x_*))} \le M. \tag{8.1}$$

For any sequence $r_n \downarrow 0$, define

$$u_n(x,t) = r_n u(x_* + r_n x, T + r_n^2 t), \qquad p_n(x,t) = r_n^2 p(x_* + r_n x, T + r_n^2 t). \tag{8.2}$$

Then the following hold.

(i) *If*

$$K = B_R \times [-S,-\sigma] \Subset \mathbb{R}^3 \times (-\infty,0), \qquad R,S<\infty, \quad \sigma>0,$$

then, for all sufficiently large n,

$$\|u_n\|_{L^\infty(K)} \le M\sigma^{-1/2}. \tag{8.3}$$

(ii) *On each such compact cylinder K, after subtracting time-dependent local pressure gauges $a_{n,K}(t)$, the sequence satisfies uniform local Seregin bounds*

$$\|u_n\|_{L_t^\infty L_x^2(K)} + \|\nabla u_n\|_{L^2(K)} + \|p_n - a_{n,K}(t)\|_{L^{3/2}(K)} \le C_K. \tag{8.4}$$

Here C_K may depend on K, M, ρ, and on the finite Leray energy level of the original solution, but it does not depend on n and does not use a terminal no-escape assumption.

(iii) *After passing to a subsequence, $(u_n, p_n - a_{n,K})$ converges on every compact $K \Subset \mathbb{R}^3 \times (-\infty,0)$ to an ancient suitable weak solution (U,Q), with*

$$u_n \to U \quad \text{strongly in } L^q(K), \qquad 1 \le q < \infty,$$

$$p_n - a_{n,K}(t) \rightharpoonup Q \quad \text{weakly in } L^{3/2}(K),$$

and

$$|U(x,t)| \le M(-t)^{-1/2}, \qquad t<0.$$

Proof. Fix $K = B_R \times [-S,-\sigma] \Subset \mathbb{R}^3 \times (-\infty,0)$. For large n,

$$r_n R < \rho/2, \qquad r_n^2 S < \rho^2/2.$$

Thus $x_* + r_n x \in B_\rho(x_*)$ and $T + r_n^2 t \in (T-\rho^2, T)$ for a.e. $(x,t) \in K$. The local pointwise Type I bound (8.1) gives

$$|u_n(x,t)| = r_n|u(x_* + r_n x, T + r_n^2 t)| \le r_n M(T-(T+r_n^2 t))^{-1/2} = M(-t)^{-1/2} \le M\sigma^{-1/2}.$$

This proves (8.3). It also gives uniform local $L_t^\infty L_x^2$ and $L_{x,t}^3$ bounds on K.

It remains to explain the pressure and dissipation bounds. Use the whole-space Leray pressure representative, which is legitimate in the finite-energy class up to addition of functions of time. Choose $\chi_R \in C_c^\infty(B_{8R})$, $\chi_R \equiv 1$ on B_{4R}, and write inside B_{4R}

$$p_n = q_n + h_n, \qquad q_n = \mathcal{R}_i \mathcal{R}_j(\chi_R u_{n,i} u_{n,j}).$$

Calderon–Zygmund boundedness and the local L^3-bound for u_n give a uniform $L^{3/2}$-bound for q_n. For h_n, take the gauge $a_{n,K}(t) = h_n(0,t)$. The harmonic pressure kernel estimate gives

$$|h_n(x,t) - a_{n,K}(t)| \le C_R \sum_{m\ge 2} 2^{-4m} \int_{2^m R<|z|<2^{m+1}R} |u_n(z,t)|^2\,dz, \qquad x \in B_{2R}.$$

For annuli whose physical image remains inside $B_\rho(x_*)$, the local Type I bound controls the L^2-mass by

$$CM^2\sigma^{-1}(2^m R)^3,$$

and the resulting series is geometric. For the outer annuli, the finite Leray energy gives

$$\int_{\mathbb{R}^3} |u_n(z,t)|^2\,dz = r_n^{-1}\int_{\mathbb{R}^3} |u(y,T+r_n^2 t)|^2\,dy,$$

while the first outer annulus has radius comparable to ρ/r_n. Hence the tail of the kernel series is bounded by

$$C_{\rho,R}\, r_n^3 \|u\|^2_{L^\infty(0,T;L^2)},$$

which is uniform in n. This proves the pressure-gauge part of (8.4).

Apply the local energy inequality for (u_n, p_n) with a cutoff equal to one on K and supported in a slightly larger compact cylinder. The local kinetic, cubic velocity, and pressure-flux terms are bounded by the preceding estimates, so $\|\nabla u_n\|_{L^2(K)} \le C_K$. The Navier–Stokes equation then bounds $\partial_t u_n$ locally in a negative Sobolev space. Aubin–Lions compactness and the pressure bound give the asserted Seregin subsequence convergence. Passing to the limit in the equations and local energy inequality gives an ancient suitable weak solution, and the pointwise Type I bound passes to the limit. □

Corollary 8.3 (Local pointwise Type I concentration sequences are admissible)**.** *Under the hypotheses of Proposition 8.2, let $r_n \downarrow 0$ be any sequence such that the rescaled velocities satisfy*

$$\int_{Q_1(0,0)} |u_n|^3\,dx\,dt \ge \eta_v > 0 \qquad \text{for all } n.$$

Then r_n is an admissible local Type I concentration sequence in the sense of Definition 8.1.

Proof. The displayed positive concentration lower bound is the first requirement in Definition 8.1. The Albritton–Barker local weak-Serrin Type I estimate implies that the local pointwise Type I envelope implies terminal velocity no-escape for every rescaled sequence [1]. This gives the second requirement in Definition 8.1. □

Lemma 8.4 (Local Type I concentration enters the Seregin extraction)**.** *Under the hypotheses of Proposition 8.2, let $r_n \downarrow 0$ be a velocity concentration sequence satisfying*

$$\int_{Q_1(0,0)} |u_n|^3\,dx\,dt \ge \eta_v > 0 \qquad \text{for all } n.$$

Then r_n is admissible by Corollary 8.3. There are $\sigma_0 \in (0,1)$ and $\eta_1 > 0$ such that

$$\int_{-1}^{-\sigma_0}\int_{B_1} |u_n|^3\,dx\,dt \ge \eta_1 \qquad \text{for all } n. \tag{8.5}$$

Consequently, the rescaled sequence retains positive local L^3 mass on a compact negative-time cylinder. In particular the associated Seregin extraction has no loss of the local L^3 mass needed for nontriviality.

Proof. The no-escape condition in Definition 8.1 gives $\sigma_0 > 0$ such that the terminal layer $B_1\times(-\sigma_0, 0)$ carries at most half of the fixed Q_1 velocity mass. This proves (8.5). The compactness part follows from Proposition 8.2 on the compact negative-time cylinder $B_1 \times [-1, -\sigma_0]$. Strong local L^3 convergence on this cylinder passes the positive mass to the ancient limit. Terminal no-escape is used only at this step.

Thus the local Type I concentration sequence is admissible for the Seregin extraction and retains positive compact L^3-mass away from the terminal slice. □

Lemma 8.5 (Local Seregin extraction from an admissible local Type I sequence)**.** *Under the hypotheses of Proposition 8.2, let $r_n \downarrow 0$ be an admissible local Type I concentration sequence in the sense of Definition 8.1. Then, after passing to a subsequence, the rescaled velocities produce a smooth bounded centered solution V of (9.3) on $\mathbb{R}^3 \times \mathbb{R}$. The solution V satisfies the normalization used for Seregin ancient limits: there are a compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$ and $\eta_0 > 0$ such that*

$$\iint_K |V|^3 \, dy \, d\tau \geq \eta_0.$$

Proof. Proposition 8.2 gives local Seregin compactness for the rescaled sequence on every compact cylinder $K \Subset \mathbb{R}^3 \times (-\infty, 0)$. The local Type I bound passes to the ancient limit as

$$|U(x,t)| \leq M(-t)^{-1/2}, \qquad t < 0.$$

Interior Serrin regularity then gives smoothness of U on $\mathbb{R}^3 \times (-\infty, 0)$ by the argument used in Lemma 13.2. The centered change of variables

$$y = \frac{x}{\sqrt{-t}}, \qquad \tau = -\log(-t), \qquad V(y,\tau) = \sqrt{-t}\, U(y\sqrt{-t}, t)$$

therefore gives a smooth bounded solution of the centered equation (9.3). The no-escape part of local admissibility and Lemma 8.4 give positive compact L^3-mass for U away from $t = 0$. On the compact image of that cylinder under $(x,t) = (e^{-\tau/2}y, -e^{-\tau})$, the factor $e^{-\tau}$ in

$$|U|^3 \, dx \, dt = e^{-\tau} |V|^3 \, dy \, d\tau$$

is bounded above and below by positive constants. Hence the positive lower bound for U becomes the displayed local L^3 nonvanishing for V. □

Theorem 8.6 (Conditional local Type I exclusion)**.** *Assume the residual-class hypothesis Hypothesis 16.1. Let (u, p) be a finite-energy suitable weak solution on $\mathbb{R}^3 \times (0, T)$. Then no singular point $z_* = (x_*, T)$ satisfying the local pointwise Type I bound (8.1) exists. More precisely, any such point would generate a bounded centered Type I Seregin limit with positive local concentration, and that limit is excluded by the Liouville alternatives established in the sequel or cited, together with Hypothesis 16.1.*

Proof. Assume Hypothesis 16.1. Suppose, toward a contradiction, that $z_* = (x_*, T)$ is a singular point satisfying (8.1). Choose the positive Q_1 velocity concentration sequence supplied by the local concentration theorem. By Corollary 8.3, this sequence satisfies terminal no-escape and is admissible. By Lemma 8.4, the local Type I sequence has local compactness and positive compact L^3-mass away from the terminal slice. By Lemma 8.5, the sequence generates a smooth bounded centered Type I Seregin limit with local L^3 nonvanishing. By the class exhaustion in Proposition 14.9, this limit lies either in one of the Liouville classes excluded by the stated Liouville theorems or in the remaining class $\mathcal{R}(\mathcal{S})$ for the collection $\mathcal{S}$ generated by the extraction. The Liouville classes are excluded by the small-amplitude, stationary, uniformly tight, and structure/decay Liouville results stated in Parts II–IV. The remaining class is empty by the residual-class hypothesis. This contradicts the local L^3 nonvanishing, so the assumed local Type I singular point cannot exist under Hypothesis 16.1. □

Corollary 8.7 (Conditional exclusion of the local Type I alternative)**.** *Assume the residual-class hypothesis Hypothesis 16.1. Then the local Type I alternative is excluded for every finite-energy singular point. Hence every finite-energy singular point belongs to the local Type II alternative in the dichotomy above.*

Proof. This is Theorem 8.6 restated in the terminology of the local Type I/local Type II dichotomy. □

Part II. Type I Blow-up Limits and the Residual Hypothesis

Overview. An admissible finite-energy Type I singularity for the three-dimensional Navier–Stokes equations is reduced to a normalized nonzero bounded ancient solution of the centered self-similar Navier–Stokes equation. The construction keeps track of pressure gauges, local compactness, stability of the local energy inequality, and the velocity concentration supplied by local Caffarelli–Kohn–Nirenberg theory. It yields a smooth bounded ancient solution for the centered equation on $\mathbb{R}^3 \times \mathbb{R}$, together with a quantitative local L^3 nonvanishing condition.

The possible normalized Seregin ancient limits are organized by the analytic hypotheses used in the class decomposition: small amplitude, stationary L^3, uniform L^3-tightness, and structure or decay assumptions covered by cited Liouville theorems. The small-amplitude case is excluded by the bounded ancient perturbative Liouville theorem in Lemma 15.3, and the stationary L^3 case by the theorem of Nečas–Růžička–Šverák. The structure/decay class contains only hypotheses for which the cited Liouville theorems force $V \equiv 0$. The uniformly L^3-tight class is closed in Theorem 25.6. The remaining class is not proved empty; its emptiness is the residual-class hypothesis Hypothesis 16.1. Consequently the Type I conclusion obtained from this extraction is conditional on that residual hypothesis within the stated admissible finite-energy class.

9. Introduction and main results

9.1. Type I blow-up limits.

Let $u : \mathbb{R}^3 \times (0, T) \to \mathbb{R}^3$ solve the incompressible Navier–Stokes equations

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \Delta u, \qquad \operatorname{div} u = 0. \tag{9.1}$$

The local regularity theory of Caffarelli–Kohn–Nirenberg [2] detects singular points through scale-invariant concentration of velocity and pressure. A terminal point $z_* = (x_*, T)$ is of Type I if

$$\limsup_{t \uparrow T} \sqrt{T - t}\, \|u(t)\|_{L^\infty(\mathbb{R}^3)} < \infty. \tag{9.2}$$

The blow-up extraction uses a selected velocity concentration sequence from the local concentration theorem and assumes only that this velocity concentration does not escape into the terminal slice after rescaling. The resulting admissibility condition is defined in Definition 10.4. It replaces a stronger all-scale compactness assumption on scale-invariant energy, pressure, and dissipation quantities by the non-escape condition needed for nontriviality of the ancient limit.

9.2. Normalized Seregin limits.

For scales $\lambda \downarrow 0$, the parabolic rescaling about z_* is

$$u^{(\lambda)}(x, t) = \lambda u(x_* + \lambda x, T + \lambda^2 t), \qquad p^{(\lambda)}(x, t) = \lambda^2 p(x_* + \lambda x, T + \lambda^2 t).$$

The rescaled time intervals exhaust $(-\infty, 0)$. The Type I bound becomes a uniform bound

$$\|u^{(\lambda)}(t)\|_{L^\infty(\mathbb{R}^3)} \le M(-t)^{-1/2}$$

on compact subintervals of $(-\infty, 0)$, and compactness produces an unrenormalized ancient solution (U, Q). The associated centered variables are

$$y = \frac{x}{\sqrt{-t}}, \qquad \tau = -\log(-t), \qquad V(y, \tau) = \sqrt{-t}\, U(y\sqrt{-t}, t).$$

They transform the ancient Navier–Stokes solution into the centered equation

$$\partial_\tau V + (V \cdot \nabla)V + \nabla \Pi = \Delta V - \frac{1}{2}(V + y \cdot \nabla V), \qquad \operatorname{div} V = 0. \tag{9.3}$$

The selected velocity concentration at the singular point persists through the extraction and gives a compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$ and a constant $\eta_0 > 0$ such that

$$\iint_K |V|^3 \, dy \, d\tau \geq \eta_0. \tag{9.4}$$

9.3. **Liouville alternatives and the remaining case.** No unconditional Liouville theorem is known for all bounded ancient solutions of (9.3). The reduction therefore uses a finite decomposition based on explicit Liouville theorems. A normalized Seregin limit is compared with the following analytic classes:

$$\mathcal{C}_{\mathrm{small}}, \qquad \mathcal{C}_{\mathrm{stat},L^3}, \qquad \mathcal{C}_{L^3-\mathrm{tight}}, \qquad \mathcal{C}_{\mathrm{sd}}, \qquad \mathcal{R}(\mathcal{S}).$$

They denote, respectively, small amplitude, stationary L^3, uniform L^3-tightness, structure or decay covered by cited Liouville theorems (sd), and the complement of these four classes inside a fixed collection $\mathcal{S}$ of normalized limits. The last set carries no additional analytic hypothesis; it is the remaining case after the stated Liouville alternatives have been removed.

9.4. **Main theorems.**

Theorem 9.1 (Normalized Type I blow-up limit)**.** *Let (u, p, z_*) be an admissible finite-energy Type I singularity. Then there exist scales $\lambda_k \downarrow 0$, pressure gauges $a_k(t)$, an ancient suitable weak solution (U, Q) on $\mathbb{R}^3 \times (-\infty, 0)$, and a centered solution (V, Π) on $\mathbb{R}^3 \times \mathbb{R}$, such that, after passing to a subsequence,*

$$u_k \to U \quad \text{strongly in } L^q_{\mathrm{loc}}(\mathbb{R}^3 \times (-\infty, 0)), \qquad 1 \leq q < \infty,$$

and

$$p_k - a_k(t) \rightharpoonup Q \quad \text{weakly in } L^{3/2}_{\mathrm{loc}}(\mathbb{R}^3 \times (-\infty, 0)).$$

The centered velocity V is smooth, bounded, solves (9.3), and satisfies the local L^3 nonvanishing condition (9.4).

For each $V \in \mathcal{S}$, the class definitions give the exhaustive alternative

$$V \in \mathcal{C}_{\mathrm{small}} \cup \mathcal{C}_{\mathrm{stat},L^3} \cup \mathcal{C}_{L^3-\mathrm{tight}} \cup \mathcal{C}_{\mathrm{sd}} \cup \mathcal{R}(\mathcal{S}).$$

This exhaustive alternative is recorded in Proposition 14.9. It is a direct consequence of the definition of the remaining class, not an independent Liouville theorem.

Theorem 9.2 (Conditional Type I exclusion)**.** *Assume the residual-class hypothesis Hypothesis 16.1. Then no admissible finite-energy Type I singularity exists.*

The argument uses one closure theorem and one explicit residual-class hypothesis. The uniformly L^3-tight class is excluded by Theorem 25.6. The remaining class is excluded only under the residual-class hypothesis Hypothesis 16.1. The structure/decay class is restricted to hypotheses covered by cited Liouville theorems.

The resulting implication is

$$\begin{gathered}\text{admissible finite-energy Type I singularity} \\ \Downarrow \\ \text{nonzero normalized Seregin limit} \\ \Downarrow \\ \text{small/stationary/tight/structure–decay/remaining alternative} \\ \Downarrow \\ \text{contradiction by the Liouville-class exclusions and the residual-class hypothesis.}\end{gathered}$$

- Small amplitude: excluded in Section 15.

- Stationary L^3: excluded by [17].
- Uniformly L^3-tight: proved in Theorem 15.6, using Theorem 25.6.
- Structure/decay: excluded by the cited theorems in Theorem 15.7.
- Remaining class: excluded under the residual-class hypothesis Hypothesis 16.1.

9.5. **Dependencies and conditional hypothesis.** The proof of the extraction theorem uses the local velocity concentration result stated as Proposition 10.7; no other result from the local concentration theory enters the blow-up construction. The uniformly L^3-tight Liouville theorem and the residual-class hypothesis are not used to construct the ancient limit. They are invoked only after the class exhaustion. The statement Theorem 15.6 follows from the endpoint L^3 closure in Theorem 25.6, and Hypothesis 16.1 is the residual-class hypothesis. The structure/decay class is restricted by definition to assumptions covered by the cited Liouville theorems. Thus the final Type I conclusion is conditional on the admissibility condition in Corollary 6.6, the extraction theorem, the Liouville-class exclusions, and the residual-class hypothesis.

9.6. **Organization.** Section 10 introduces suitable solutions, scale-invariant quantities, the admissibility condition, scaling, pressure gauges, and local concentration. Section 11 proves the compactness theorem for Type I rescalings away from the terminal time and constructs the ancient suitable weak limit. Section 12 proves persistence of concentration and nontriviality. Section 13 derives the centered equation and defines normalized Seregin limits and collections. Section 14 defines the Liouville classes and the remaining class. Section 15 proves the class exclusions and the uniformly tight theorem. Section 16 proves the conditional remaining-case criterion. Appendix 17 gives the elementary symmetries of the centered equation.

10. Suitable solutions, scale-invariant quantities, and local concentration

For $z_0 = (x_0, t_0)$, write
$$Q_r(z_0) = B_r(x_0) \times (t_0 - r^2, t_0).$$
When $z_0 = (0, 0)$, write $Q_r = Q_r(0, 0)$. The spatial average of f over $B_r(x_0)$ is
$$(f)_{B_r(x_0)}(t) = \frac{1}{|B_r|} \int_{B_r(x_0)} f(x, t)\, dx.$$
Throughout this part, $\mathbb{P}$ denotes the Helmholtz–Leray projection onto divergence-free vector fields on $\mathbb{R}^3$.

Definition 10.1 (Terminal singular point)**.** Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (0, T)$. A terminal point $z_* = (x_*, T)$ is regular if there are $r > 0$ and $K < \infty$ such that
$$\operatorname*{ess\,sup}_{Q_r(z_*)} |u| \le K.$$
It is singular otherwise.

Definition 10.2 (Suitable weak solution)**.** Let $\Omega \subset \mathbb{R}^3 \times \mathbb{R}$ be open. A pair (u, p) is a suitable weak solution of (9.1) in Ω if
$$u \in L^\infty_t L^2_x(\Omega') \cap L^2_t \dot{H}^1_x(\Omega'), \qquad p \in L^{3/2}(\Omega')$$

for every compact cylinder $\Omega' \Subset \Omega$, if (u,p) solves (9.1) distributionally, $\operatorname{div} u = 0$, and if the local energy inequality

$$\begin{aligned}
&\int_{\mathbb{R}^3} |u(x,t_2)|^2 \phi(x,t_2)\,dx + 2\int_{t_1}^{t_2}\int_{\mathbb{R}^3} |\nabla u|^2 \phi\,dx\,dt \\
&\le \int_{\mathbb{R}^3} |u(x,t_1)|^2 \phi(x,t_1)\,dx + \int_{t_1}^{t_2}\int_{\mathbb{R}^3} |u|^2(\partial_t\phi + \Delta\phi)\,dx\,dt \\
&\quad + \int_{t_1}^{t_2}\int_{\mathbb{R}^3} (|u|^2 + 2p)u\cdot\nabla\phi\,dx\,dt
\end{aligned} \tag{10.1}$$

holds for every non-negative $\phi \in C_c^\infty(\Omega)$ and almost every $t_1 < t_2$ for which the displayed terms are defined.

This is the Caffarelli–Kohn–Nirenberg notion of suitability [2]; the global finite-energy background is Leray's Leray–Hopf class [5].

Definition 10.3 (Suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0,T)$)**.** A suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0,T)$ is a suitable weak solution on $\mathbb{R}^3 \times (0,T)$ satisfying

$$u \in L^\infty(0,T;L^2(\mathbb{R}^3)) \cap L^2(0,T;\dot H^1(\mathbb{R}^3)), \qquad u \in C_w([0,T);L^2(\mathbb{R}^3)),$$

and the global Leray energy inequality. The pressure is taken in $L^{3/2}_{\mathrm{loc}}(\mathbb{R}^3 \times (0,T))$, modulo functions of time.

Define

$$A(u;z_0,r) = r^{-1} \operatorname*{ess\,sup}_{t_0-r^2<t<t_0} \int_{B_r(x_0)} |u(x,t)|^2\,dx, \tag{10.2}$$

$$C(u;z_0,r) = r^{-2}\iint_{Q_r(z_0)} |u|^3\,dx\,dt, \qquad D(p;z_0,r) = r^{-2}\iint_{Q_r(z_0)} |p-(p)_{B_r(x_0)}(t)|^{3/2}\,dx\,dt, \tag{10.3}$$

$$E(u;z_0,r) = r^{-1}\iint_{Q_r(z_0)} |\nabla u|^2\,dx\,dt, \tag{10.4}$$

Definition 10.4 (Admissible finite-energy Type I singularity)**.** A triple (u,p,z_*), with $z_* = (x_*,T)$, is an admissible finite-energy Type I singularity if:

(i) (u,p) is a suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0,T)$;
(ii) $z_* = (x_*,T)$ is a singular point;
(iii) the Type I bound (9.2) holds;
(iv) there are scales $\lambda_k \downarrow 0$ and a constant $\eta_v > 0$ such that

$$C(u;z_*,\lambda_k) \ge \eta_v \qquad \text{for every } k;$$

(v) for the corresponding rescaled velocities

$$u_k(x,t) = \lambda_k u(x_* + \lambda_k x, T + \lambda_k^2 t),$$

the velocity no-escape condition holds:

$$\lim_{\sigma\downarrow 0}\sup_k \int_{-\sigma}^0 \int_{B_1} |u_k(x,t)|^3\,dx\,dt = 0. \tag{10.5}$$

The sequence $\{\lambda_k\}$ is called an admissible concentration sequence at z_*.

Remark 10.5. Item (iv) is the velocity-only concentration supplied by the local concentration theorem; see Proposition 10.7. Thus, for a singular point, the additional admissibility requirement is the no-escape condition (10.5). It is the sole terminal-slice compactness assumption used to prove nontriviality of the ancient limit. No uniform bound on $A + C + D + E$ over all terminal scales is assumed.

Lemma 10.6 (Uniform kinetic energy implies no escape)**.** *Let (u, p) satisfy the Type I bound (9.2), and let $\lambda_k \downarrow 0$. Suppose that*

$$\sup_k A(u; z_*, \lambda_k) \leq B < \infty.$$

Then the corresponding rescaled velocities satisfy the no-escape condition (10.5).

Proof. Choose $M < \infty$ and $t_0 < T$ such that

$$\|u(t)\|_{L^\infty(\mathbb{R}^3)} \leq M(T - t)^{-1/2}, \qquad t_0 < t < T.$$

For k sufficiently large, $T + \lambda_k^2 t > t_0$ whenever $-1 < t < 0$. Hence

$$\|u_k(t)\|_{L^\infty(B_1)} \leq M(-t)^{-1/2}, \qquad -1 < t < 0.$$

By scale invariance of A,

$$\operatorname*{ess\,sup}_{-1<t<0} \int_{B_1} |u_k(x, t)|^2 \, dx = A(u; z_*, \lambda_k) \leq B.$$

Therefore, for $0 < \sigma < 1$,

$$\begin{aligned} \int_{-\sigma}^0 \int_{B_1} |u_k|^3 \, dx \, dt &\leq \int_{-\sigma}^0 \|u_k(t)\|_{L^\infty(B_1)} \int_{B_1} |u_k(x, t)|^2 \, dx \, dt \\ &\leq MB \int_{-\sigma}^0 (-t)^{-1/2} \, dt = 2MB\sigma^{1/2}. \end{aligned}$$

Taking the supremum over the tail $k \geq k_0$ and then letting $\sigma \downarrow 0$ gives (10.5) for the tail of the sequence. For each of the finitely many remaining indices $1 \leq k < k_0$, the rescaled velocity is defined on $B_1 \times (-\delta_k, 0)$ for some $\delta_k > 0$, and belongs to L^3 there. Therefore absolute continuity of the L^3-integral gives

$$\lim_{\sigma \downarrow 0} \max_{1 \leq k < k_0} \int_{-\sigma}^0 \int_{B_1} |u_k|^3 \, dx \, dt = 0.$$

Combining the finite initial segment with the tail proves the full supremum in (10.5). □

Proposition 10.7 (Local velocity concentration)**.** *Let (u, p) be suitable in a parabolic neighborhood of $z_* = (x_*, T)$. If z_* is singular, then there exist $\lambda_k \downarrow 0$ and $\eta_v > 0$ such that*

$$C(u; z_*, \lambda_k) \geq \eta_v \qquad \text{for every } k.$$

Proof. This is the velocity-only concentration corollary of the local concentration result in Theorem 7.3. That result follows from the velocity ε-regularity criterion: if

$$\limsup_{r \downarrow 0} C(u; z_*, r) = 0,$$

then u is bounded in a backward parabolic cylinder ending at z_*, contradicting singularity of z_*. □

Lemma 10.8 (Scaling)**.** *Let (u,p) be suitable in a neighborhood of (x_*,T). For $\lambda>0$, set*

$$u^\lambda(x,t)=\lambda u(x_*+\lambda x,T+\lambda^2 t),\qquad p^\lambda(x,t)=\lambda^2 p(x_*+\lambda x,T+\lambda^2 t).$$

Then (u^λ,p^λ) is suitable on the rescaled domain. Moreover A,C,D,E are invariant under this scaling.

Proof. Suitability of the rescaled pair and preservation of pressure gauges follow by applying (10.1) to the test function

$$\phi^\lambda(X,S)=\phi\left(\frac{X-x_*}{\lambda},\frac{S-T}{\lambda^2}\right)$$

and then changing variables $(X,S)=(x_*+\lambda x,T+\lambda^2 t)$. The differential identities

$$\partial_t u^\lambda=\lambda^3(\partial_t u)(X,S),\quad (u^\lambda\cdot\nabla)u^\lambda=\lambda^3((u\cdot\nabla)u)(X,S),$$
$$\nabla p^\lambda=\lambda^3(\nabla p)(X,S),\qquad \Delta u^\lambda=\lambda^3(\Delta u)(X,S)$$

give the rescaled equation and the divergence condition.

For the pressure average,

$$(p^\lambda)_{B_r}(t)=\lambda^2(p)_{B_{\lambda r}(x_*)}(S).$$

Consequently

$$\iint_{Q_r}|u^\lambda|^3\,dx\,dt=\lambda^{-2}\iint_{Q_{\lambda r}(z_*)}|u|^3\,dX\,dS$$

and

$$\iint_{Q_r}|p^\lambda-(p^\lambda)_{B_r}(t)|^{3/2}\,dx\,dt=\lambda^{-2}\iint_{Q_{\lambda r}(z_*)}|p-(p)_{B_{\lambda r}(x_*)}(S)|^{3/2}\,dX\,dS.$$

Multiplication by r^{-2} therefore gives invariance of C and D. The remaining two quantities have the same critical dimensional balance:

$$\int_{B_r}|u^\lambda(x,t)|^2\,dx=\lambda^{-1}\int_{B_{\lambda r}(x_*)}|u(X,S)|^2\,dX,$$
$$\iint_{Q_r}|\nabla u^\lambda|^2\,dx\,dt=\lambda^{-1}\iint_{Q_{\lambda r}(z_*)}|\nabla u|^2\,dX\,dS.$$

Multiplication by r^{-1} on the rescaled cylinder is the same as multiplication by $(\lambda r)^{-1}$ on the original cylinder. Hence A and E are invariant as well. □

Lemma 10.9 (Pressure gauges do not change the equations)**.** *Let (v,q) be suitable in a cylinder Q, and let $\alpha(t)\in L^{3/2}_{\mathrm{loc}}$ depend only on time. Then $(v,q-\alpha(t))$ satisfies the same distributional Navier–Stokes equation, the same divergence condition, and the same local energy inequality.*

Proof. In the distributional equation, replacing q by $q-\alpha(t)$ adds $-\nabla\alpha(t)=0$. In the local energy inequality the only new term is

$$-2\int \alpha(t)v(x,t)\cdot\nabla\phi(x,t)\,dx\,dt.$$

The product is integrable on the support of ϕ: by suitability, $v\in L^\infty_t L^2_x$ locally, and

$$\left|\int v(\cdot,t)\cdot\nabla\phi(\cdot,t)\,dx\right|\le \|v(t)\|_{L^2(\operatorname{supp}\phi(t))}\|\nabla\phi(t)\|_{L^2}.$$

For almost every t, $\operatorname{div}v(\cdot,t)=0$ in distributions and $\phi(\cdot,t)$ is compactly supported, so

$$\int v(x,t)\cdot\nabla\phi(x,t)\,dx=0.$$

Thus the additional term vanishes. □

11. The Type I blow-up sequence and compactness

11.1. The rescaled sequence. Choose $M < \infty$ so that, after decreasing $T_0 < T$ if necessary,

$$\|u(t)\|_{L^\infty(\mathbb{R}^3)} \le \frac{M}{\sqrt{T-t}}, \qquad T_0 < t < T. \tag{11.1}$$

For $\lambda_k \downarrow 0$, define

$$u_k(x,t) = \lambda_k u(x_* + \lambda_k x, T + \lambda_k^2 t), \qquad p_k(x,t) = \lambda_k^2 p(x_* + \lambda_k x, T + \lambda_k^2 t). \tag{11.2}$$

Then u_k is defined on $\mathbb{R}^3 \times (-T/\lambda_k^2, 0)$, and these domains exhaust $\mathbb{R}^3 \times (-\infty, 0)$.

11.2. Inherited Type I bounds.

Lemma 11.1 (Inherited Type I bound)**.** *For every compact interval $I \Subset (-\infty, 0)$,*

$$\sup_k \|u_k\|_{L^\infty(\mathbb{R}^3 \times I)} < \infty.$$

More precisely, for every fixed $t < 0$ and all sufficiently large k,

$$\|u_k(t)\|_{L^\infty(\mathbb{R}^3)} \le \frac{M}{\sqrt{-t}}.$$

Proof. Since $T - (T + \lambda_k^2 t) = \lambda_k^2(-t)$, (11.1) gives, for all large k,

$$\|u_k(t)\|_\infty = \lambda_k \|u(T + \lambda_k^2 t)\|_\infty \le \lambda_k \frac{M}{\sqrt{\lambda_k^2(-t)}} = \frac{M}{\sqrt{-t}}.$$

Taking the supremum over $t \in I$ gives the interval statement. □

11.3. Local energy, pressure, and time-derivative estimates.

Lemma 11.2 (Pressure gauges and local pressure bounds)**.** *After fixing the whole-space Leray pressure representative below, set*

$$a_k(t) = (p_k)_{B_1}(t)$$

whenever the rescaled solution is defined. Then for every

$$Q = B_R \times [-S, -\sigma] \Subset \mathbb{R}^3 \times (-\infty, 0)$$

one has

$$\sup_k \|p_k - a_k(t)\|_{L^{3/2}(Q)} < \infty.$$

Proof. For large k, the original times $T + \lambda_k^2 t$, $t \in [-S, -\sigma]$, lie in the interval where (11.1) holds. On those slices $u(t) \in L^2(\mathbb{R}^3) \cap L^\infty(\mathbb{R}^3)$, hence

$$u_i(t)u_j(t) \in L^1(\mathbb{R}^3) \cap L^{3/2}(\mathbb{R}^3).$$

Use the whole-space Leray pressure representative

$$p^L = \mathcal{R}_i\mathcal{R}_j(u_i u_j),$$

modulo functions of time. The standard convention is

$$\widehat{\mathcal{R}_i f}(\xi) = i\xi_i|\xi|^{-1}\widehat{f}(\xi),$$

so that $\mathcal{R}_i\mathcal{R}_j$ has multiplier $-\xi_i\xi_j|\xi|^{-2}$ and $-\Delta(\mathcal{R}_i\mathcal{R}_j f) = \partial_i\partial_j f$. Thus this is the pressure reconstruction from $-\Delta p = \partial_i\partial_j(u_i u_j)$ and the Calderon–Zygmund theory of Riesz transforms [10, 19]. Since $u(t) \in L^2(\mathbb{R}^3) \cap L^\infty(\mathbb{R}^3)$, interpolation gives $u(t) \in L^3(\mathbb{R}^3)$, hence $u_i(t)u_j(t) \in L^{3/2}(\mathbb{R}^3)$ and $p^L(t) \in L^{3/2}(\mathbb{R}^3)$ for a.e. t.

The given pressure and p^L differ by a function of time. Indeed, applying the Helmholtz projection to the whole-space weak momentum equation gives

$$\partial_t u - \Delta u + \mathbb{P}\operatorname{div}(u \otimes u) = 0$$

in solenoidal distributions. The complementary gradient part of $\operatorname{div}(u \otimes u)$ is precisely $-\nabla p^L$ with the convention above. Therefore the same velocity field solves the distributional equation with p^L in place of p. Subtracting the two formulations gives

$$\nabla(p - p^L) = 0 \quad \text{in } \mathcal{D}'(\mathbb{R}^3)$$

for a.e. time. By the de Rham theorem on $\mathbb{R}^3$, $p - p^L$ is spatially constant for a.e. time. Replacing the pressure by p^L is therefore a time-dependent gauge change and does not affect suitability by Lemma 10.9.

Fix $\chi_R \in C_c^\infty(B_{8R})$ with $\chi_R = 1$ on B_{4R}. For $t \in [-S, -\sigma]$, choose the representative so that, in B_{2R},

$$p_k = q_k + h_k, \qquad q_k = \mathcal{R}_i\mathcal{R}_j(\chi_R u_{k,i} u_{k,j}),$$

and $h_k(\cdot, t)$ is harmonic in B_{4R} for a.e. t. This choice is made before $a_k(t) = (p_k)_{B_1}(t)$ is evaluated. The harmonicity holds because $(1 - \chi_R)u_{k,i}u_{k,j}$ is supported outside B_{4R}, so the localized Riesz potential generated by this exterior part has zero Laplacian in B_{4R}. Calderon–Zygmund boundedness gives, for a.e. t,

$$\|q_k(t)\|_{L^{3/2}(B_{2R})} \le C\|u_k(t)\|^2_{L^3(B_{8R})} \le C(R, \sigma, M).$$

Integrating over $[-S, -\sigma]$ gives a uniform $L^{3/2}_{x,t}$-bound for q_k. Let $c_{k,R}(t) = h_k(0, t)$, where $h_k(\cdot, t)$ is represented by its smooth harmonic representative. For $x \in B_R$, the kernel estimate

$$|K_{ij}(x - z) - K_{ij}(-z)| \le CR|z|^{-4}, \qquad |z| > 4R,$$

implies

$$|h_k(x, t) - c_{k,R}(t)| \le CR \sum_{m=2}^{\infty} (2^m R)^{-4} \int_{2^m R < |z| < 2^{m+1} R} |u_k(z, t)|^2 \, dz.$$

Using Lemma 11.1,

$$\int_{2^m R < |z| < 2^{m+1} R} |u_k(z, t)|^2 \, dz \le CM^2 \sigma^{-1} (2^m R)^3,$$

and therefore

$$|h_k(x, t) - c_{k,R}(t)| \le CM^2 \sigma^{-1} \sum_{m=2}^{\infty} 2^{-m}, \qquad x \in B_R,$$

which implies

$$\|h_k(t) - c_{k,R}(t)\|_{L^\infty(B_R)} \le C(M, \sigma).$$

Thus

$$\sup_k \|p_k - c_{k,R}(t)\|_{L^{3/2}(B_R \times [-S, -\sigma])} < \infty.$$

If $R < 1$, enlarge R to 1. Otherwise $B_1 \subset B_R$. Jensen's inequality gives, for a.e. t,

$$|a_k(t) - c_{k,R}(t)|^{3/2} \le |B_1|^{-1} \int_{B_1} |p_k - c_{k,R}(t)|^{3/2} \, dx.$$

Integrating in time and adding constants on B_R gives the desired bound for $p_k - a_k(t)$. $\square$

Lemma 11.3 (Compatibility of pressure gauges)**.** *Let $B_1 \subset B_R$, $I \Subset (-\infty, 0)$, and suppose*

$$p_k - c_k(t) \quad \textit{is bounded in } L^{3/2}(B_R \times I)$$

for some time-dependent gauges $c_k(t)$. If $a_k(t) = (p_k)_{B_1}(t)$, then $p_k - a_k(t)$ is bounded in $L^{3/2}(B_{R'} \times I)$ for every $R' \leq R$. Changing between such gauges changes the pressure only by a function of time and therefore does not affect the equation, the local energy inequality, or the centered equation.

Proof. For a.e. t, Jensen's inequality gives

$$|a_k(t) - c_k(t)|^{3/2} \leq |B_1|^{-1} \int_{B_1} |p_k - c_k(t)|^{3/2}\, dx.$$

Integrating in t and using $B_{R'} \subset B_R$ yields

$$\|a_k - c_k\|^{3/2}_{L^{3/2}(I)} \leq |B_1|^{-1} \|p_k - c_k(t)\|^{3/2}_{L^{3/2}(B_R \times I)}.$$

Thus

$$\|p_k - a_k(t)\|_{L^{3/2}(B_{R'} \times I)} \leq \|p_k - c_k(t)\|_{L^{3/2}(B_{R'} \times I)} + |B_{R'}|^{2/3} \|a_k - c_k\|_{L^{3/2}(I)}$$

and the right-hand side is bounded by

$$C(R') \|p_k - c_k(t)\|_{L^{3/2}(B_R \times I)}.$$

This proves the asserted bound. The last assertion is Lemma 10.9, and centering transforms the time-dependent gauge into another time-dependent gauge. □

Lemma 11.4 (Local energy and dissipation)**.** *Let*

$$Q = B_R \times [-S, -\sigma] \Subset \mathbb{R}^3 \times (-\infty, 0).$$

Then

$$\sup_k \|u_k\|_{L^\infty_t L^2_x(Q)} + \sup_k \|\nabla u_k\|_{L^2(Q)} < \infty.$$

Proof. The $L^\infty_t L^2_x$ estimate follows from

$$\|u_k(t)\|_{L^2(B_R)} \leq |B_R|^{1/2} M \sigma^{-1/2}, \qquad t \in [-S, -\sigma].$$

Choose $\phi \in C_c^\infty(B_{2R} \times [-2S, -\sigma/2])$, $0 \leq \phi \leq 1$, with $\phi = 1$ on Q. Apply the local energy inequality to $(u_k, p_k - a_k(t))$, using Lemma 11.2 on the larger cylinder. For a.e. admissible times and then by monotone approximation of temporal cutoffs,

$$2 \iint |\nabla u_k|^2 \phi \leq \iint |u_k|^2 |\partial_t \phi + \Delta \phi| + \iint (|u_k|^2 + 2|p_k - a_k(t)|) |u_k| |\nabla \phi|.$$

The first term is bounded because $\partial_t \phi + \Delta \phi$ is bounded and u_k is uniformly bounded in L^∞ on the finite-measure support of ϕ. The cubic velocity term satisfies

$$\iint |u_k|^3 |\nabla \phi| \leq \|\nabla \phi\|_\infty |\operatorname{supp} \nabla \phi|\, \|u_k\|^3_{L^\infty(\operatorname{supp} \nabla \phi)} \leq C(Q, \phi, M, \sigma).$$

For the pressure term, Hölder's inequality on $\operatorname{supp} \nabla \phi$ gives

$$\iint |p_k - a_k(t)|\, |u_k|\, |\nabla \phi| \leq \|p_k - a_k(t)\|_{L^{3/2}} \|u_k \nabla \phi\|_{L^3},$$

and both factors are uniformly bounded by Lemma 11.2 and Lemma 11.1. Since $\phi = 1$ on Q, the dissipation bound follows. □

Lemma 11.5 (Time derivative bound)**.** *For every compact cylinder $Q \Subset \mathbb{R}^3 \times (-\infty, 0)$,*

$$\partial_t u_k \quad \textit{is bounded in } L^{3/2}_t W^{-1,3/2}_x(Q).$$

Proof. The equation is

$$\partial_t u_k = \Delta u_k - \operatorname{div}(u_k \otimes u_k) - \nabla(p_k - a_k(t)).$$

Let $Q = B_R \times [-S, -\sigma]$. The term Δu_k is bounded in $L^2_t H^{-1}_x(Q)$ by Lemma 11.4; since $W^{1,3}_0(B_R) \subset H^1_0(B_R)$ and the time interval is finite, this is also an $L^{3/2}_t W^{-1,3/2}_x$-bound. Explicitly, for $\zeta \in W^{1,3}_0(B_R)$,

$$|\langle \Delta u_k, \zeta \rangle| = \left| \int_{B_R} \nabla u_k : \nabla \zeta \, dx \right| \le C(R) \|\nabla u_k\|_{L^2(B_R)} \|\zeta\|_{W^{1,3}(B_R)}.$$

The L^2_t-bound on ∇u_k, together with the finite length of $[-S, -\sigma]$, gives the claimed $L^{3/2}_t$-bound.

For the nonlinear term, for $\zeta \in W^{1,3}_0(B_R)$,

$$\left| \int_{B_R} (u_k \otimes u_k) : \nabla \zeta \, dx \right| \le \|u_k \otimes u_k\|_{L^{3/2}(B_R)} \|\nabla \zeta\|_{L^3(B_R)}.$$

Moreover

$$\|u_k \otimes u_k\|_{L^{3/2}(Q)} \le \|u_k\|_{L^\infty(Q)} \|u_k\|_{L^{3/2}(Q)} \le C(Q).$$

Thus $\operatorname{div}(u_k \otimes u_k)$ is bounded in $L^{3/2}_t W^{-1,3/2}_x(Q)$. For the pressure term,

$$\left| \int_{B_R} (p_k - a_k(t)) \operatorname{div} \zeta \, dx \right| \le \|p_k - a_k(t)\|_{L^{3/2}(B_R)} \|\nabla \zeta\|_{L^3(B_R)}.$$

Lemma 11.2 gives

$$p_k - a_k(t) \quad \text{bounded in } L^{3/2}(Q).$$

Hence $\nabla(p_k - a_k(t))$ is bounded in $L^{3/2}_t W^{-1,3/2}_x(Q)$. Summing the three terms gives the result. □

11.4. **Compactness away from the terminal time.**

Proposition 11.6 (Compactness)**.** *After passing to a subsequence,*

$$u_k \to U$$

strongly in $L^q_{\mathrm{loc}}(\mathbb{R}^3 \times (-\infty, 0))$ *for every finite* q*, and*

$$p_k - a_k(t) \rightharpoonup Q \quad \textit{weakly in } L^{3/2}_{\mathrm{loc}}(\mathbb{R}^3 \times (-\infty, 0)).$$

Proof. Fix $Q_0 \Subset \mathbb{R}^3 \times (-\infty, 0)$. By Lemma 11.4, u_k is bounded in $L^2_t H^1_x(Q_0)$, and by Lemma 11.5, $\partial_t u_k$ is bounded in $L^{3/2}_t W^{-1,3/2}_x(Q_0)$. On bounded balls,

$$H^1 \Subset L^2 \hookrightarrow W^{-1,3/2}.$$

The Aubin–Lions–Simon theorem [11, 14, 18] gives relative compactness in $L^2(Q_0)$. Hence, after passing to a subsequence, $u_k \to U$ strongly in $L^2(Q_0)$ and a.e. on Q_0.

The inherited Type I bound gives a common $L^\infty(Q_0)$ bound, and the limit belongs to the same L^∞ ball by a.e. convergence. For $q \ge 2$,

$$\|u_k - U\|^q_{L^q(Q_0)} \le (2 \sup_j \|u_j\|_{L^\infty(Q_0)} + 2\|U\|_{L^\infty(Q_0)})^{q-2} \|u_k - U\|^2_{L^2(Q_0)}.$$

For $1 \le q < 2$, Hölder's inequality on the finite-measure set Q_0 gives

$$\|u_k - U\|_{L^q(Q_0)} \le |Q_0|^{1/q - 1/2} \|u_k - U\|_{L^2(Q_0)} \to 0.$$

Choose an exhaustion $Q^{(m)} \Subset \mathbb{R}^3 \times (-\infty, 0)$ by compact cylinders. The preceding argument gives a subsequence converging on $Q^{(1)}$; extracting successively on $Q^{(m)}$ and taking the diagonal subsequence gives strong local convergence on every compact cylinder.

For pressure, Lemma 11.2 gives boundedness of $p_k - a_k(t)$ in $L^{3/2}(Q^{(m)})$ for every m. Since $L^{3/2}$ is reflexive, each restriction has a weakly convergent subsequence. The same diagonal extraction gives a single limit Q in $L^{3/2}_{\mathrm{loc}}$, because the weak limits agree on overlaps by uniqueness of distributional limits. □

11.5. **The ancient suitable weak limit.**

Lemma 11.7 (Limit equation)**.** *The limit (U, Q) solves*

$$\partial_t U + (U \cdot \nabla)U + \nabla Q = \Delta U, \qquad \operatorname{div} U = 0$$

in distributions on $\mathbb{R}^3 \times (-\infty, 0)$.

Proof. For $\varphi \in C_c^\infty(\mathbb{R}^3 \times (-\infty, 0); \mathbb{R}^3)$, the weak formulation for $(u_k, p_k - a_k(t))$ is

$$\iint [-u_k \cdot \partial_t \varphi - (u_k \otimes u_k) : \nabla\varphi + (p_k - a_k(t)) \operatorname{div} \varphi + \nabla u_k : \nabla\varphi] \, dx \, dt = 0.$$

The linear time term passes by strong L^2_{loc}-convergence of u_k. The pressure term passes by weak $L^{3/2}_{\mathrm{loc}}$-convergence of $p_k - a_k(t)$, since $\operatorname{div} \varphi \in L^3$. The viscous term passes by weak L^2_{loc}-convergence of ∇u_k, which follows from the uniform local dissipation bound. The nonlinear term passes because

$$\|u_k \otimes u_k - U \otimes U\|_{L^1(\operatorname{supp}\varphi)} \le \|u_k - U\|_{L^2}\big(\|u_k\|_{L^2} + \|U\|_{L^2}\big) \to 0.$$

Testing with gradients of scalar functions gives $\operatorname{div} U = 0$. □

Lemma 11.8 (Stability of suitability)**.** *The pair (U, Q) is a suitable weak solution on $\mathbb{R}^3 \times (-\infty, 0)$.*

Proof. Use the distributional form of the local energy inequality. For every non-negative $\phi \in C_c^\infty(\mathbb{R}^3 \times (-\infty, 0))$,

$$2\iint |\nabla u_k|^2 \phi \le \iint |u_k|^2(\partial_t \phi + \Delta\phi) + \iint (|u_k|^2 + 2P_k) u_k \cdot \nabla\phi, \qquad P_k = p_k - a_k(t).$$

This follows from (10.1) by applying (10.1) with temporal cutoffs which increase to 1 on the time projection of $\operatorname{supp}\phi$. The monotone convergence theorem passes the nonnegative dissipation term to the displayed distributional form.

The terms with $|u_k|^2$ converge strongly in L^1 because

$$\||u_k|^2 - |U|^2\|_{L^1(\operatorname{supp}\phi)} \le \|u_k - U\|_{L^2}\big(\|u_k\|_{L^2} + \|U\|_{L^2}\big) \to 0.$$

The cubic flux converges strongly since $u_k \to U$ strongly in L^3_{loc}, hence

$$|u_k|^2 u_k \to |U|^2 U \quad \text{in } L^1_{\mathrm{loc}}.$$

For the pressure flux, $u_k \cdot \nabla\phi \to U \cdot \nabla\phi$ strongly in L^3, while $P_k \rightharpoonup Q$ weakly in $L^{3/2}$. Therefore

$$\iint P_k u_k \cdot \nabla\phi \to \iint Q \, U \cdot \nabla\phi.$$

Finally, $\nabla u_k \rightharpoonup \nabla U$ in L^2_{loc}, and

$$w \mapsto \iint |w|^2 \phi$$

is weakly lower semicontinuous because $\phi \ge 0$ and the expression equals $\|\sqrt{\phi}\, w\|^2_{L^2}$. Passing to the limit gives the local energy inequality for (U, Q) in distributional form. The endpoint form in Definition 10.2 is recovered by applying the distributional inequality to standard time cutoffs approximating characteristic functions of (t_1, t_2); weak L^2_{loc}-continuity of local energy-class solutions gives the endpoint traces for a.e. $t_1 < t_2$. □

Lemma 11.9 (Inherited ancient Type I bound)**.** *For every compact interval $I = [-S, -\sigma] \Subset (-\infty, 0)$,*

$$\|U\|_{L^\infty(\mathbb{R}^3 \times I)} \le M\sigma^{-1/2}.$$

Proof. Lemma 11.1 gives

$$\|u_k\|_{L^\infty(\mathbb{R}^3 \times I)} \le M\sigma^{-1/2}.$$

Fix $R < \infty$. Since $u_k \to U$ a.e. on $B_R \times I$, the essential supremum of U on $B_R \times I$ cannot exceed $M\sigma^{-1/2}$. If it did, $|u_k|$ would exceed this bound by a positive amount on a positive measure subset for large k. Letting $R \to \infty$ through integers gives the result on $\mathbb{R}^3 \times I$. □

Proposition 11.10 (Ancient suitable Type I limit)**.** *The pair (U, Q) is an ancient suitable weak solution on $\mathbb{R}^3 \times (-\infty, 0)$ satisfying the interval Type I bound in Lemma 11.9.*

Proof. Combine Lemmas 11.7 to 11.9. □

12. Persistence of concentration and nontriviality

12.1. **Persistence of concentration away from the terminal time.** The persistence of concentration is tracked through the velocity concentration sequence in Definition 10.4, supplied by Proposition 10.7. The no-escape condition says that this velocity mass does not concentrate entirely in the terminal layer $t = 0$ after rescaling.

12.2. **Scale selection.**

Lemma 12.1 (Scale selection from local concentration)**.** *Let $\lambda_k \downarrow 0$ be an admissible concentration sequence in the sense of Definition 10.4. Then there are $\sigma \in (0, 1)$, a compact cylinder*

$$K_0 = B_1 \times (-1, -\sigma) \Subset \mathbb{R}^3 \times (-\infty, 0),$$

and $\eta_1 > 0$ such that

$$\iint_{K_0} |u_k|^3 \, dx \, dt \ge \eta_1 \tag{12.1}$$

for all k.

Proof. By scale invariance of C,

$$\int_{-1}^{0} \int_{B_1} |u_k|^3 \, dx \, dt = C(u; z_*, \lambda_k) \ge \eta_v.$$

The no-escape condition (10.5) gives $\sigma \in (0, 1)$ such that

$$\sup_k \int_{-\sigma}^{0} \int_{B_1} |u_k|^3 \, dx \, dt \le \frac{\eta_v}{2}.$$

Therefore

$$\int_{-1}^{-\sigma} \int_{B_1} |u_k|^3 \, dx \, dt \ge \frac{\eta_v}{2}$$

for every k. The assertion follows with $\eta_1 = \eta_v/2$. □

12.3. Nonzero ancient limits.

Proposition 12.2 (Nonzero ancient limit)**.** *For the scales chosen in Lemma 12.1, the limit U in Proposition 11.10 is nonzero. More precisely, there are a compact cylinder $K_0 \Subset \mathbb{R}^3 \times (-\infty, 0)$ and $\eta_1 > 0$ such that*

$$\iint_{K_0} |U|^3 \, dx \, dt \geq \eta_1.$$

Proof. Let $K_0 = B_1 \times (-1, -\sigma)$ and $\eta_1 > 0$ be given by Lemma 12.1. Since $K_0 \Subset \mathbb{R}^3 \times (-\infty, 0)$, the compactness theorem gives, after passing to the subsequence used to define U,

$$u_k \to U \qquad \text{strongly in } L^3(K_0).$$

Hence

$$\int_{K_0} |U|^3 \, dx \, dt = \lim_{k \to \infty} \int_{K_0} |u_k|^3 \, dx \, dt \geq \eta_1.$$

□

Thus concentration generated by the singular point propagates through the compactness limit. The velocity concentration and no-escape condition are the inputs that prevent the ancient limit from being identically zero.

13. Centered variables and normalized Seregin limits

13.1. The centered self-similar equation.

Set

$$y = \frac{x}{\sqrt{-t}}, \qquad \tau = -\log(-t), \qquad V(y, \tau) = \sqrt{-t}\, U(y\sqrt{-t}, t),$$

and

$$\Pi(y, \tau) = (-t) Q(y\sqrt{-t}, t),$$

up to addition of a function of τ.

Lemma 13.1 (Renormalized equation)**.** *The pair (V, Π) solves (9.3).*

Proof. Write $s = -t = e^{-\tau}$, $x = s^{1/2} y$, and

$$U(x, t) = s^{-1/2} V(y, \tau).$$

The identities are first obtained in distributions by testing the equation for U against the inverse change of variables. After Lemma 13.2, the same identities hold classically for the smooth representative. Because $s_t = -1$, $\tau_t = s^{-1}$, and $y_t = (2s)^{-1} y$,

$$\partial_t U = s^{-3/2} \left(\partial_\tau V + \frac{1}{2} V + \frac{1}{2} y \cdot \nabla V \right),$$

$$(U \cdot \nabla_x) U = s^{-3/2} (V \cdot \nabla) V, \qquad \Delta_x U = s^{-3/2} \Delta V, \qquad \nabla_x Q = s^{-3/2} \nabla \Pi.$$

Substitution into the equation for U gives (9.3), and the divergence condition is preserved by the spatial scaling. □

13.2. **Boundedness, smoothness, and local energy.**

Lemma 13.2 (Smoothness)**.** *The ancient solution U is smooth on $\mathbb{R}^3 \times (-\infty, 0)$, and (V, Π) is smooth on $\mathbb{R}^3 \times \mathbb{R}$ after fixing a pressure gauge.*

Proof. On each compact cylinder $Q_0 \Subset \mathbb{R}^3 \times (-\infty, 0)$, Lemma 11.9 gives $U \in L^\infty(Q_0)$. Hence $U \in L^s_t L^r_x(Q_0)$ for every finite r, s. Choose $r = s = 10$, so

$$\frac{2}{s} + \frac{3}{r} = \frac{1}{2} < 1.$$

Serrin's interior regularity criterion [9, 19] applies and gives local Hölder regularity of U on a slightly smaller cylinder. Since Q_0 was arbitrary, this removes every possible interior singular point in $\mathbb{R}^3 \times (-\infty, 0)$. The bootstrap starts from

$$\partial_t U - \Delta U + \nabla Q = -\operatorname{div}(U \otimes U), \qquad \operatorname{div} U = 0,$$

with $U \otimes U$ locally bounded after Serrin regularity. Interior Stokes estimates on nested cylinders give $W^{2,1}_q$-regularity for every finite q. Parabolic Sobolev embedding then improves the Hölder regularity of ∇U, and differentiating the equation and iterating gives

$$U \in C^\infty(\mathbb{R}^3 \times (-\infty, 0)).$$

The pressure is recovered locally from $-\Delta Q = \partial_i \partial_j (U_i U_j)$ plus a harmonic function and is smooth after fixing a time-dependent gauge. The centered change of variables is a smooth diffeomorphism for $t < 0$, so V and Π are smooth. □

Corollary 13.3 (Classical Type I bound)**.** *For the smooth representative from Lemma 13.2,*

$$\|U(t)\|_{L^\infty(\mathbb{R}^3)} \le \frac{M}{\sqrt{-t}}, \qquad t < 0.$$

Proof. Fix $t_0 < 0$. Choose $\delta_n \downarrow 0$ with

$$I_n = [t_0 - \delta_n, t_0 + \delta_n] \Subset (-\infty, 0).$$

Lemma 11.9 gives

$$\|U\|_{L^\infty(\mathbb{R}^3 \times I_n)} \le M(-t_0 - \delta_n)^{-1/2}.$$

Since the representative is continuous, the essential spacetime bound holds pointwise. Letting $n \to \infty$ gives the estimate at t_0. □

Lemma 13.4 (Renormalized boundedness and nontriviality)**.** *The centered field satisfies*

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \le M.$$

Moreover (9.4) *holds for some compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$ and $\eta_0 > 0$.*

Proof. The L^∞ bound follows from Corollary 13.3:

$$|V(y, \tau)| = \sqrt{-t}|U(y\sqrt{-t}, t)| \le M.$$

For nontriviality, Proposition 12.2 gives a positive local L^3 lower bound on a compact cylinder in (x, t). Under $(x, t) = (e^{-\tau/2} y, -e^{-\tau})$,

$$dx\, dt = e^{-5\tau/2}\, dy\, d\tau, \qquad U = e^{\tau/2} V,$$

and hence

$$|U|^3\, dx\, dt = e^{-\tau} |V|^3\, dy\, d\tau.$$

On the compact image cylinder the factor $e^{-\tau}$ is bounded above and below by positive constants, so the positive lower bound becomes (9.4), with a possibly different η_0. □

Lemma 13.5 (Local energy bounds in centered variables)**.** *Let V be obtained from an admissible finite-energy Type I singularity. For every $R < \infty$ and compact interval $I \subset \mathbb{R}$,*

$$\int_I \int_{B_R} |V|^2 \, dy \, d\tau + \int_I \int_{B_R} |\nabla V|^2 \, dy \, d\tau < \infty.$$

After subtracting a function of τ, the associated pressure satisfies

$$\Pi - b(\tau) \in L^{3/2}(B_R \times I).$$

Proof. The unrenormalized limit (U, Q) is suitable and has local energy, dissipation, and pressure bounds on compact negative-time cylinders by Lemmas 11.2 and 11.4 and Proposition 11.10; the bounds for the limit follow from the weak lower semicontinuity and weak compactness already used in Lemma 11.8. On each compact (y, τ)-cylinder, the map

$$(y, \tau) \mapsto (x, t) = (e^{-\tau/2} y, -e^{-\tau})$$

has Jacobian $dx \, dt = e^{-5\tau/2} \, dy \, d\tau$, with

$$V = e^{-\tau/2} U, \qquad \nabla_y V = e^{-\tau} \nabla_x U, \qquad \Pi = e^{-\tau} Q.$$

If $I = [\tau_1, \tau_2]$, then $e^{-\tau}$ is bounded above and below by positive constants on I. Therefore

$$\int_I \int_{B_R} |V|^2 \, dy \, d\tau \le C(I) \int_{-e^{-\tau_1}}^{-e^{-\tau_2}} \int_{B_{C(I)R}} |U|^2 \, dx \, dt,$$

and similarly

$$\int_I \int_{B_R} |\nabla_y V|^2 \, dy \, d\tau \le C(I) \int_{-e^{-\tau_1}}^{-e^{-\tau_2}} \int_{B_{C(I)R}} |\nabla_x U|^2 \, dx \, dt.$$

For the pressure, subtract a spatial-average gauge $a(t)$ for Q on the corresponding x-ball and set $b(\tau) = e^{-\tau} a(-e^{-\tau})$. Then $\Pi - b = e^{-\tau}(Q - a)$, and the same bounded-weight change of variables gives $\Pi - b \in L^{3/2}(B_R \times I)$. □

Remark 13.6. Lemma 13.5 is local. It does not give uniform L^3-tightness in y, nor a global $L^\infty_\tau L^3_y$ bound. This is the reason for retaining the remaining case.

13.3. **Normalized Seregin ancient limits.**

Definition 13.7 (Admissible Type I extraction)**.** An admissible Type I extraction is one of the following two blow-up procedures:

(i) the finite-energy Type I extraction in Definition 10.4 and Theorem 9.1;
(ii) the local pointwise Type I extraction in Definition 8.1 and Lemma 8.5.

In both cases the conclusion is a smooth bounded solution of the centered equation (9.3) with compact local L^3 nonvanishing.

Definition 13.8 (Normalized Seregin ancient limit)**.** A smooth bounded solution (V, Π) of (9.3) is a normalized Seregin ancient limit if it is generated by an admissible Type I extraction in the sense of Definition 13.7. Equivalently, it is a centered Type I blow-up limit satisfying the local L^3 nonvanishing normalization (9.4).

Definition 13.9 (Collection of normalized Seregin limits)**.** A collection $\mathcal{S}$ is any nonempty set of normalized Seregin ancient limits generated by admissible Type I extractions. When needed, $\mathcal{S}(u, p, z_*)$ denotes a collection associated with a fixed singular point z_*.

Remark 13.10. The local L^3 nonvanishing condition is part of the normalization, not an additional hypothesis. It is inherited from the selected velocity concentration and the no-escape condition through Lemma 12.1 and Proposition 12.2 for the finite-energy extraction, and through Lemma 8.5 for the local pointwise Type I extraction. Thus every normalized Seregin ancient limit is nonzero in the sense of (9.4).

Proof of Theorem 9.1. Choose an admissible concentration sequence $\lambda_k \downarrow 0$ from Definition 10.4, and form the rescaled sequence (11.2). The compact ancient limit is Proposition 11.10; the strong and weak convergence are Proposition 11.6. The no-escape argument in Lemma 12.1, followed by strong L^3-compactness away from $t = 0$, gives nontriviality through Proposition 12.2. The pointwise Type I bound is Corollary 13.3, and the centered equation, smoothness, boundedness, and local L^3 nonvanishing are Lemmas 13.1, 13.2 and 13.4. □

14. Liouville classes and the remaining class

Fix a collection $\mathcal{S}$ of normalized Seregin ancient limits. Complements below are taken relative to $\mathcal{S}$. Structural conditions involving the unrenormalized ancient field refer to

$$U(x,t) = (-t)^{-1/2} V(x/\sqrt{-t}, -\log(-t)).$$

Definition 14.1 (Small-amplitude class)**.** Let $\varepsilon_0 > 0$ be the small-data Liouville constant for bounded ancient solutions of (9.3). We say $V \in \mathcal{C}_{\mathrm{small}}$ if

$$\|V\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} \le \varepsilon_0.$$

Definition 14.2 (Stationary L^3 class)**.** We say $V \in \mathcal{C}_{\mathrm{stat},L^3}$ if

$$V(y,\tau) = W(y) \quad \text{and} \quad W \in L^3(\mathbb{R}^3).$$

Definition 14.3 (Uniformly L^3-tight class)**.** We say $V \in \mathcal{C}_{L^3-\mathrm{tight}}$ if

$$\lim_{R\to\infty} \sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V(y,\tau)|^3 \, dy = 0. \tag{14.1}$$

Every $V \in \mathcal{S}$ is bounded. Theorem 25.6 shows that, for such limits, the tail condition (14.1) gives the endpoint sequence-L^3 control needed after physical pullback; no separate $L^\infty_\tau L^3_y$ hypothesis is required for the tight-class exclusion.

Definition 14.4 (Liouville-covered structure and decay class)**.** A normalized Seregin limit V belongs to $\mathcal{C}_{\mathrm{sd}}$ if its physical ancient representative U satisfies at least one of the following analytic hypotheses:

(i) U is axisymmetric without swirl;

(ii) U is axisymmetric and satisfies the pointwise scale-invariant bound

$$|U(x,t)| \le \frac{C}{r}$$

for some $C < \infty$, where r is the cylindrical radius;

(iii) U is axisymmetric and

$$rU_\theta \in L^\infty_t L^p_x(\mathbb{R}^3 \times (-\infty, 0))$$

for some $1 \le p < \infty$;

(iv) U is axisymmetric, periodic in the axial variable, and

$$rU_\theta \in L^\infty(\mathbb{R}^3 \times (-\infty, 0));$$

(v) the associated rescaled vorticity satisfies the perturbative Gallay–Wayne weighted-vorticity hypotheses used in Theorem 32.1;

Remark 14.5. Only hypotheses for which a cited Liouville theorem gives $U \equiv 0$ are included in $\mathcal{C}_{\mathrm{sd}}$. Broader qualitative descriptions are not included unless they imply one of the listed hypotheses.

Lemma 14.6 (Structural quantities under centering)**.** *The centered change of variables preserves axisymmetry and absence of swirl. If (r, z, t) and (ρ, Z, τ) are related by*

$$\rho = \frac{r}{\sqrt{-t}}, \qquad Z = \frac{z}{\sqrt{-t}}, \qquad \tau = -\log(-t),$$

then the swirl variables satisfy

$$\rho V_\theta(\rho, Z, \tau) = r U_\theta(r, z, t).$$

Proof. The map $x \mapsto y = x/\sqrt{-t}$ is radial in the cylindrical variables and does not mix the e_r, e_θ, e_z frame. Since $V(y,\tau) = \sqrt{-t}\, U(x,t)$, the θ-components obey $V_\theta = \sqrt{-t}\, U_\theta$, and the displayed identities follow. $\square$

Definition 14.7 (Remaining class)**.** The remaining class is

$$\mathcal{R}(\mathcal{S}) = \mathcal{S} \setminus \left(\mathcal{C}_{\mathrm{small}} \cup \mathcal{C}_{\mathrm{stat},L^3} \cup \mathcal{C}_{L^3-\mathrm{tight}} \cup \mathcal{C}_{\mathrm{sd}}\right).$$

Remark 14.8. The first four classes are not intended to be disjoint. The remaining class is the complement of their union, so overlaps among the Liouville classes cause no ambiguity.

Proposition 14.9 (Exhaustion by defined classes)**.** *Every $V \in \mathcal{S}$ belongs either to one of the four Liouville classes or to $\mathcal{R}(\mathcal{S})$.*

Proof. Let

$$\mathcal{U} = \mathcal{C}_{\mathrm{small}} \cup \mathcal{C}_{\mathrm{stat},L^3} \cup \mathcal{C}_{L^3-\mathrm{tight}} \cup \mathcal{C}_{\mathrm{sd}}.$$

For $V \in \mathcal{S}$, either $V \in \mathcal{U}$, in which case it lies in at least one Liouville class, or $V \in \mathcal{S} \setminus \mathcal{U} = \mathcal{R}(\mathcal{S})$. $\square$

15. Liouville hypotheses and class exclusions

Lemma 15.1 (Centered Stokes kernel estimates)**.** *Let*

$$L = \Delta - \frac{1}{2} y \cdot \nabla - \frac{1}{2}, \qquad S(s) = e^{sL}.$$

There is a universal constant C_0 such that, for all $s > 0$,

$$\|S(s)f\|_{L^\infty} \le e^{-s/2} \|f\|_{L^\infty}$$

and

$$\|S(s)\mathbb{P}\nabla \cdot F\|_{L^\infty} \le C_0 e^{-s/2}(1 - e^{-s})^{-1/2} \|F\|_{L^\infty}.$$

Proof. The first estimate follows from the explicit Ornstein–Uhlenbeck kernel:

$$S(s)f(y) = e^{-s/2} \int_{\mathbb{R}^3} G_{1-e^{-s}}(e^{-s/2}y - z) f(z)\, dz,$$

where G_a is the heat kernel with variance a. The kernel has total mass one. For the projected-divergence estimate, write the corresponding kernel as the Oseen projected-gradient kernel after the same change of variables. More explicitly, the semigroup for $\Delta - \frac{1}{2} y \cdot \nabla$ is

$$(e^{s(\Delta - \frac{1}{2} y \cdot \nabla)} g)(y) = (e^{(1-e^{-s})\Delta} g)(e^{-s/2} y).$$

Applying this to the distribution $g = \mathbb{P}\nabla \cdot F$ and using the projected-gradient Oseen kernel estimate

$$\|e^{\theta\Delta}\mathbb{P}\nabla \cdot F\|_\infty \le C\theta^{-1/2} \|F\|_\infty, \qquad \theta > 0,$$

gives the displayed estimate with $\theta = 1 - e^{-s}$, together with the extra factor $e^{-s/2}$ from the zeroth-order term in L. The projected-gradient Oseen estimate follows from the L^1-bound

$$\|\nabla O(\cdot, \theta)\|_{L^1} \leq C\theta^{-1/2}$$

for the gradient of the Oseen kernel $O(\cdot, \theta)$. This is the bounded-data Stokes estimate used by Kato [12]; it does not require boundedness of the Leray projector itself on L^∞. $\square$

Lemma 15.2 (Mild representation for normalized limits)**.** *Every normalized Seregin ancient limit satisfies, for all $\Theta > 0$,*

$$V(\tau) = S(\Theta)V(\tau - \Theta) - \int_0^\Theta S(\Theta - \sigma)\mathbb{P}\nabla \cdot (V \otimes V)(\tau - \Theta + \sigma)\, d\sigma.$$

Proof. Normalized Seregin limits are smooth and bounded by Lemmas 13.2 and 13.4. On a finite interval $[\tau - \Theta, \tau]$, the tensor $V \otimes V$ is bounded, so Lemma 15.1 makes the Duhamel integral finite in L^∞. In the sense of solenoidal distributions, applying the Leray projector to (9.3) removes the pressure and gives

$$\partial_\tau V = LV - \mathbb{P}\nabla \cdot (V \otimes V)$$

on every finite interval. Since V is smooth, the map

$$s \mapsto S(\tau - s)V(s)$$

is differentiable as a tempered distribution for $\tau - \Theta < s < \tau$, and

$$\frac{d}{ds}\big(S(\tau - s)V(s)\big) = -S(\tau - s)\mathbb{P}\nabla \cdot (V \otimes V)(s).$$

Integrating this identity from $\tau - \Theta$ to τ gives the stated Duhamel formula. $\square$

Lemma 15.3 (Small bounded ancient Liouville theorem)**.** *There is $\varepsilon_0 > 0$ such that every bounded mild ancient solution of* (9.3) *with*

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq \varepsilon_0$$

is identically zero.

Proof. Write the projected centered equation as

$$\partial_\tau V = LV - \mathbb{P}\nabla \cdot (V \otimes V), \qquad L = \Delta - \frac{1}{2}y \cdot \nabla - \frac{1}{2}.$$

Let $S(s) = e^{sL}$. By Lemma 15.1, the nonlinear kernel is integrable on $(0, \infty)$; set

$$A = C_0 \int_0^\infty e^{-s/2}(1 - e^{-s})^{-1/2}\, ds < \infty.$$

For fixed τ and $T > 0$, the mild identity on $[\tau - T, \tau]$ gives

$$V(\tau) = S(T)V(\tau - T) - \int_0^T S(T - \sigma)\mathbb{P}\nabla \cdot (V \otimes V)(\tau - T + \sigma)\, d\sigma.$$

If $M = \|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}$, then

$$\|V(\tau)\|_\infty \leq e^{-T/2}M + AM^2.$$

Letting $T \to \infty$ and taking the supremum in τ gives

$$M \leq AM^2.$$

If $0 < M < 1/A$, this is impossible. Choose $\varepsilon_0 < 1/A$; then $M \leq \varepsilon_0$ implies $M = 0$. $\square$

Lemma 15.4 (Stationary centered profiles)**.** *If $V(y,\tau) = W(y)$ is a smooth solution of* (9.3)*, then after subtracting a function of τ from the pressure, W solves*

$$(W\cdot\nabla)W + \nabla P = \Delta W - \frac{1}{2}(W + y\cdot\nabla W), \qquad \operatorname{div} W = 0.$$

Proof. Since V is independent of τ, $\partial_\tau V = 0$. Set

$$F(y) = \Delta W - \frac{1}{2}(W + y\cdot\nabla W) - (W\cdot\nabla)W.$$

The centered equation gives $\nabla_y \Pi(\cdot,\tau) = F$ for a.e. τ. Fix one such time τ_0 and put $P(y) = \Pi(y,\tau_0)$. Then $\nabla_y(\Pi(\cdot,\tau) - P) = 0$ for a.e. τ, so $\Pi(y,\tau) - P(y) = b(\tau)$ is spatially constant. Subtracting this gauge from Π leaves the equation unchanged and yields the displayed stationary self-similar equation for W. □

Theorem 15.5 (Small and stationary class exclusions)**.** *Let V be a normalized Seregin ancient limit. Then*

$$V \notin \mathcal{C}_{\mathrm{small}}, \qquad V \notin \mathcal{C}_{\mathrm{stat},L^3}.$$

Proof. If $V \in \mathcal{C}_{\mathrm{small}}$, then Lemmas 15.2 and 15.3 give $V \equiv 0$, contradicting the local L^3 nonvanishing condition (9.4).

If $V \in \mathcal{C}_{\mathrm{stat},L^3}$, then $V(y,\tau) = W(y)$ with $W \in L^3(\mathbb{R}^3)$. By Lemma 15.4, W satisfies the stationary self-similar profile equation after a pressure gauge change. The theorem of Nečas–Růžička–Šverák [17] gives $W \equiv 0$. Again (9.4) gives the contradiction. □

Theorem 15.6 (Uniformly tight L^3 closure)**.** *No nonzero normalized Seregin ancient limit belongs to $\mathcal{C}_{L^3-\mathrm{tight}}$.*

Proof. This is the tight-class Liouville theorem proved in Theorem 25.6 and Corollary 26.4. Theorem 25.6 proves that, for bounded centered ancient solutions, the tail-tightness condition gives the endpoint sequence-L^3 control after physical pullback. It then applies the Albritton–Barker endpoint ancient Liouville theorem and contradicts the compact velocity normalization Proposition 12.2. □

Theorem 15.7 (Liouville-covered structure and decay exclusions)**.** *No nonzero normalized Seregin ancient limit belongs to $\mathcal{C}_{\mathrm{sd}}$.*

Proof. By Definition 14.4, membership in $\mathcal{C}_{\mathrm{sd}}$ means that the physical ancient representative U satisfies one of the listed analytic hypotheses. The axisymmetric no-swirl and pointwise $1/r$ cases are excluded by Koch–Nadirashvili–Seregin–Šverák [13]. The finite L^p circulation case is excluded by Lei–Zhang–Zhao [16], and the axial-periodic bounded-circulation case is excluded by Lei–Ren–Zhang [15]. The perturbative Gallay–Wayne weighted-vorticity hypotheses are excluded by the corresponding Liouville theorem in Theorem 32.1. Each alternative gives $V \equiv 0$, contradicting the local L^3 nonvanishing condition (9.4). Hence no nonzero normalized Seregin ancient limit lies in $\mathcal{C}_{\mathrm{sd}}$. □

Proposition 15.8 (Exclusion of the Liouville classes)**.** *No normalized Seregin ancient limit can belong to*

$$\mathcal{C}_{\mathrm{small}} \cup \mathcal{C}_{\mathrm{stat},L^3} \cup \mathcal{C}_{L^3-\mathrm{tight}} \cup \mathcal{C}_{\mathrm{sd}}.$$

Proof. The small and stationary L^3 classes are excluded by Theorem 15.5. The uniformly tight class is excluded by Theorem 15.6. The structure/decay class is excluded by Theorem 15.7. □

16. The remaining-case criterion

Hypothesis 16.1 (Residual-class hypothesis)**.** *For every collection $\mathcal{S}$ of normalized Seregin ancient limits generated by admissible Type I extractions,*

$$\mathcal{R}(\mathcal{S}) = \varnothing.$$

Corollary 16.2 (Rigidity in the remaining case)**.** *Assume Hypothesis 16.1. For every collection $\mathcal{S}$ of normalized Seregin ancient limits generated by admissible Type I extractions, every element of $\mathcal{R}(\mathcal{S})$ is identically zero.*

Proof. The assertion follows directly from Hypothesis 16.1. □

Proposition 16.3 (Equivalence on normalized Seregin limits)**.** *For collections of normalized Seregin limits, the residual-class hypothesis Hypothesis 16.1 implies Corollary 16.2. Conversely, the corresponding rigidity assertion implies emptiness of the remaining class.*

Proof. If Hypothesis 16.1 holds, the rigidity statement is vacuous. Conversely, assume Corollary 16.2. If some normalized collection had $V \in \mathcal{R}(\mathcal{S})$, the rigidity hypothesis would give $V \equiv 0$, directly contradicting the local L^3 nonvanishing condition (9.4). Hence the remaining class is empty. □

Proof of Theorem 9.2. Assume Hypothesis 16.1. Suppose that (u, p, z_*) is an admissible finite-energy Type I singularity. By Theorem 9.1, there exists a normalized nonzero Seregin ancient limit V. Let $\mathcal{S}$ be any collection of normalized limits containing V.

By Proposition 14.9, either V belongs to one of the Liouville classes or $V \in \mathcal{R}(\mathcal{S})$. The first alternative is impossible by Proposition 15.8. The second alternative is impossible by Hypothesis 16.1. Hence the assumed admissible finite-energy Type I singularity cannot exist. □

The preceding theorem is conditional on the residual-class hypothesis for the stated admissible finite-energy class. The compactness construction, pressure normalization, and class exhaustion are established above, with the local velocity concentration result recorded in Proposition 10.7. The structure/decay class is restricted to the assumptions in Definition 14.4; any other structured behavior lies in the remaining class and is addressed only by Hypothesis 16.1. No whole-space critical-norm estimate on the original solution is used in the conditional closure step.

17. Elementary symmetries of the centered equation

Lemma 17.1 (Logarithmic-time translation)**.** *If V solves (9.3), then*

$$V_{\tau_0}(y, \tau) = V(y, \tau + \tau_0)$$

also solves (9.3). If

$$U^\lambda(x, t) = \lambda U(\lambda x, \lambda^2 t),$$

then the renormalized field associated with U^λ is

$$V^\lambda(y, \tau) = V(y, \tau - 2 \log \lambda).$$

Proof. The first statement follows because (9.3) is autonomous in τ. For the second, put $s = -t = e^{-\tau}$. Then

$$\sqrt{s}\, U^\lambda(y\sqrt{s}, -s) = \lambda \sqrt{s}\, U(\lambda y \sqrt{s}, -\lambda^2 s) = V(y, -\log(\lambda^2 s)) = V(y, \tau - 2 \log \lambda).$$

□

Proposition 17.2 (Dilation and translation are not symmetries)**.** *The centered equation* (9.3) *is autonomous in τ, but it is not invariant under*

$$V(y,\tau) \mapsto \alpha V(\alpha y, \alpha^2 \tau)$$

unless $\alpha = 1$. It is also not invariant under constant translations

$$V(y,\tau) \mapsto V(y-a,\tau)$$

unless $a = 0$.

Proof. For scaling, set $Y = \alpha y$, $T = \alpha^2 \tau$, and $\widetilde{V}(y,\tau) = \alpha V(Y,T)$, $\widetilde{\Pi}(y,\tau) = \alpha^2 \Pi(Y,T)$. The non-drift part scales with factor α^3, while

$$-\frac{1}{2}(\widetilde{V} + y \cdot \nabla \widetilde{V}) = -\frac{\alpha}{2}(V + Y \cdot \nabla_Y V)(Y,T).$$

Using the equation for V, the remaining term in the equation for $\widetilde{V}$ is

$$\frac{\alpha - \alpha^3}{2}(V + Y \cdot \nabla_Y V)(Y,T).$$

If $\alpha \neq 1$, this forces $V + Y \cdot \nabla V = 0$. Along each ray,

$$\frac{d}{dr}\big(rV(r\omega, T)\big) = 0.$$

Boundedness and smoothness at the origin force the constant to be zero, hence $V \equiv 0$. Thus a nonzero centered solution has no such scaling symmetry.

For translations, all translation-invariant terms transform by composition, but the drift satisfies

$$y \cdot \nabla V(y-a,\tau) = (y-a) \cdot \nabla V(y-a,\tau) + a \cdot \nabla V(y-a,\tau).$$

The extra term is absent from the original equation. Thus spatial translation is not a symmetry of the fixed centered equation unless $a = 0$, apart from exceptional fixed solutions satisfying $a \cdot \nabla V \equiv 0$. $\square$

Part III. Uniformly Tight Ancient Dynamics

Overview. Bounded ancient solutions of the three-dimensional Navier–Stokes equations are considered in backward self-similar variables under uniform L^3-tightness. The analytic compactness framework shows that bounded mild ancient solutions are smooth, locally compact under time translation, and that the common $L^\infty_\tau L^3_y$ and tightness bounds used in compactness statements are stable under locally smooth limits. A minimal-element reduction is then obtained for closed normalized uniformly tight collections: any nonempty such collection contains a nonzero solution of minimal $L^\infty(\mathbb{R}^3 \times \mathbb{R})$ size.

For the time-translation hull of a bounded ancient solution, compact invariant hulls and invariant probability measures are constructed by Krylov–Bogolyubov averaging. These measures satisfy a stationary statistical form of the self-similar Navier–Stokes equation. Passing to the barycenter produces an explicit quadratic covariance term, isolating the obstruction to obtaining a single stationary self-similar profile.

Finally, the uniformly L^3-tight class is closed directly from its defining hypotheses. The tail condition, together with the boundedness of the ancient solution, gives the sequence-L^3 bound needed after the physical pullback. Since this pullback preserves the critical L^3-norm on each time slice, the endpoint ancient Liouville theorem of Albritton–Barker then forces the pullback, and hence the centered solution, to vanish. Therefore the uniformly L^3-tight class of the Type I ancient-limit reduction is excluded without any compact-hull Lyapunov hypothesis.

18. Introduction and main results

18.1. **The uniformly tight ancient class.** Consider the three-dimensional incompressible Navier–Stokes equations in backward self-similar variables. The centered velocity V and pressure P solve

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \Delta V - \frac{1}{2}(V + y \cdot \nabla V), \qquad \operatorname{div} V = 0 \tag{18.1}$$

on $\mathbb{R}^3 \times \mathbb{R}$. In Theorem 9.1, admissible Type I singularities are reduced to nonzero normalized ancient solutions of (18.1). The uniformly L^3-tight class is the class in which

$$\lim_{R\to\infty} \sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V(y,\tau)|^3 \, dy = 0. \tag{18.2}$$

The compactness statements also record the common bound

$$\sup_{\tau\in\mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} < \infty, \tag{18.3}$$

but the endpoint exclusion of the tight class uses only boundedness and the tail condition (18.2).

This condition on the centered limit is distinct from the original-solution condition

$$u \in L^\infty(0,T; L^3(\mathbb{R}^3)).$$

For finite-energy suitable weak solutions, the original-solution assumption $u \in L^\infty_t L^3_x$ near a putative singular time is incompatible with singularity by the theorem of Escauriaza–Seregin–Šverák [3]. Uniform L^3-tightness is imposed on centered ancient solutions after blow-up.

18.2. **Minimal elements and compact hulls.** The first reduction is variational:

$$\text{nonempty tight normalized class} \quad \Longrightarrow \quad L^\infty\text{-minimal ancient element.}$$

The normalization is a fixed local velocity nonvanishing condition. It prevents minimizing sequences from losing all mass by time translation. Once a minimal element V_* is selected, its time-translation hull is

$$\mathcal{H}(V_*) = \overline{\{\Theta_s V_* : \ s \in \mathbb{R}\}}^{C^\infty_{\mathrm{loc}}}, \qquad (\Theta_s V)(y,\tau) = V(y,\tau+s).$$

The hull is compact because bounded mild ancient solutions are locally smooth with estimates depending only on their common L^∞ bound.

18.3. **Invariant measures and covariance.** Every compact invariant hull K carries invariant probability measures. Indeed, Krylov–Bogolyubov averages

$$\mu_T = \frac{1}{T}\int_0^T \delta_{\Theta_s W_0}\, ds$$

have weak subsequential limits. Such invariant measures satisfy a stationary statistical form of (18.1). If

$$\overline{W}(y) = \int_K W(y,0)\, d\mu(W),$$

then the averaged nonlinearity decomposes as

$$\int_K W_i W_j \, d\mu = \overline{W}_i \, \overline{W}_j + C_{ij}.$$

Thus the barycenter solves the stationary self-similar equation only up to the explicit covariance term C. This invariant-measure construction does not prove that a recurrent compact hull contains a single stationary profile.

18.4. **Endpoint L^3 closure of the tight class.** The compact-hull and invariant-measure constructions identify the covariance obstruction in a purely local topology, but they are not needed to close the uniformly L^3-tight class. The tail condition already gives the sequence-L^3 information required by the endpoint ancient Liouville theorem, because the centered solution is bounded. Under the physical change of variables

$$u(x,t) = (-t)^{-1/2} V\left(\frac{x}{\sqrt{-t}}, -\log(-t)\right), \qquad t<0,$$

the L^3-norm is invariant:

$$\|u(t)\|_{L^3_x} = \|V(-\log(-t))\|_{L^3_y}.$$

Consequently every tail-tight centered ancient solution produces a physical mild ancient solution with bounded L^3-norm along a sequence $t_k \downarrow -\infty$. The endpoint theorem of Albritton–Barker [1] then gives $u \equiv 0$, and the change of variables gives $V \equiv 0$.

The endpoint closure uses only the boundedness and tail tightness already present in the ancient solution class and introduces no additional estimate, recurrence object, or Lyapunov functional. The compactness and invariant-measure results establish the local smooth compactness framework, the existence of minimal elements and compact trajectory hulls, and the stationary statistical identity with its covariance term. The endpoint L^3 exclusion of the uniformly tight class is then obtained by the physical pullback and the Albritton–Barker ancient Liouville theorem.

Thus the uniformly L^3-tight class is closed by its own hypotheses. The compact-hull construction describes recurrent dynamics and the covariance obstruction, but no compact-recurrence or Lyapunov assumption enters the tight-class exclusion in Theorem 25.6.

18.5. **Main results.** The first main theorem is the analytic compactness and closure statement.

Theorem 18.1 (Compactness and closure of uniformly tight ancient solutions)**.** *Let $\{V_n\}$ be a sequence of bounded mild ancient solutions of* (18.1)*. Assume*

$$\sup_n \|V_n\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} < \infty.$$

Then a subsequence converges locally smoothly to a bounded mild ancient solution V.

If in addition there are $L < \infty$ and a modulus $\omega : [1, \infty) \to [0, \infty)$, $\omega(R) \to 0$, such that

$$\sup_n \sup_{\tau\in\mathbb{R}} \|V_n(\tau)\|_{L^3(\mathbb{R}^3)} \le L$$

and

$$\sup_n \sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V_n(y,\tau)|^3\, dy \le \omega(R) \qquad (R \ge 1),$$

then

$$\sup_{\tau\in\mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} \le L, \qquad \sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V(y,\tau)|^3\, dy \le \omega(R),$$

and in particular V is uniformly L^3-tight.

Compatible pressure representatives also converge locally weakly modulo time-dependent gauges under the additional $L^\infty_\tau L^3_y$ assumption; see Lemma 20.3.

The second main theorem is the minimal-element reduction.

Theorem 18.2 (Minimal element in a normalized uniformly tight class)**.** *Let A be a nonempty collection of bounded mild ancient solutions satisfying a common L^∞ bound, a common $L^\infty_\tau L^3_y$ bound, a common uniform L^3-tightness modulus, and a fixed local nonvanishing normalization*

$$\sup_{\tau\in\mathbb{R}} \int_{B_{R_0}} |V(y,\tau)|^3\, dy \ge \eta_0 > 0.$$

Assume A is invariant under time translation and closed under locally smooth limits which retain the normalization. Then A contains an element V_* minimizing

$$\|V\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})}$$

among all $V \in A$. In particular $V_* \not\equiv 0$.

The third main theorem is the endpoint L^3 tight-class Liouville criterion.

Theorem 18.3 (Uniformly tight Liouville criterion)**.** *Let A be a collection of bounded mild ancient solutions of* (18.1)*. Assume there is a modulus $\omega : [1, \infty) \to [0, \infty)$, $\omega(R) \to 0$, such that*

$$\sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V(y,\tau)|^3\, dy \le \omega(R) \qquad \text{for every } V \in A,\ R \ge 1,$$

and assume that A has a fixed local nonvanishing normalization:

$$\sup_{\tau\in\mathbb{R}} \int_{B_{R_0}} |V(y,\tau)|^3\, dy \ge \eta_0 > 0 \qquad \text{for every } V \in A.$$

Then

$$A = \varnothing.$$

The proof of the last theorem is summarized by the implications

$$\begin{gathered}
\text{uniformly } L^3\text{-tight locally nonzero class } A \\
\Downarrow \quad \text{boundedness plus tail tightness} \\
\sup_k \|V(\tau_k)\|_{L^3(\mathbb{R}^3)} < \infty \quad \text{for a sequence } \tau_k \to -\infty \\
\Downarrow \quad \text{physical pullback and critical } L^3\text{-invariance} \\
\text{mild ancient Navier–Stokes solution } u \\
\text{with } \sup_k \|u(t_k)\|_{L^3(\mathbb{R}^3)} < \infty \quad \text{for some } t_k \downarrow -\infty \\
\Downarrow \quad \text{Albritton–Barker endpoint theorem} \\
u = 0 \\
\Downarrow \\
\text{contradiction with local velocity nonvanishing in } A.
\end{gathered}$$

18.6. **Relation to the Type I reduction.** Theorem 9.1 reduces admissible Type I singularities to normalized Seregin ancient limits and organizes such limits into small, stationary L^3, uniformly tight, structured/decay, and residual classes. The tight-class exclusion requires no additional compact-dynamical hypothesis: the uniformly L^3-tight class contains no nonzero normalized ancient limit.

18.7. **Organization.** Section 19 fixes the centered equation, mild formulation, time translation, and local smooth topology. Section 20 proves smoothing, pressure reconstruction, stability, and compactness. Section 21 proves closure of uniform L^3-tightness and recalls stationary L^3 rigidity. Section 22 proves the minimal-element reduction. Section 23 constructs compact trajectory hulls and minimal invariant subsets. Section 24 constructs invariant measures and identifies the covariance obstruction. Section 25 proves the physical pullback identities, applies the endpoint L^3 ancient Liouville theorem, and states the consequence used in the Type I reduction.

19. Centered ancient solutions and the local smooth topology

19.1. **The centered equation.** Throughout this part V denotes a centered ancient velocity and P its pressure. An element of a compact hull is denoted by W, and a stationary profile by $w(y)$. Define

$$LV = \Delta V - \frac{1}{2} y \cdot \nabla V - \frac{1}{2} V.$$

After applying the Helmholtz–Leray projection $\mathbb{P}$, the centered equation becomes

$$\partial_\tau V = LV - \mathbb{P}\operatorname{div}(V \otimes V). \tag{19.1}$$

19.2. **The Ornstein–Uhlenbeck–Stokes semigroup.** Let $S(a) = e^{aL}$, $a > 0$. It is given by

$$(S(a)f)(y) = e^{-a/2} \int_{\mathbb{R}^3} \frac{\exp\left(-\frac{|z-e^{-a/2}y|^2}{4(1-e^{-a})}\right)}{[4\pi(1-e^{-a})]^{3/2}} f(z)\, dz. \tag{19.2}$$

The factor $e^{-a/2}$ represents the damping caused by the term $-V/2$.

Lemma 19.1 (Centered Stokes kernel estimate)**.** *For $a > 0$ and $\phi \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$,*

$$\|(S(a)\mathbb{P}\operatorname{div})^* \phi\|_{L^1(\mathbb{R}^3)} \le C e^{-a/2}(1 - e^{-a})^{-1/2} \|\phi\|_{L^1(\mathbb{R}^3)}. \tag{19.3}$$

Consequently the Duhamel term in (19.4) is well-defined by duality for bounded V.

Proof. Set $\lambda = e^{-a/2}$ and $\theta = 1 - e^{-a}$. From (19.2),

$$S(a)f(y) = e^{-a/2}(e^{\theta\Delta} f)(\lambda y).$$

Thus

$$S(a)^*\phi(z) = e^{-a/2}\lambda^{-3}(e^{\theta\Delta}\psi)(z), \qquad \psi(x) = \phi(x/\lambda),$$

and $\|\psi\|_{L^1} = \lambda^3\|\phi\|_{L^1}$. The Leray projection is a homogeneous Fourier multiplier of degree zero, so it commutes with this dilation. The classical heat–Stokes kernel estimate

$$\|(e^{\theta\Delta}\mathbb{P}\,\mathrm{div})^*\phi\|_{L^1} \le C\theta^{-1/2}\|\phi\|_{L^1}, \qquad \theta > 0,$$

follows from the Oseen kernel bound; see [21, 22, 19]. Substituting $\theta = 1 - e^{-a}$ and using

$$e^{-a/2}\lambda^{-3}\|\psi\|_{L^1} = e^{-a/2}\|\phi\|_{L^1}$$

gives (19.3). Since $(1 - e^{-a})^{-1/2}$ is integrable at $a = 0$, the duality pairing in (19.4) is integrable on finite time intervals for $V \in L^\infty$. □

19.3. **Riesz transform convention.** The convention is that R_i has Fourier symbol $i\xi_i/|\xi|$. Thus R_iR_j has symbol $-\xi_i\xi_j/|\xi|^2$, and

$$P = R_iR_j(F_{ij}) \quad \Longleftrightarrow \quad -\Delta P = \partial_i\partial_j F_{ij}.$$

Pressures are determined only modulo functions of time unless a whole-space Leray gauge is explicitly chosen.

19.4. **Bounded mild ancient solutions.**

Definition 19.2 (Bounded mild ancient solution)**.** A vector field $V : \mathbb{R}^3 \times \mathbb{R} \to \mathbb{R}^3$ is a bounded mild ancient solution of (18.1) if

$$V \in L^\infty(\mathbb{R}^3 \times \mathbb{R}), \qquad \mathrm{div}\, V = 0$$

in distributions, and for every $\sigma < \tau$,

$$V(\tau) = S(\tau - \sigma)V(\sigma) - \int_\sigma^\tau S(\tau - s)\mathbb{P}\,\mathrm{div}(V \otimes V)(s)\, ds \tag{19.4}$$

in the weak-* topology of $L^\infty(\mathbb{R}^3)$.

The Duhamel term in (19.4) is interpreted by duality. For $\phi \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$,

$$\langle S(\tau - s)\mathbb{P}\,\mathrm{div}(V \otimes V)(s), \phi\rangle = \langle V \otimes V(s), (S(\tau - s)\mathbb{P}\,\mathrm{div})^*\phi\rangle\,,$$

and Lemma 19.1 makes the scalar integrand absolutely integrable on $s \in (\sigma, \tau)$.

19.5. **Time translation.** For $s \in \mathbb{R}$ define

$$(\Theta_s V)(y, \tau) = V(y, \tau + s).$$

The equation, the mild formulation, the L^∞ norm, and the $L^\infty_\tau L^3_y$ and tail bounds used below are invariant under Θ_s. Indeed, replacing (σ, τ) by $(\sigma + s, \tau + s)$ in (19.4) and changing variables in the time integral gives the mild formula for $\Theta_s V$.

19.6. The locally smooth topology.

Definition 19.3 (Local smooth topology)**.** The locally smooth topology on $C^\infty_{\mathrm{loc}}(\mathbb{R}^3\times\mathbb{R};\mathbb{R}^3)$ is generated by the seminorms

$$\|V\|_{m,N} = \max_{|\alpha|+j\le m} \sup_{\substack{|y|\le N\\ |\tau|\le N}} |\partial_y^\alpha \partial_\tau^j V(y,\tau)|, \qquad m,N\in\mathbb{N}.$$

One compatible metric is

$$d(U,V) = \sum_{m,N=1}^\infty 2^{-(m+N)} \frac{\|U-V\|_{m,N}}{1+\|U-V\|_{m,N}}.$$

Thus $V_n\to V$ locally smoothly if and only if the convergence is in $C^m(K)$ for every compact cylinder $K\Subset\mathbb{R}^3\times\mathbb{R}$ and every $m\ge 0$.

20. Smoothing, pressure reconstruction, and compactness

20.1. Local smoothing.

Lemma 20.1 (Smoothing of bounded mild solutions)**.** *Every bounded mild ancient solution is smooth on $\mathbb{R}^3\times\mathbb{R}$ and satisfies the projected equation (19.1) classically. Moreover, for each compact cylinder $K\Subset\mathbb{R}^3\times\mathbb{R}$ and each integer $m\ge 0$,*

$$\|V\|_{C^m(K)} \le C\big(m, K, \|V\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})}\big).$$

The same estimate holds uniformly for any family with a common L^∞ bound.

Proof. Fix a compact time interval $J\Subset\mathbb{R}$ containing the time projection of K, and set $t=-e^{-\tau}$. On the corresponding compact interval $t(J)\Subset(-\infty,0)$, define

$$U(x,t) = (-t)^{-1/2} V(x/\sqrt{-t}, -\log(-t)).$$

The map $(x,t)\mapsto(x/\sqrt{-t},-\log(-t))$ is a smooth diffeomorphism between $\mathbb{R}^3\times t(J)$ and $\mathbb{R}^3\times J$, with all derivatives bounded on compact subcylinders.

The centered mild formula (19.4) transforms, under this parabolic change of variables, into the usual projected Navier–Stokes mild formula on every compact subinterval of $t(J)$:

$$U(t) = e^{(t-s)\Delta}U(s) - \int_s^t e^{(t-r)\Delta}\mathbb{P}\,\mathrm{div}(U\otimes U)(r)\,dr, \qquad s<t.$$

This transformation uses only the projected mild equation: the Leray projection is homogeneous of degree zero and commutes with the parabolic dilation, while the Ornstein–Uhlenbeck kernel in (19.2) becomes the heat kernel with time parameter $t-s$. No pressure formulation is used at this stage. The L^∞ norm of U on $\mathbb{R}^3\times t(J)$ is bounded by a constant depending only on J and $\|V\|_{L^\infty}$.

The standard local smoothing theorem for bounded projected mild Navier–Stokes solutions [12, 22] gives first local derivative bounds away from the initial time in any mild representation. In the present ancient setting one may choose the initial time strictly before the compact subcylinder under consideration. Iterating the differentiated Duhamel formula and then applying the interior parabolic bootstrap for the Stokes system [24, 19] gives uniform bounds for all derivatives of U on compact subcylinders of $\mathbb{R}^3\times t(J)$. The mild formula may then be differentiated in time, so U solves the projected Navier–Stokes equation classically. Returning to (y,τ) gives the asserted estimates for V and the classical projected equation (19.1). The argument is uniform for families with a common L^∞ bound. $\square$

20.2. **Pressure reconstruction.**

Lemma 20.2 (Local pressure reconstruction)**.** *Let V be a smooth bounded solution of the projected centered equation (19.1). Then for every ball B_R and compact interval $I \Subset \mathbb{R}$ there exists*

$$P_R \in C^\infty(B_R \times I),$$

unique up to an additive function of time, such that

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P_R = \Delta V - \frac{1}{2}(V + y \cdot \nabla V), \qquad \operatorname{div} V = 0,$$

on $B_R \times I$.

Proof. Set

$$F = \Delta V - \frac{1}{2}(V + y \cdot \nabla V) - \partial_\tau V - (V \cdot \nabla)V.$$

Since $(V \cdot \nabla)V = \operatorname{div}(V \otimes V)$ and V satisfies (19.1), self-adjointness of the Leray projection gives

$$\int_I \int_{B_R} F \cdot \Phi \, dy \, d\tau = 0$$

for every $\Phi \in C_c^\infty(B_R \times I; \mathbb{R}^3)$ satisfying $\operatorname{div}_y \Phi = 0$. The local de Rham theorem, applied in the spatial variables with τ as a parameter, gives a scalar distribution P_R on $B_R \times I$, unique up to a distribution depending only on time, such that $\nabla_y P_R = F$. Since F is smooth, all spatial derivatives of P_R of positive order are smooth. Fixing, for instance, the spatial mean of P_R over B_R to be zero for each τ selects a representative which is smooth on $B_R \times I$. This proves the local pressure formulation. □

Lemma 20.3 (Leray pressure and local gauges)**.** *Let V be a bounded mild ancient solution satisfying*

$$\sup_{\tau \in \mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Then the whole-space Leray pressure

$$P = R_i R_j (V_i V_j)$$

belongs to $L^\infty(\mathbb{R}; L^{3/2}(\mathbb{R}^3))$. Moreover, if $V_n \to V$ locally smoothly, the V_n have a common $L^\infty_\tau L^3_y$ bound, and

$$P_n = R_i R_j (V_{n,i} V_{n,j}),$$

then on every compact cylinder $B_R \times [\tau_1, \tau_2]$ there are bounded measurable gauges $a_n(\tau)$ and $a(\tau)$ such that, after passing to a subsequence,

$$P_n - a_n(\tau) \rightharpoonup P - a(\tau) \quad \text{weakly in } L^{3/2}(B_R \times [\tau_1, \tau_2]).$$

Equivalently, after replacing the limiting pressure by an equivalent representative modulo a function of time, the weak limit may be denoted by P.

Proof. The first assertion follows from the Calderon–Zygmund boundedness of Riesz transforms on $L^{3/2}(\mathbb{R}^3)$ [10]:

$$\|P(\tau)\|_{L^{3/2}} \le C \|V_i(\tau) V_j(\tau)\|_{L^{3/2}} \le C \|V(\tau)\|_{L^3}^2.$$

For the local stability statement, choose $\chi \in C_c^\infty(B_{4R})$ with $\chi \equiv 1$ on B_{3R}. Decompose on each time slice

$$P_n = R_i R_j (\chi V_{n,i} V_{n,j}) + R_i R_j ((1 - \chi) V_{n,i} V_{n,j}).$$

The local term converges strongly in $L^q(B_{2R} \times [\tau_1, \tau_2])$ for every finite q. This is because $V_n \to V$ smoothly on $\overline{B}_{4R} \times [\tau_1, \tau_2]$, so $\chi V_{n,i} V_{n,j} \to \chi V_i V_j$ strongly in every finite L^q, and the Riesz transforms are Calderon–Zygmund operators on L^q, $1 < q < \infty$.

It remains to control the exterior part modulo constants. Let K_{ij} be the convolution kernel of R_iR_j. For $x \in B_R$ and $|z| > 4R$,

$$|K_{ij}(x-z) - K_{ij}(-z)| \le C_R|z|^{-4}.$$

Define the exterior gauges

$$b_n(\tau) = \int_{\mathbb{R}^3} K_{ij}(-z)(1-\chi(z))V_{n,i}(z,\tau)V_{n,j}(z,\tau)\,dz$$

and $b(\tau)$ by the same formula with V in place of V_n. The cutoff removes the singularity at the origin, and the integral is absolutely convergent at infinity because, on the dyadic annulus $A_m = \{2^mR < |z| < 2^{m+1}R\}$,

$$\int_{A_m} |z|^{-3}|V_n(z,\tau)|^2\,dz \le C(2^mR)^{-3}\|V_n(\tau)\|^2_{L^3(A_m)}|A_m|^{1/3} \le CL^2(2^mR)^{-2}.$$

The same dyadic estimate shows that b_n and b are bounded uniformly on $[\tau_1,\tau_2]$. The maps $\tau \mapsto b_n(\tau)$ and $\tau \mapsto b(\tau)$ are measurable, because they are limits of measurable truncated integrals. After subtracting $b_n(\tau)$, the exterior contribution on B_R is

$$E_n(x,\tau) = \int_{\mathbb{R}^3} [K_{ij}(x-z) - K_{ij}(-z)](1-\chi(z))V_{n,i}(z,\tau)V_{n,j}(z,\tau)\,dz.$$

The same dyadic estimate, now using $|z|^{-4}$, gives

$$\|E_n(\cdot,\tau)\|_{L^\infty(B_R)} \le C(R)L^2$$

uniformly in n and τ. Splitting the exterior integral into $\{4R < |z| < R'\}$ and $\{|z| > R'\}$, the first part converges by local smooth convergence, while the second part is bounded uniformly by CL^2/R'. Letting first $n \to \infty$ and then $R' \to \infty$ identifies the local limit of E_n with the corresponding exterior contribution for V. Therefore, with $a_n = b_n$ and $a = b$, the decompositions of $P_n - a_n$ and $P - a$ have local parts converging strongly and exterior parts converging distributionally with a uniform local $L^{3/2}$ bound. Hence

$$P_n - a_n(\tau) \rightharpoonup P - a(\tau) \quad \text{weakly in } L^{3/2}(B_R \times [\tau_1,\tau_2]).$$

□

20.3. Stability under locally smooth limits.

Lemma 20.4 (Closure of the mild formulation)**.** *Let V_n be bounded mild ancient solutions satisfying*

$$\sup_n \|V_n\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} < \infty.$$

If $V_n \to V$ locally smoothly, then V is a bounded mild ancient solution of (18.1).

Proof. The common L^∞ bound passes to V by pointwise convergence on compact subsets. Divergence freeness passes distributionally. Fix $\sigma < \tau$ and test (19.4) for V_n against $\phi \in C_c^\infty(\mathbb{R}^3;\mathbb{R}^3)$. The endpoint terms are

$$\langle V_n(\tau),\phi\rangle, \qquad \langle V_n(\sigma), S(\tau-\sigma)^*\phi\rangle.$$

The first converges by local smooth convergence. For the second, note that $S(\tau-\sigma)^*\phi \in L^1(\mathbb{R}^3)$. Approximate this L^1 function by a compactly supported smooth function and use the common L^∞ bound to control the L^1 approximation error uniformly in n.

For the Duhamel term, write

$$\langle S(\tau-s)\mathbb{P}\operatorname{div}(V_n\otimes V_n)(s),\phi\rangle = \langle V_n(s)\otimes V_n(s), (S(\tau-s)\mathbb{P}\operatorname{div})^*\phi\rangle\,.$$

For fixed s, the adjoint test belongs to L^1; pairing convergence again follows by compact approximation in L^1, local smooth convergence, and the common L^∞ bound on $V_n \otimes V_n$. The approximation error is uniform in n for this fixed s. Moreover Lemma 19.1 gives the integrable domination

$$C \sup_n \|V_n\|_{L^\infty}^2 e^{-(\tau-s)/2}(1-e^{-(\tau-s)})^{-1/2}\|\phi\|_{L^1}, \qquad s \in (\sigma,\tau).$$

Dominated convergence in s passes the Duhamel integral to the limit. Thus V satisfies (19.4). □

20.4. **Compactness of bounded families.**

Proposition 20.5 (Compactness of bounded ancient families). *Let $\{V_n\}$ be bounded mild ancient solutions satisfying*

$$\sup_n \|V_n\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} < \infty.$$

Then a subsequence converges locally smoothly to a bounded mild ancient solution V.

Proof. By Lemma 20.1, the sequence is bounded in every C^m-norm on every compact cylinder. Arzela–Ascoli on a countable exhaustion $B_N \times [-N,N]$, followed by a diagonal extraction over (m,N), gives a subsequence converging locally smoothly to a smooth limit. Lemma 20.4 shows that the limit is again a bounded mild ancient solution. □

21. Uniform L^3-tightness and stationary L^3 rigidity

21.1. **Uniform L^3-tightness.**

Definition 21.1 (Uniform L^3-tightness). A bounded ancient solution V is uniformly L^3-tight if

$$\lim_{R\to\infty} \sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V(y,\tau)|^3\, dy = 0.$$

For the compactness statements in this section we sometimes assume, in addition, a common $L^\infty_\tau L^3_y$ bound. The endpoint Liouville argument in Section 25 does not take this bound as a separate hypothesis; it obtains the required sequence estimate from boundedness and the tail condition above.

21.2. **Closedness under locally smooth convergence.**

Lemma 21.2 (Closure of uniform L^3-tightness). *Let V_n be bounded mild ancient solutions satisfying*

$$\sup_n \sup_{\tau\in\mathbb{R}} \|V_n(\tau)\|_{L^3(\mathbb{R}^3)} \le L < \infty$$

and assume that they have a common tightness modulus: for some $\omega : [1,\infty) \to [0,\infty)$ with $\omega(R) \to 0$,

$$\sup_n \sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V_n(y,\tau)|^3\, dy \le \omega(R), \qquad R \ge 1.$$

If $V_n \to V$ locally smoothly, then

$$\sup_{\tau\in\mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} \le L$$

and

$$\sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V(y,\tau)|^3\, dy \le \omega(R), \qquad R \ge 1.$$

In particular V is uniformly L^3-tight.

Proof. Fix τ. On B_R, local smooth convergence gives

$$\int_{B_R} |V(y,\tau)|^3\,dy = \lim_{n\to\infty}\int_{B_R} |V_n(y,\tau)|^3\,dy \le L^3.$$

Letting $R\to\infty$ and using monotone convergence gives $\|V(\tau)\|_{L^3} \le L$. Since τ was arbitrary, the same bound is uniform in time.

For the tail bound, fix $1 \le R < R'$. On the bounded annulus $\{R < |y| < R'\}$, dominated convergence gives

$$\int_{R<|y|<R'} |V(y,\tau)|^3\,dy = \lim_{n\to\infty}\int_{R<|y|<R'} |V_n(y,\tau)|^3\,dy \le \omega(R).$$

Letting $R'\to\infty$ and using monotone convergence gives

$$\int_{|y|>R} |V(y,\tau)|^3\,dy \le \omega(R).$$

The estimate is independent of τ, so taking the supremum in time gives the asserted tail bound. Since $\omega(R)\to 0$, V is uniformly L^3-tight. □

Proof of Theorem 18.1. The compactness assertion is Proposition 20.5. The $L^\infty_\tau L^3_y$ and tightness closure assertions follow from Lemma 21.2 with the displayed constants L and ω. □

21.3. **Stationary time-translate limits.** The stationary self-similar profile equation used below is

$$(w\cdot\nabla)w + \nabla p = \Delta w - \frac{1}{2}(w + y\cdot\nabla w), \qquad \operatorname{div} w = 0. \tag{21.1}$$

Theorem 21.3 (Stationary limits under uniform L^3-tightness)**.** *Let V be uniformly L^3-tight. Let $\tau_n\in\mathbb{R}$, and suppose*

$$\Theta_{\tau_n} V \to W$$

locally smoothly. If W is stationary, then $W\equiv 0$.

Proof. The sequence $\Theta_{\tau_n}V$ has the same L^3 bound and tightness modulus as V. By Lemma 21.2, the limit W belongs to $L^\infty_\tau L^3_y$. If W is stationary, write $W(y,\tau) = w(y)$. Then $w\in L^3(\mathbb{R}^3)$.

By Lemma 20.4, W is a bounded mild ancient solution. The smoothness from Lemma 20.1 and the local pressure lemma Lemma 20.2 give smooth local pressures p_N on B_N at time 0. Since $\partial_\tau W = 0$, each p_N satisfies

$$(w\cdot\nabla)w + \nabla p_N = \Delta w - \frac{1}{2}(w + y\cdot\nabla w) \quad \text{on } B_N.$$

On overlaps the gradients of the local pressures agree, so the pressures differ only by constants. Adjusting those constants and passing to the compatible union gives a scalar distribution p on $\mathbb{R}^3$ for which (21.1) holds in distributions. The stationary L^3 rigidity theorem Theorem 21.4 applies and gives $w=0$. □

21.4. **Stationary self-similar rigidity.**

Theorem 21.4 (Stationary self-similar rigidity)**.** *Let $w\in C^\infty(\mathbb{R}^3;\mathbb{R}^3)\cap L^3(\mathbb{R}^3)$ solve (21.1) in distributions for some scalar distribution p, with $\operatorname{div} w = 0$. Then $w\equiv 0$.*

Proof. This is the theorem of Nečas, Růžička, and Šverák [17], applied to the stationary Leray self-similar system. The smooth distributional formulation stated here is a special case of their weak $L^3(\mathbb{R}^3)$ theorem. □

22. Normalized collections and minimal ancient solutions

22.1. Closed normalized collections.

Definition 22.1 (Closed normalized collection)**.** A collection A of bounded mild ancient solutions is called a closed normalized collection if:

(i) there is $M < \infty$ such that

$$\|V\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} \le M \qquad \text{for all } V \in A;$$

(ii) there are $R_0 > 0$ and $\eta_0 > 0$ such that

$$\sup_{\tau\in\mathbb{R}} \int_{B_{R_0}} |V(y,\tau)|^3\,dy \ge \eta_0 \qquad \text{for all } V \in A; \tag{22.1}$$

(iii) A is invariant under time translation:

$$V \in A,\ s \in \mathbb{R} \quad \Longrightarrow \quad \Theta_s V \in A;$$

(iv) if $V_n \in A$, $V_n \to V$ locally smoothly, and V satisfies the same (R_0, η_0)-nonvanishing condition (22.1), then $V \in A$.

22.2. Time-centering.

Lemma 22.2 (Time-centering)**.** *Let A be a closed normalized collection. If $V \in A$, then for every $\delta > 0$ there exists $s \in \mathbb{R}$ such that*

$$\int_{B_{R_0}} |\Theta_s V(y,0)|^3\,dy \ge \eta_0 - \delta.$$

Proof. By (22.1), the supremum in time of the local L^3 mass is at least η_0. Choose s within δ of this supremum. Since $(\Theta_s V)(y,0) = V(y,s)$, the displayed estimate follows. Time-translation invariance keeps $\Theta_s V$ in A. ☐

22.3. Stability of local nonvanishing.

Lemma 22.3 (Stability of local nonvanishing)**.** *If $V_n \to V$ locally smoothly and*

$$\int_{B_{R_0}} |V_n(y,0)|^3\,dy \ge \eta_0 - o(1),$$

then

$$\int_{B_{R_0}} |V(y,0)|^3\,dy \ge \eta_0.$$

Proof. Local smooth convergence implies uniform convergence on $\overline{B}_{R_0} \times \{0\}$, hence convergence in $L^3(B_{R_0})$. The map $f \mapsto \int_{B_{R_0}} |f|^3$ is continuous on $L^3(B_{R_0})$, and the claim follows by passing to the limit. ☐

22.4. Lower semicontinuity of size.

Lemma 22.4 (Lower semicontinuity of L^∞ size)**.** *If $V_n \to V$ locally smoothly and*

$$\sup_n \|V_n\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} < \infty,$$

then

$$\|V\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} \le \liminf_{n\to\infty} \|V_n\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})}.$$

Proof. Let L_0 be the liminf on the right. If $|V(y_0,\tau_0)| > L_0+\varepsilon$, choose a subsequence with $\|V_n\|_{L^\infty} \to L_0$. Local uniform convergence at (y_0,τ_0) gives $|V_n(y_0,\tau_0)| > L_0+\varepsilon/2$ for all large n, contradicting $\|V_n\|_{L^\infty} \to L_0$. Hence $|V| \le L_0$ pointwise. Since the smooth representative has pointwise supremum equal to its essential L^∞ norm, the result follows. □

22.5. **Existence of a minimal element.**

Theorem 22.5 (Existence of a minimal ancient solution)**.** *Let A be nonempty and closed normalized. Set*

$$m = \inf_{V\in A} \|V\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})}.$$

Then $0 < m < \infty$, and there exists $V_* \in A$ such that

$$\|V_*\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} = m.$$

In particular $V_* \not\equiv 0$.

Proof. The common L^∞ bound gives $m < \infty$. The local nonvanishing gives a positive lower bound:

$$\eta_0 \le \sup_{\tau\in\mathbb{R}} \int_{B_{R_0}} |V(y,\tau)|^3\,dy \le |B_{R_0}|\|V\|^3_{L^\infty(\mathbb{R}^3\times\mathbb{R})},$$

so

$$m \ge (\eta_0/|B_{R_0}|)^{1/3} > 0.$$

Choose $V_n \in A$ with

$$m \le \|V_n\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} \le m + \frac{1}{n}.$$

By Lemma 22.2, after replacing V_n by a time translate, which stays in A and preserves the L^∞ norm, we may assume

$$\int_{B_{R_0}} |V_n(y,0)|^3\,dy \ge \eta_0 - \frac{1}{n}.$$

The common L^∞ bound and Proposition 20.5 give a locally smooth subsequential limit V_*, which is a bounded mild ancient solution. By Lemma 22.3,

$$\int_{B_{R_0}} |V_*(y,0)|^3\,dy \ge \eta_0,$$

so V_* retains the fixed (R_0,η_0)-normalization. Closedness of A then gives $V_* \in A$. Finally, Lemma 22.4 gives

$$\|V_*\|_{L^\infty} \le \liminf_{n\to\infty} \|V_n\|_{L^\infty} = m.$$

Since m is the infimum over A and $V_* \in A$, equality holds. The positive lower bound on m rules out $V_* \equiv 0$. □

22.6. **The uniformly tight normalized class.**

Definition 22.6 (Uniformly tight normalized collection)**.** Fix $M, L, R_0, \eta_0 > 0$ and a modulus

$$\omega : [1,\infty) \to [0,\infty), \qquad \omega(R) \to 0.$$

A collection A of bounded mild ancient solutions is called a uniformly tight normalized collection with data $(M, L, \omega, R_0, \eta_0)$ if:

(1) for every $V \in A$,

$$\|V\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} \le M;$$

(2) for every $V \in A$,

$$\sup_{\tau\in\mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} \le L;$$

(3) for every $V \in A$ and every $R \geq 1$,
$$\sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V(y,\tau)|^3\, dy \leq \omega(R);$$
(4) for every $V \in A$,
$$\sup_{\tau\in\mathbb{R}} \int_{B_{R_0}} |V(y,\tau)|^3\, dy \geq \eta_0;$$
(5) A is invariant under time translation;
(6) A is closed under locally smooth limits retaining item (4).

Proposition 22.7 (Uniformly tight normalized collections are closed normalized)**.** *Every uniformly tight normalized collection is a closed normalized collection.*

Proof. The L^∞ bound, local nonvanishing, time-translation invariance, and closedness clause in Definition 22.1 are part of Definition 22.6. It remains to verify that the stated L^3 bound and fixed tightness modulus are preserved under locally smooth limits. This follows from Lemma 21.2. □

Proof of Theorem 18.2. By Proposition 22.7, the collection is closed normalized. Theorem 22.5 gives the asserted nonzero minimizer. □

23. Compact trajectory hulls

23.1. Trajectory hulls.

Definition 23.1 (Trajectory hull)**.** For a bounded mild ancient solution V, its trajectory hull is
$$\mathcal{H}(V) = \overline{\{\Theta_s V :\ s \in \mathbb{R}\}}^{C^\infty_{\text{loc}}}.$$

23.2. Compactness of bounded hulls.

Theorem 23.2 (Compact trajectory hull)**.** *If V is a bounded mild ancient solution, then $\mathcal{H}(V)$ is compact in the locally smooth topology and invariant under every time translation Θ_s. Every $W \in \mathcal{H}(V)$ is a bounded mild ancient solution with*
$$\|W\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} \leq \|V\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})}.$$

Proof. Every translate $\Theta_s V$ is a bounded mild ancient solution with the same L^∞ bound as V. Hence every sequence of translates has a locally smoothly convergent subsequence by Proposition 20.5. If $(W_n) \subset \mathcal{H}(V)$, choose translates $\Theta_{s_n} V$ with $d(W_n, \Theta_{s_n} V) < 1/n$. A subsequence of $\Theta_{s_n} V$ converges locally smoothly, and the corresponding subsequence of W_n has the same limit. Thus $\mathcal{H}(V)$ is sequentially compact. Since the locally smooth topology is metrizable, it is compact.

The L^∞ estimate and the mild equation pass to elements of the hull by local smooth convergence and Lemma 20.4. For invariance, note that Θ_t maps the orbit of V onto itself. If $W_n \to W$ locally smoothly, then $\Theta_t W_n \to \Theta_t W$ locally smoothly because time translation only shifts the compact time interval on which the seminorms are evaluated. Hence $\Theta_t \mathcal{H}(V) = \mathcal{H}(V)$. □

Corollary 23.3 (Uniform tightness on hulls)**.** *If V is uniformly L^3-tight with bound L and modulus ω, then every $W \in \mathcal{H}(V)$ satisfies*
$$\sup_{\tau\in\mathbb{R}} \|W(\tau)\|_{L^3(\mathbb{R}^3)} \leq L$$
and
$$\sup_{\tau\in\mathbb{R}} \int_{|y|>R} |W(y,\tau)|^3\, dy \leq \omega(R), \qquad R \geq 1.$$

The same estimates hold uniformly on every compact invariant subset $K \subset \mathcal{H}(V)$.

Proof. Every translate of V has the same L^3 bound and tightness modulus. Applying Lemma 21.2 to any sequence of translates converging locally smoothly to $W \in \mathcal{H}(V)$ gives the displayed estimates for W. Taking the supremum over $W \in K \subset \mathcal{H}(V)$ gives the final assertion. □

23.3. Compact invariant sets.

Definition 23.4 (Compact invariant set)**.** A nonempty compact subset K of the locally smooth space of bounded mild ancient solutions is invariant if

$$\Theta_t K = K \qquad \text{for every } t \in \mathbb{R}.$$

It is minimal if it contains no proper nonempty compact invariant subset.

Lemma 23.5 (Continuity of the translation action)**.** *Let K be a compact invariant set. For each $t \in \mathbb{R}$, $\Theta_t : K \to K$ is a homeomorphism, and the action*

$$\mathbb{R} \times K \ni (t, W) \mapsto \Theta_t W \in K$$

is jointly continuous.

Proof. For fixed t, convergence $W_n \to W$ locally smoothly implies

$$\|\Theta_t W_n - \Theta_t W\|_{C^m(B_R \times [-T,T])} = \|W_n - W\|_{C^m(B_R \times [-T+t,T+t])} \to 0.$$

Thus Θ_t is continuous, and its inverse is Θ_{-t}. If $t_n \to t$ and $W_n \to W$, then on any fixed compact cylinder the difference

$$\Theta_{t_n} W_n - \Theta_t W$$

is bounded by a term involving $W_n - W$ on a slightly larger compact cylinder and a term involving the uniform continuity of finitely many derivatives of W on that larger cylinder. Both terms tend to zero, proving joint continuity. □

23.4. Minimal compact invariant subsets.

Proposition 23.6 (Minimal compact invariant subsets)**.** *Let K be a nonempty compact invariant set. Then K contains a nonempty compact minimal invariant subset.*

Proof. Let $\mathcal{F}$ be the family of nonempty compact invariant subsets of K, ordered by reverse inclusion. For a chain $\{K_\alpha\}$, the intersection $\bigcap_\alpha K_\alpha$ is nonempty by compactness and the finite intersection property, since the chain is totally ordered by ordinary inclusion. The intersection is compact and invariant: invariance follows from $\Theta_t K_\alpha = K_\alpha$ for all α and the fact that Θ_t is a homeomorphism. Thus every chain has an upper bound in the reverse-inclusion order. Zorn's lemma gives a maximal element for that order, equivalently a minimal compact invariant subset under ordinary inclusion. □

23.5. The zero trajectory.

Lemma 23.7 (The zero trajectory)**.** *The singleton $\{0\}$ is compact and invariant. If a minimal compact invariant set contains the zero trajectory, then it equals $\{0\}$.*

Proof. The zero solution satisfies $\Theta_t 0 = 0$ for all t. Therefore $\{0\}$ is a nonempty compact invariant subset of any invariant set containing zero. Minimality forces equality. □

24. Invariant measures and the covariance obstruction

24.1. **Krylov–Bogolyubov averages.** The following is the standard Krylov–Bogolyubov averaging argument [23, 20].

Theorem 24.1 (Krylov–Bogolyubov averages)**.** *Let K be a compact invariant set and $W_0 \in K$. For $T > 0$, define*

$$\mu_T = \frac{1}{T}\int_0^T \delta_{\Theta_s W_0}\, ds.$$

Then every sequence $T_n \to \infty$ has a subsequence such that

$$\mu_{T_n} \rightharpoonup \mu$$

weakly as Borel probability measures on K, and every such limit μ is invariant:

$$(\Theta_t)_{\#}\mu = \mu \qquad \textit{for all } t \in \mathbb{R}.$$

If K is minimal, then every invariant Borel probability measure on K has support equal to K.

Proof. The orbit map $s \mapsto \Theta_s W_0$ is continuous by Lemma 23.5; thus μ_T is the push-forward of normalized Lebesgue measure on $[0, T]$. Since K is compact and metrizable, the space of Borel probability measures on K is weakly sequentially compact [20]. Hence μ_{T_n} has a weakly convergent subsequence.

For $F \in C(K)$ and $t > 0$,

$$\int_K F(\Theta_t W)\, d\mu_T(W) - \int_K F(W)\, d\mu_T(W) = \frac{1}{T}\int_T^{T+t} F(\Theta_s W_0)\, ds - \frac{1}{T}\int_0^t F(\Theta_s W_0)\, ds.$$

The absolute value is at most $2t\|F\|_{C(K)}/T$. Letting $T = T_{n_k} \to \infty$ gives invariance for $t > 0$; negative times follow by applying the positive-time result to $F \circ \Theta_t$.

If K is minimal and μ is invariant, then $\operatorname{supp}\mu$ is nonempty, closed, and invariant. Indeed, if U is a neighborhood of $\Theta_t W$, then $\Theta_{-t}U$ is a neighborhood of W, and invariance shows $\mu(U) = \mu(\Theta_{-t}U) > 0$ whenever $W \in \operatorname{supp}\mu$. Hence $\Theta_t(\operatorname{supp}\mu) \subset \operatorname{supp}\mu$. Applying the same argument with $-t$ gives the reverse inclusion, so $\operatorname{supp}\mu$ is invariant. It is nonempty because $\mu(K) = 1$, and it is compact because it is closed in compact K. Minimality forces $\operatorname{supp}\mu = K$. □

24.2. **Support on minimal hulls.**

Corollary 24.2 (Nonzero support)**.** *Let K be a minimal compact invariant set with $K \neq \{0\}$. Then zero is not in K, and every invariant probability measure on K has support equal to K.*

Proof. If zero were in K, Lemma 23.7 would give $K = \{0\}$, contrary to the assumption. The support conclusion is the minimal-support part of Theorem 24.1. □

24.3. **Stationary statistical identity.**

Theorem 24.3 (Stationary statistical form)**.** *Let K be a compact invariant set and μ an invariant Borel probability measure on K. Then for every solenoidal $\phi \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$,*

$$\begin{aligned} 0 = \int_K \Bigg[&\int_{\mathbb{R}^3} W(y,0)\cdot\left(\Delta\phi + \phi + \frac{1}{2}y\cdot\nabla\phi\right) dy \\ &+ \int_{\mathbb{R}^3} W_i(y,0)W_j(y,0)\partial_j\phi_i(y)\, dy\Bigg] d\mu(W). \end{aligned} \tag{24.1}$$

Proof. Define

$$F(W) = \int_{\mathbb{R}^3} W(y,0)\cdot\phi(y)\,dy.$$

Local smooth convergence makes F continuous on K: if $W_n \to W$, then $W_n(\cdot,0) \to W(\cdot,0)$ uniformly on the compact support of ϕ. Invariance gives

$$\int_K \frac{F(\Theta_h W) - F(W)}{h}\,d\mu(W) = 0 \qquad (0 < |h| \le 1).$$

Choose R with $\operatorname{supp}\phi \subset B_R$. On $\overline{B}_R \times [-1,1]$, compactness of K in the locally smooth topology gives

$$M_\phi = \sup_{W\in K} \|\partial_\tau W\|_{L^\infty(\overline{B}_R\times[-1,1])} < \infty.$$

For $0 < |h| \le 1$,

$$\frac{F(\Theta_h W) - F(W)}{h} = \int_{\mathbb{R}^3} \left(\frac{1}{h}\int_0^h \partial_\tau W(y,s)\,ds\right)\cdot\phi(y)\,dy,$$

and the absolute value is bounded by $|B_R|\|\phi\|_{L^\infty} M_\phi$, uniformly in W and h. For each fixed W, smoothness gives uniform convergence of the inner average to $\partial_\tau W(y,0)$ on $\operatorname{supp}\phi$. Dominated convergence on K yields

$$\int_K \int_{\mathbb{R}^3} \partial_\tau W(y,0)\cdot\phi(y)\,dy\,d\mu(W) = 0.$$

For each W, the smooth local equation (18.1) at time zero, tested against solenoidal ϕ, gives

$$\begin{aligned}\int_{\mathbb{R}^3} \partial_\tau W\cdot\phi\,dy = \int_{\mathbb{R}^3} W\cdot\Delta\phi\,dy + \int_{\mathbb{R}^3} W\cdot\phi\,dy + \frac{1}{2}\int_{\mathbb{R}^3} W\cdot(y\cdot\nabla\phi)\,dy \\ + \int_{\mathbb{R}^3} W_iW_j\partial_j\phi_i\,dy.\end{aligned}$$

The pressure term vanishes because $\operatorname{div}\phi = 0$; the drift identity uses

$$\int y_j\partial_j W_i\,\phi_i\,dy = -3\int W\cdot\phi\,dy - \int W\cdot(y\cdot\nabla\phi)\,dy.$$

The right-hand side is a continuous bounded function of $W \in K$. Averaging over K gives (24.1). □

24.4. Barycenter and covariance.

Theorem 24.4 (Barycenter and covariance)**.** *Let K and μ be as in Theorem 24.3. Define*

$$\overline{W}(y) = \int_K W(y,0)\,d\mu(W)$$

and

$$C_{ij}(y) = \int_K W_i(y,0)W_j(y,0)\,d\mu(W) - \overline{W}_i(y)\overline{W}_j(y).$$

Then $\overline{W}$ and C are C^∞_{loc}, $\overline{W}$ is solenoidal, and for every solenoidal $\phi \in C_c^\infty(\mathbb{R}^3;\mathbb{R}^3)$,

$$\int_{\mathbb{R}^3} \overline{W}\cdot\left(\Delta\phi + \phi + \frac{1}{2}y\cdot\nabla\phi\right)dy + \int_{\mathbb{R}^3} \overline{W}_i\overline{W}_j\partial_j\phi_i\,dy + \int_{\mathbb{R}^3} C_{ij}\partial_j\phi_i\,dy = 0. \tag{24.2}$$

In particular, $\overline{W}$ is a weak stationary self-similar solution against solenoidal test fields if and only if

$$\int_{\mathbb{R}^3} C_{ij}\partial_j\phi_i\,dy = 0$$

for every solenoidal test field ϕ.

Proof. Compactness of K in the locally smooth topology gives, for every compact ball and every derivative order, a uniform bound on $\partial_y^\alpha W(y,0)$ for $W \in K$. Hence one may differentiate $\overline{W}$ under the integral sign. More explicitly, for every multiindex α and every y in a fixed compact ball,

$$\partial_y^\alpha \overline{W}(y) = \int_K \partial_y^\alpha W(y,0)\, d\mu(W),$$

because the difference quotients are dominated by the next derivative seminorm, uniformly on K. The same argument applies to the second moment

$$M_{ij}(y) = \int_K W_i(y,0) W_j(y,0)\, d\mu(W),$$

using Leibniz' rule and the uniform bounds on derivatives of W. This gives C^∞_{loc} regularity of $\overline{W}$, M, and $C = M - \overline{W} \otimes \overline{W}$. Since every W is divergence free, Fubini against scalar test functions gives $\operatorname{div} \overline{W} = 0$: for $\eta \in C_c^\infty(\mathbb{R}^3)$,

$$\int_{\mathbb{R}^3} \overline{W} \cdot \nabla \eta \, dy = \int_K \int_{\mathbb{R}^3} W(y,0) \cdot \nabla \eta(y)\, dy\, d\mu(W) = 0.$$

For each compactly supported test field ϕ, the maps

$$W \mapsto \int_{\mathbb{R}^3} W(y,0) \cdot \phi(y)\, dy, \qquad W \mapsto \int_{\mathbb{R}^3} W_i(y,0) W_j(y,0) \partial_j \phi_i(y)\, dy$$

are continuous and bounded on K, again by compactness in C^∞_{loc}. Thus all integrals below are finite, and Fubini's theorem applies.

Applying Theorem 24.3 and passing the linear term through the K-integral gives the first integral in (24.2). Passing the nonlinear term through the integral gives

$$\int_K W_i W_j \, d\mu = \overline{W}_i \overline{W}_j + C_{ij}.$$

Substitution yields (24.2). Comparing this identity with the pressure-free stationary weak formulation for $\overline{W}$ gives the stated equivalence with vanishing covariance contribution. ☐

25. Endpoint L^3 closure of the uniformly tight class

25.1. **Physical pullback.** Let V be a smooth centered ancient velocity and let P be a compatible local pressure. For $t < 0$ define

$$u(x,t) = (-t)^{-1/2} V\left(\frac{x}{\sqrt{-t}}, -\log(-t)\right), \tag{25.1}$$

and

$$p(x,t) = (-t)^{-1} P\left(\frac{x}{\sqrt{-t}}, -\log(-t)\right). \tag{25.2}$$

Equivalently,

$$V(y,\tau) = e^{-\tau/2} u(e^{-\tau/2} y, -e^{-\tau}), \qquad P(y,\tau) = e^{-\tau} p(e^{-\tau/2} y, -e^{-\tau}).$$

Lemma 25.1 (Pullback solves physical Navier–Stokes)**.** *Let V be a bounded mild ancient solution of* (18.1)*. Then the pullback u in* (25.1) *is a smooth divergence-free solution of*

$$\partial_t u + (u \cdot \nabla) u + \nabla p = \Delta u, \qquad \operatorname{div} u = 0, \tag{25.3}$$

on $\mathbb{R}^3 \times (-\infty, 0)$, with pressure p locally given by (25.2)*. Moreover, for every $T < 0$,*

$$\|u\|_{L^\infty(\mathbb{R}^3 \times (-\infty, T])} \le (-T)^{-1/2} \|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}.$$

Proof. By Lemma 20.1, V is smooth. By Lemma 20.2, on every compact parabolic cylinder there is a smooth pressure representative P, unique up to a function of τ, for which (18.1) holds classically. Adding a function of τ to P only adds a function of t to p, and hence does not affect ∇p.

Set
$$\lambda(t) = (-t)^{-1/2}, \qquad y = \lambda(t)x, \qquad \tau = -\log(-t).$$
Then $\dot{\tau} = \lambda^2$ and
$$\partial_t y = \frac{1}{2}\lambda^2 y, \qquad \dot{\lambda} = \frac{1}{2}\lambda^3.$$
Using $u = \lambda V(y, \tau)$, one obtains
$$\partial_t u = \lambda^3 \left(\partial_\tau V + \frac{1}{2} y \cdot \nabla V + \frac{1}{2} V \right),$$
while
$$(u \cdot \nabla_x) u = \lambda^3 (V \cdot \nabla_y) V, \qquad \Delta_x u = \lambda^3 \Delta_y V, \qquad \nabla_x p = \lambda^3 \nabla_y P.$$
Substituting these identities into (18.1) gives (25.3). The divergence identity follows similarly:
$$\operatorname{div}_x u = \lambda^2 \operatorname{div}_y V = 0.$$
Finally, if $t \le T < 0$, then $(-t)^{-1/2} \le (-T)^{-1/2}$, and the stated L^∞ bound follows directly from (25.1). $\square$

25.2. **Critical L^3 invariance.**

Lemma 25.2 (Invariance of the critical norm)**.** *Let u be the physical pullback of V. Then, for every $t < 0$,*
$$\|u(t)\|_{L^3(\mathbb{R}^3)} = \|V(-\log(-t))\|_{L^3(\mathbb{R}^3)}. \tag{25.4}$$

Proof. With $y = x/\sqrt{-t}$, (25.1) gives
$$|u(x,t)|^3 = (-t)^{-3/2} |V(y, -\log(-t))|^3, \qquad dx = (-t)^{3/2} dy.$$
Multiplying these two identities and integrating over $\mathbb{R}^3$ gives (25.4). $\square$

Lemma 25.3 (Tail tightness gives an endpoint sequence)**.** *Let V be a bounded function on $\mathbb{R}^3 \times \mathbb{R}$ satisfying*
$$\lim_{R\to\infty} \sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V(y,\tau)|^3 \, dy = 0.$$
Then, for every sequence $\tau_k \in \mathbb{R}$,
$$\sup_k \|V(\tau_k)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Proof. Let $M = \|V\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})}$. Choose $R \ge 1$ such that
$$\sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V(y,\tau)|^3 \, dy \le 1.$$
Then, for every k,
$$\int_{\mathbb{R}^3} |V(y,\tau_k)|^3 \, dy \le M^3 |B_R| + \sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V(y,\tau)|^3 \, dy \le M^3 |B_R| + 1.$$
This gives the claimed endpoint sequence bound. $\square$

25.3. **Mildness after finite terminal-time restriction.**

Lemma 25.4 (Physical pullbacks are mild ancient solutions on finite intervals)**.** *Let V be a bounded mild ancient solution of* (18.1)*, and let u be its physical pullback. For every $T < 0$, the restriction of u to $\mathbb{R}^3 \times (-\infty, T]$ is a bounded mild ancient solution of the physical Navier–Stokes equations after translating the terminal time T to 0.*

Proof. By Lemma 25.1, u is a smooth bounded classical solution on $\mathbb{R}^3 \times (-\infty, T]$. Let $s < t \le T$. The classical equation on the finite slab $[s, t]$ gives, for every solenoidal $\phi \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$, the weak Duhamel identity

$$\begin{aligned}\langle u(t), \phi\rangle &= \langle e^{(t-s)\Delta} u(s), \phi\rangle \\ &\quad - \int_s^t \left\langle e^{(t-r)\Delta} \mathbb{P} \operatorname{div}(u \otimes u)(r), \phi \right\rangle dr.\end{aligned}$$

Equivalently,

$$u(t) = e^{(t-s)\Delta} u(s) - \int_s^t e^{(t-r)\Delta} \mathbb{P} \operatorname{div}(u \otimes u)(r)\, dr$$

in the weak-* topology of $L^\infty(\mathbb{R}^3)$. The heat–Stokes kernel bound used in Lemma 19.1 makes the integral finite on every finite interval because u is bounded on $\mathbb{R}^3 \times (-\infty, T]$. Thus the restriction is a bounded mild ancient solution. The time translate $u_T(x, \sigma) = u(x, \sigma + T)$, $\sigma \le 0$, has the same property on $(-\infty, 0]$. □

25.4. **Endpoint L^3 ancient Liouville theorem.**

Theorem 25.5 (Albritton–Barker endpoint ancient Liouville theorem)**.** *Let u be a bounded mild ancient solution of the three-dimensional Navier–Stokes equations on $\mathbb{R}^3 \times (-\infty, 0]$. If there exists a sequence $s_k \downarrow -\infty$ such that*

$$\sup_k \|u(s_k)\|_{L^3(\mathbb{R}^3)} < \infty,$$

then $u \equiv 0$.

Proof. This is the endpoint ancient Liouville theorem of Albritton–Barker [1]. The formulation above is the translated terminal-time version needed here. □

Theorem 25.6 (Tail-tight centered ancient solutions vanish)**.** *Let V be a bounded mild ancient solution of* (18.1)*. Assume*

$$\lim_{R\to\infty} \sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V(y, \tau)|^3\, dy = 0.$$

Then $V \equiv 0$.

Proof. Let u be the physical pullback of V. Fix $T < 0$. By Lemma 25.4, after translating the terminal time T to 0, the restriction of u to $(-\infty, T]$ is a bounded mild ancient solution. Choose $t_k = -e^k$. For all sufficiently large k, $t_k < T$, and $\tau_k = -\log(-t_k) = -k$. By Lemma 25.3,

$$\sup_k \|V(-k)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Hence (25.4) gives

$$\|u(t_k)\|_{L^3(\mathbb{R}^3)} = \|V(-k)\|_{L^3(\mathbb{R}^3)}$$

uniformly along the same sequence, after discarding the finitely many indices for which $t_k \ge T$. Set $u_T(x, \sigma) = u(x, \sigma + T)$ for $\sigma \le 0$ and $s_k = t_k - T$. Then $s_k \downarrow -\infty$ and

$$\|u_T(s_k)\|_{L^3(\mathbb{R}^3)} = \|u(t_k)\|_{L^3(\mathbb{R}^3)}$$

is uniformly bounded. Applying Theorem 25.5 to u_T gives $u_T \equiv 0$ on $\mathbb{R}^3 \times (-\infty, 0]$, equivalently $u = 0$ on $\mathbb{R}^3 \times (-\infty, T]$. Since $T < 0$ was arbitrary, $u = 0$ on $\mathbb{R}^3 \times (-\infty, 0)$. The inverse change of variables then gives

$$V(y,\tau) = e^{-\tau/2} u(e^{-\tau/2} y, -e^{-\tau}) = 0$$

for every (y, τ). □

26. Tight-Class Liouville Criterion and Type I Reduction

26.1. Normalized tight collections.

Theorem 26.1 (No nonempty normalized uniformly tight collection)**.** *No uniformly tight normalized collection in the sense of Definition 22.6 is nonempty.*

Proof. Suppose A is such a collection and choose $V \in A$. By the tightness modulus in item (3) of Definition 22.6, V satisfies the tail condition in Theorem 25.6. Hence Theorem 25.6 gives $V \equiv 0$. This contradicts the local velocity normalization in the same definition:

$$\sup_{\tau\in\mathbb{R}} \int_{B_{R_0}} |V(y,\tau)|^3 \, dy \geq \eta_0 > 0.$$

Hence A is empty. □

Proof of Theorem 18.3. If A were nonempty, choose $V \in A$. The common tail modulus gives the tail condition in Theorem 25.6, so $V \equiv 0$. This contradicts the fixed local nonvanishing normalization in Theorem 18.3. □

26.2. Use in the Type I reduction.

Definition 26.2 (Tight-class Liouville consequence)**.** The tight-class Liouville consequence is the assertion that no normalized Seregin ancient limit generated by an admissible Type I extraction and satisfying the compact velocity nonvanishing normalization of the form in Proposition 12.2 is uniformly L^3-tight.

Lemma 26.3 (From compact velocity nonvanishing to local velocity normalization)**.** *Let V be a smooth bounded normalized Seregin ancient limit. Suppose that the extraction gives a compact velocity lower bound: there are a compact set $K \Subset \mathbb{R}^3 \times \mathbb{R}$ and $\eta_I > 0$ such that*

$$\iint_K |V(y,\tau)|^3 \, dy \, d\tau \geq \eta_I.$$

Then there exist $R_0 > 0$ and $\eta_0 > 0$ such that

$$\sup_{\tau\in\mathbb{R}} \int_{B_{R_0}} |V(y,\tau)|^3 \, dy \geq \eta_0.$$

Proof. Choose $R_0 > 0$ and a compact interval $I = [a, b]$, $b > a$, such that $K \subset B_{R_0} \times I$. Then

$$\eta_I \leq \int_a^b \int_{B_{R_0}} |V(y,\tau)|^3 \, dy \, d\tau \leq (b-a) \sup_{\tau\in\mathbb{R}} \int_{B_{R_0}} |V(y,\tau)|^3 \, dy.$$

Thus the conclusion holds with $\eta_0 = \eta_I/(b-a)$. □

Corollary 26.4 (Tight-class consequence for the Type I reduction)**.** *Definition 26.2 holds.*

Proof. Suppose, contrary to Definition 26.2, that the Seregin extraction produces a normalized ancient limit V in the uniformly L^3-tight class from an admissible Type I extraction. By the definition of an admissible Type I extraction, V carries compact velocity nonvanishing: there are $K \Subset \mathbb{R}^3 \times \mathbb{R}$ and $\eta_I > 0$ with

$$\iint_K |V(y,\tau)|^3 \, dy \, d\tau \geq \eta_I.$$

In particular $V \not\equiv 0$. Since V is assumed uniformly L^3-tight, it satisfies the tail condition

$$\lim_{R\to\infty} \sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V(y,\tau)|^3 \, dy = 0.$$

Theorem 25.6 then gives $V \equiv 0$, contradicting the compact velocity lower bound above. Hence no normalized Seregin ancient limit generated by an admissible Type I extraction can belong to the uniformly L^3-tight class. □

The tight-class consequence used in the Type I reduction is therefore proved without the residual-class hypothesis. The proof relies only on boundedness, the tail-tightness condition of the uniformly L^3-tight class, the local-in-time physical pullback, and the endpoint ancient Liouville theorem of Albritton–Barker.

Part IV. Structured Ancient Solution Exclusions

Overview. The final part concerns bounded ancient solutions of the three-dimensional Navier–Stokes equations arising from Type I blow-up arguments. Explicit PDE hypotheses are stated under which such solutions are spatially constant; in the Type I class, these constants vanish. The perturbative weighted-vorticity Lyapunov rigidity used by the conditional Type I reduction is also proved.

For the unforced equations, the cited axisymmetric Liouville theorems cover the following cases: absence of swirl, pointwise scale-invariant control, finite $L^\infty_t L^p_x$ control of the swirl variable $\Gamma = ru_\theta$, and axial periodicity with bounded Γ. When one of these hypotheses holds for a nonzero Type I blow-up limit, conclusions up to Galilean transformations reduce to the zero solution and contradict the nontriviality of the limit.

The perturbative part uses the Lyapunov rigidity argument coming from Gallay–Wayne's small-data decay theorem for the three-dimensional rescaled vorticity equation. It yields a coercive Lyapunov functional forcing any ancient trajectory that stays in the perturbative weighted-vorticity ball to be zero. The section concludes with a summary of the Liouville hypotheses and cited PDE results.

27. Introduction

27.1. **Ancient solutions from Type I blow-up.** The Type I blow-up argument in Theorem 9.1 produces, after parabolic rescaling around a Type I singular point, a nonzero ancient solution (U, P) of

$$\partial_t U + (U\cdot\nabla)U + \nabla P = \Delta U, \qquad \nabla\cdot U = 0, \qquad t<0, \tag{27.1}$$

satisfying the Type I ancient bound

$$\sup_{t<0} \sqrt{-t}\,\|U(t)\|_{L^\infty(\mathbb{R}^3)} < \infty. \tag{27.2}$$

Thus every result proving $U \equiv 0$ for an ancient solution satisfying (27.2) has the same consequence in the Type I problem:

$$\text{PDE hypothesis} \quad\Longrightarrow\quad U\equiv 0 \quad\Longrightarrow\quad \text{contradiction with nonzero blow-up limit.}$$

The hypotheses treated in this part are precisely those covered by a cited Liouville theorem or by the perturbative argument in Section 31:

$$\text{axisymmetric Liouville hypotheses,} \qquad \text{small weighted rescaled vorticity.}$$

No conclusion is asserted for decaying or symmetry-restricted ancient solutions outside the stated hypotheses. Each conclusion is tied to an explicit PDE hypothesis and a specified analytic result.

For the Type I application in Theorem 9.1, the exclusions used are precisely the cases in which the stated PDE hypothesis and the cited or proved Liouville theorem imply $U \equiv 0$. Ancient solutions outside these hypotheses are treated separately.

27.2. **Axisymmetric Liouville hypotheses.** Several unforced axisymmetric cases are covered by cited Liouville theorems. Koch–Nadirashvili–Seregin–Šverák cover the no-swirl case and the pointwise scale-invariant condition $|u| \le C/r$. The swirl-bearing finite-L^p condition

$$\Gamma = ru_\theta \in L^\infty_t L^p_x, \qquad 1\le p<\infty,$$

is covered by Lei–Zhang–Zhao. The z-periodic case with bounded Γ is covered by Lei–Ren–Zhang.

For Type I ancient limits, a constant conclusion after the relevant Galilean reduction must be converted into zero in the blow-up frame. The proof time-shifts away from the terminal time $t = 0$, applies the bounded ancient Liouville theorem, patches the resulting constants on backward intervals, and uses (27.2): if $U(x,t) \equiv c$, then

$$\sup_{t<0} \sqrt{-t}\, \|U(t)\|_\infty = \sup_{t<0} \sqrt{-t}\, |c|$$

is finite only when $c = 0$.

27.3. **Lyapunov hypotheses.** The Lyapunov result uses a forward Lyapunov functional in the Gallay–Wayne perturbative weighted-vorticity regime. If w solves the rescaled three-dimensional vorticity equation and remains in the Gallay–Wayne small ball of $L^2_\sigma(m)$, $m > 7/2$, then the functional

$$\mathcal{L}_m(w_0) = \sup_{\tau \ge 0} e^{2\tau} \|\Phi_\tau w_0\|_m^2 \tag{27.3}$$

is coercive and contracts along the forward semiflow. Sending the starting time to $-\infty$ then forces every ancient trajectory in the small ball to be zero.

27.4. **Main results.** The first result states the unforced axisymmetric Liouville consequences relevant to the Type I argument.

Theorem 27.1 (Axisymmetric Liouville consequences for Type I ancient solutions)**.** *Let U be a smooth Type I ancient solution of* (27.1) *satisfying* (27.2)*. If U satisfies any of the following axisymmetric assumptions, then $U \equiv 0$:*

(i) *axisymmetric without swirl;*

(ii) *axisymmetric and satisfying a pointwise scale-invariant bound*

$$|U(x,t)| \le \frac{C}{r};$$

(iii) *axisymmetric with*

$$\Gamma = rU_\theta \in L^\infty_t L^p_x\big(\mathbb{R}^3 \times (-\infty, 0)\big), \qquad 1 \le p < \infty;$$

(iv) *axisymmetric, periodic in the physical axial variable z, and with*

$$\Gamma = rU_\theta \in L^\infty\big(\mathbb{R}^3 \times (-\infty, 0)\big).$$

Consequently none of these assumptions can hold for a nonzero Type I blow-up limit.

Theorem 27.2 (Perturbative weighted-vorticity Lyapunov rigidity)**.** *Let $m > \frac{7}{2}$. Let $w(\tau)$ be an ancient mild solution of the Gallay–Wayne rescaled vorticity equation*

$$\partial_\tau w = \Lambda w - (v \cdot \nabla) w + (w \cdot \nabla) v, \qquad \nabla \cdot w = 0, \tag{27.4}$$

with v recovered by the three-dimensional Biot–Savart law. Assume

$$\sup_{\tau \in \mathbb{R}} \|w(\tau)\|_{L^2_\sigma(m)} \le \rho_m,$$

where ρ_m is the Gallay–Wayne small-data radius. Then

$$w \equiv 0.$$

Here "ancient mild" means that every forward time-shift of w is the Gallay–Wayne mild solution launched from its value at the starting time.

Theorem 27.3 (Consequences for Type I blow-up limits)**.** *Let U be a nonzero Type I blow-up limit. Then U cannot satisfy any of the axisymmetric assumptions in Theorem 27.1; nor can its corresponding rescaled vorticity satisfy the hypotheses of Theorem 27.2.*

Only the explicit hypotheses in the theorem are used.

27.5. **Relation to the preceding parts.** Theorem 9.1 reduces Type I singularities to nonzero ancient limits satisfying (27.2). Theorem 25.6 treats uniformly L^3-tight ancient solutions. The axisymmetric and perturbative weighted-vorticity exclusions used by the Type I argument are proved in Sections 30 and 31. Cited Liouville theorems rule out the axisymmetric cases, and the Lyapunov argument handles the perturbative weighted-vorticity case. Theorem 32.1 does not depend on Theorem 25.6.

27.6. **Summary of hypotheses.**

Hypothesis	Conclusion	Analytic ingredient
axisymmetric no swirl	$U \equiv 0$	KNSS
pointwise $\lvert u\rvert \le C/r$	$U \equiv 0$	KNSS
$\Gamma \in L^\infty_t L^p_x$, $1 \le p < \infty$	$U \equiv 0$	LZZ
z-periodic bounded Γ	$U \equiv 0$	LRZ
small weighted vorticity	$w \equiv 0$	Gallay–Wayne

27.7. **Organization.** Section 28 introduces the Type I ancient setup, self-similar variables, the invariance facts, and the time-shift lemma. Section 29 fixes axisymmetric notation and the cited Liouville theorems. Section 30 proves the unforced axisymmetric Liouville consequences. Section 31 proves the Gallay–Wayne Lyapunov rigidity, and Section 32 summarizes the consequences.

28. Ancient Type I limits and self-similar variables

28.1. **Type I ancient solutions.**

Definition 28.1 (Type I ancient solution)**.** A smooth mild solution U of (27.1) on $\mathbb{R}^3 \times (-\infty, 0)$ is called Type I ancient if (27.2) holds. Mild is understood in the standard velocity formulation [12, 19]: for every $-\infty < s < t < 0$,

$$U(t) = e^{(t-s)\Delta}U(s) - \int_s^t e^{(t-\sigma)\Delta}\mathbb{P}\nabla \cdot (U(\sigma) \otimes U(\sigma))\, d\sigma \tag{28.1}$$

in distributions, where $\mathbb{P}$ is the Leray projector.

28.2. **Nonzero Type I blow-up limits.**

Definition 28.2 (Nonzero Type I blow-up limit)**.** A Type I ancient solution U is called a nonzero Type I blow-up limit if it arises as the limit of parabolic rescalings around a Type I singular point and the limiting solution is not identically zero. The arguments below require only the conclusion $U \not\equiv 0$.

Lemma 28.3 (Nontriviality of blow-up limits)**.** *A nonzero Type I blow-up limit is not identically zero.*

Proof. The definition includes the condition $U \not\equiv 0$. □

28.3. **Backward self-similar variables.** For $t < 0$, define

$$V(y,\tau) = \sqrt{-t}\, U(x,t), \qquad Q(y,\tau) = (-t)P(x,t), \qquad y = \frac{x}{\sqrt{-t}}, \qquad \tau = -\log(-t). \tag{28.2}$$

Equivalently,

$$x = e^{-\tau/2}y, \qquad t = -e^{-\tau}.$$

Lemma 28.4 (Self-similar equation)**.** *Let (U, P) be a smooth solution of (27.1), and define (V, Q) by (28.2). Then V is defined on $\mathbb{R}^3 \times \mathbb{R}$ and satisfies*

$$\partial_\tau V + (V \cdot \nabla)V + \nabla Q = \Delta V - \frac{1}{2}(V + y \cdot \nabla V), \qquad \nabla \cdot V = 0. \tag{28.3}$$

Moreover (27.2) is equivalent to

$$\sup_{\tau \in \mathbb{R}} \|V(\tau)\|_{L^\infty(\mathbb{R}^3)} < \infty.$$

Proof. Let $a = \sqrt{-t} = e^{-\tau/2}$. Then $V = aU$, $y = x/a$, and

$$\partial_t \tau = \frac{1}{-t} = \frac{1}{a^2}, \qquad \partial_t y = \frac{y}{2a^2}, \qquad \partial_t(a^{-1}) = \frac{1}{2a^3}.$$

Since $U = a^{-1}V$, differentiating at fixed x gives

$$\partial_t U = a^{-3}\left(\partial_\tau V + \frac{1}{2} y \cdot \nabla V + \frac{1}{2} V\right).$$

Spatial derivatives scale as

$$\nabla_x U = a^{-2}\nabla_y V, \qquad \Delta_x U = a^{-3}\Delta_y V,$$

and the pressure scaling gives

$$\nabla_x P = a^{-3}\nabla_y Q.$$

Thus

$$(U \cdot \nabla_x)U = a^{-3}(V \cdot \nabla_y)V.$$

Multiplying (27.1) by a^3 yields (28.3); the divergence equation follows from $\nabla_x \cdot U = a^{-2}\nabla_y \cdot V$. The spatial map $y \mapsto x = e^{-\tau/2}y$ is a bijection for each τ, and

$$\|V(\tau)\|_\infty = \sqrt{-t}\,\|U(t)\|_\infty.$$

Taking suprema over $t < 0$, equivalently $\tau \in \mathbb{R}$, gives the last assertion. $\square$

28.4. **Preservation of axisymmetry and swirl variables.** For axisymmetric flows we write

$$x = (r\cos\theta, r\sin\theta, z), \qquad U = U_r e_r + U_\theta e_\theta + U_z e_z,$$

and in centered variables

$$y = (\rho\cos\theta, \rho\sin\theta, Z), \qquad V = V_\rho e_\rho + V_\theta e_\theta + V_Z e_Z.$$

The scalar swirl variables are

$$\Gamma_U(r, z, t) = rU_\theta(r, z, t), \qquad \Gamma_V(\rho, Z, \tau) = \rho V_\theta(\rho, Z, \tau).$$

Lemma 28.5 (Preservation of axisymmetric structure)**.** *Axisymmetry and absence of swirl are preserved under (28.2). If U is axisymmetric, then*

$$\Gamma_V(\rho, Z, \tau) = \Gamma_U(r, z, t), \qquad (r, z) = \sqrt{-t}\,(\rho, Z).$$

Fixed physical z-periodicity is preserved by physical time shifts, but under the centered change of variables it becomes a τ-dependent period in the Z-coordinate.

Proof. The self-similar map is a scalar spatial dilation and a scalar multiplication of the velocity, so it commutes with rotations around the symmetry axis. The cylindrical basis is unchanged by radial dilation, and

$$V_\theta(\rho, Z, \tau) = \sqrt{-t}\,U_\theta(r, z, t), \qquad r = \sqrt{-t}\rho, \quad z = \sqrt{-t}Z.$$

Thus $U_\theta \equiv 0$ if and only if $V_\theta \equiv 0$, and

$$\Gamma_V = \rho V_\theta = \rho\sqrt{-t}\,U_\theta = rU_\theta = \Gamma_U.$$

If U has physical axial period L, then V has period $L/\sqrt{-t} = Le^{\tau/2}$ in Z, not a fixed period. $\square$

28.5. **Time shifts away from the singular time.**

Lemma 28.6 (Time shifts away from the singular time)**.** *Let U be a smooth Type I ancient solution and fix $T < 0$. Then*

$$W_T(x,s) = U(x,T+s), \qquad \Pi_T(x,s) = P(x,T+s), \qquad s < 0,$$

is a smooth bounded mild ancient solution of (27.1) *on $\mathbb{R}^3 \times (-\infty, 0)$. Moreover the axisymmetric, no-swirl, L^p-swirl, bounded-Γ, and physical z-periodic properties are preserved.*

Proof. Time translation preserves the differential equation because $\partial_s W_T(x,s) = \partial_t U(x,T+s)$ and all spatial derivatives are unchanged. For the mild formulation, apply (28.1) to U between the physical times $T + s < T + t < 0$:

$$U(T+t) = e^{(t-s)\Delta}U(T+s) - \int_{T+s}^{T+t} e^{(T+t-\eta)\Delta}\mathbb{P}\nabla \cdot (U(\eta) \otimes U(\eta))\, d\eta.$$

With $\eta = T + \sigma$, this becomes the mild formula for W_T on the interval (s,t). For $s < 0$, $T + s < T < 0$, so the Type I bound gives

$$\|W_T(s)\|_\infty = \|U(T+s)\|_\infty \le \frac{1}{\sqrt{-T}} \sup_{\eta<0} \sqrt{-\eta}\, \|U(\eta)\|_\infty.$$

The invariance assertions follow because only the time variable has been translated. In particular

$$\Gamma_{W_T}(r,z,s) = \Gamma_U(r,z,T+s),$$

so $L^\infty_s L^p_x$ and L^∞ bounds for Γ restrict to the shifted solution, and a fixed physical axial period remains the same. $\square$

28.6. **PDE hypotheses.**

The arguments are stated directly in terms of hypotheses on ancient solutions. The axisymmetric hypotheses are:

(i) axisymmetry and $U_\theta \equiv 0$;
(ii) axisymmetry and the pointwise bound $|U(x,t)| \le C/r$ for $r > 0$;
(iii) axisymmetry and

$$\Gamma = rU_\theta \in L^\infty_t L^p_x\big(\mathbb{R}^3 \times (-\infty,0)\big), \qquad 1 \le p < \infty;$$

(iv) axisymmetry, periodicity in the physical axial variable z, and $\Gamma \in L^\infty$.

The additional hypothesis used later is the Gallay–Wayne small weighted-vorticity condition.

29. Axisymmetric notation and cited Liouville theorems

29.1. **Cylindrical variables and the swirl scalar.**

For an axisymmetric solution u, write

$$u = u_r e_r + u_\theta e_\theta + u_z e_z, \qquad \Gamma = ru_\theta, \qquad b = u_r e_r + u_z e_z.$$

When Γ is placed in $L^p(\mathbb{R}^3)$, it is regarded as the corresponding function of $x = (r\cos\theta, r\sin\theta, z)$, independent of θ:

$$\|\Gamma(t)\|^p_{L^p(\mathbb{R}^3)} = 2\pi \int_{\mathbb{R}} \int_0^\infty |\Gamma(r,z,t)|^p r\, dr\, dz, \qquad 1 \le p < \infty.$$

29.2. **Constant solutions and Galilean reduction.**

Definition 29.1 (Galilean reduction to a constant)**.** A velocity field is constant up to a Galilean transformation if a Galilean transform sends it to a constant vector field. In the Type I blow-up frame only the following consequence is used: when a cited Liouville theorem gives a constant velocity field in that frame, the Type I bound forces the constant to be zero.

Lemma 29.2 (Spatially constant mild solutions)**.** *Let W be a smooth mild solution of (27.1) on $\mathbb{R}^3 \times (-\infty, 0)$. If $W(x,t) = b(t)$ for all x and t, then b is constant in time.*

Proof. Insert $W(x,t) = b(t)$ into (28.1). The heat semigroup leaves constant vector fields fixed, and $\nabla \cdot (b(\sigma) \otimes b(\sigma)) = 0$. Thus $b(t) = b(s)$ for every $s < t < 0$. □

Lemma 29.3 (Patching constants on backward intervals)**.** *Let $U : \mathbb{R}^3 \times (-\infty, 0) \to \mathbb{R}^3$ be a function. Assume that for every $T < 0$ there exists $c_T \in \mathbb{R}^3$ such that*

$$U(x,t) = c_T \qquad (x \in \mathbb{R}^3,\ t < T).$$

Then there is a single $c \in \mathbb{R}^3$ such that

$$U(x,t) = c \qquad (x \in \mathbb{R}^3,\ t < 0).$$

Proof. If $T_1 < T_2 < 0$, the two constants agree on the nonempty interval $(-\infty, T_1)$. Hence all c_T are equal. For any $t < 0$, choose T with $t < T < 0$. □

Lemma 29.4 (Constant Type I ancient solutions vanish)**.** *Let U be a Type I ancient solution. If $U(x,t) \equiv c$, then $c = 0$.*

Proof. The Type I bound gives

$$\sqrt{-t}\,|c| \le \sup_{s<0} \sqrt{-s}\,\|U(s)\|_\infty < \infty \qquad (t < 0).$$

Letting $t \to -\infty$ gives $c = 0$. □

29.3. **Cited Liouville theorems.** The following Liouville theorems are used in the stated forms.

Theorem 29.5 (KNSS no-swirl Liouville theorem)**.** *Every bounded ancient mild axisymmetric no-swirl solution of the unforced Navier–Stokes equations is constant up to a Galilean transformation. In the form needed for the Type I argument, the solution is a spatially constant axial drift.*

Proof. This is the no-swirl Liouville theorem of Koch–Nadirashvili–Seregin–Šverák [13], stated in the form needed for the Type I vanishing argument. □

Theorem 29.6 (KNSS pointwise scale-invariant condition)**.** *Every bounded ancient mild axisymmetric solution satisfying*

$$|u(x,t)| \le \frac{C}{r} \qquad (r > 0,\ t < 0)$$

is constant up to a Galilean transformation. In the form needed here, it is identically zero.

Proof. This is the pointwise Liouville theorem of [13], stated in the form used here. □

Theorem 29.7 (Lei–Zhang–Zhao)**.** *Let u be a bounded ancient mild axisymmetric solution. If*

$$\Gamma = r u_\theta \in L^\infty_t L^p_x\big(\mathbb{R}^3 \times (-\infty, 0)\big), \qquad 1 \le p < \infty,$$

then u is a constant vector field.

Proof. This is the ancient axisymmetric L^p-swirl Liouville theorem of Lei–Zhang–Zhao [30]. □

Theorem 29.8 (Lei–Ren–Zhang)**.** *Let u be a bounded ancient mild axisymmetric solution, periodic in z, with*
$$\Gamma = ru_\theta \in L^\infty\big(\mathbb{R}^3 \times (-\infty, 0)\big).$$
Then u is a constant axial vector field.

Proof. This is the periodic bounded-swirl theorem of Lei–Ren–Zhang [29]. Their statement is on $\mathbb{R}^2 \times \mathbb{T}^1$; a physical z-periodic whole-space solution descends to the quotient, and a Navier–Stokes scaling normalizes the period. □

30. Unforced axisymmetric Liouville consequences

30.1. No swirl.

Proposition 30.1 (No-swirl Type I vanishing)**.** *Let U be a smooth Type I ancient solution. Assume U is axisymmetric and $U_\theta \equiv 0$. Then $U \equiv 0$.*

Proof. Fix $T < 0$ and set $W_T(x, s) = U(x, T + s)$. By Lemma 28.6, W_T is a bounded ancient mild axisymmetric no-swirl solution. Applying Theorem 29.5 gives that W_T is a spatially constant axial drift, say $W_T(x, s) = b_T(s)e_z$. By Lemma 29.2, $b_T(s)$ is constant in s. Thus $U(x, t) = c_T$ for $t < T$. Since $T < 0$ is arbitrary, Lemma 29.3 gives $U(x, t) = c$ for all $t < 0$. Finally, Lemma 29.4 gives $c = 0$. □

30.2. Pointwise scale-invariant control.

Proposition 30.2 (Pointwise $1/r$ Type I vanishing)**.** *Let U be a smooth Type I ancient solution. Assume U is axisymmetric and satisfies*
$$|U(x, t)| \le \frac{C}{r} \qquad (r > 0,\ t < 0).$$
Then $U \equiv 0$.

Proof. Fix $T < 0$. The shifted solution $W_T(x, s) = U(x, T + s)$ is bounded, ancient, axisymmetric, and satisfies the same pointwise bound $|W_T| \le C/r$. Theorem 29.6 gives $W_T \equiv 0$. Hence $U \equiv 0$ on $(-\infty, T)$. Since $T < 0$ is arbitrary, $U \equiv 0$. □

30.3. Finite L^p control of Γ.

Proposition 30.3 (Finite-L^p swirl Type I vanishing)**.** *Let U be a smooth Type I ancient solution. Assume U is axisymmetric and, for some $1 \le p < \infty$,*
$$\Gamma = rU_\theta \in L^\infty_t L^p_x\big(\mathbb{R}^3 \times (-\infty, 0)\big).$$
Then $U \equiv 0$.

Proof. Fix $T < 0$. By Lemma 28.6, $W_T(x, s) = U(x, T+s)$ is a bounded ancient mild axisymmetric solution and
$$\Gamma_{W_T} \in L^\infty_s L^p_x\big(\mathbb{R}^3 \times (-\infty, 0)\big).$$
Theorem 29.7 gives that W_T is a constant vector field. Patch these constants in T by Lemma 29.3 and then use Lemma 29.4. □

30.4. **Periodic bounded swirl.**

Proposition 30.4 (Periodic bounded-swirl Type I vanishing)**.** *Let U be a smooth Type I ancient solution. Assume U is axisymmetric, periodic in the physical axial variable z, and satisfies*

$$\Gamma = rU_\theta \in L^\infty\big(\mathbb{R}^3 \times (-\infty, 0)\big).$$

Then $U \equiv 0$.

Proof. Fix $T < 0$. By Lemma 28.6, the shifted solution W_T is bounded, ancient, axisymmetric, physically z-periodic, and has bounded Γ. Theorem 29.8 gives that W_T is a constant axial vector field. The constants patch as T varies, and the Type I bound eliminates the resulting constant by Lemma 29.4. □

30.5. **Proof of the axisymmetric Liouville theorem.**

Proof of Theorem 27.1. The four cases are precisely the hypotheses of Propositions 30.1 to 30.4. Each proposition gives $U \equiv 0$. This contradicts Lemma 28.3 when U is a nonzero Type I blow-up limit. □

31. Perturbative weighted-vorticity Lyapunov rigidity

31.1. **The rescaled vorticity equation.** In this section τ is the forward self-similar time used in the Gallay–Wayne vorticity formulation. The ancient trajectories are indexed by all $\tau \in \mathbb{R}$, and the forward semiflow acts by increasing τ.

Let $w = \nabla \times v$. The rescaled vorticity equation is

$$\partial_\tau w = \Lambda w - (v \cdot \nabla)w + (w \cdot \nabla)v, \qquad \nabla \cdot w = 0, \tag{31.1}$$

where

$$\Lambda w = \Delta w + \frac{1}{2}\xi \cdot \nabla w + w, \tag{31.2}$$

and v is recovered from w by the three-dimensional Biot–Savart law. For $m \geq 0$, set

$$L^2_\sigma(m) = \left\{ f \in L^2(\mathbb{R}^3; \mathbb{R}^3) : \nabla \cdot f = 0, \quad \|f\|_m < \infty \right\},$$

where

$$\|f\|_m^2 = \int_{\mathbb{R}^3} (1 + |\xi|)^{2m} |f(\xi)|^2 \, d\xi.$$

31.2. **Gallay–Wayne small-data decay.**

Theorem 31.1 (Gallay–Wayne small-data decay)**.** *Let $0 < \mu \leq 1$ and $m > 2\mu + \frac{1}{2}$. There exist constants $\rho_m > 0$ and $C_m \geq 1$ such that if $\|w_0\|_m \leq \rho_m$, then (31.1) has a unique global mild solution $w(\tau) = \Phi_\tau w_0$ in $C([0,\infty); L^2_\sigma(m))$ and*

$$\|\Phi_\tau w_0\|_m \leq C_m e^{-\mu\tau} \|w_0\|_m, \qquad \tau \geq 0.$$

In particular, if $m > \frac{7}{2}$, the choice $\mu = 1$ is admissible.

Proof. This is the Gallay–Wayne weighted small-data theorem for the rescaled three-dimensional vorticity equation [27]. The conclusions needed below are existence, uniqueness, the semiflow property, and the displayed exponential decay. □

31.3. **The Lyapunov functional.**

Definition 31.2 (Perturbative weighted-vorticity condition)**.** Fix $m > \frac{7}{2}$, and let ρ_m be the radius in Theorem 31.1. An ancient vorticity trajectory satisfies the perturbative weighted-vorticity condition with exponent m if it is a function

$$w \in C(\mathbb{R}; L^2_\sigma(m))$$

that solves (31.1) for all $\tau \in \mathbb{R}$ and satisfies

$$\sup_{\tau \in \mathbb{R}} \|w(\tau)\|_m \le \rho_m.$$

Solving for all τ means that every time shift is compatible with the Gallay–Wayne forward mild semiflow.

Proposition 31.3 (Semiflow Lyapunov functional)**.** *Fix $m > \frac{7}{2}$. For $\|w_0\|_m \le \rho_m$, define*

$$\mathcal{L}_m(w_0) = \sup_{\tau \ge 0} e^{2\tau} \|\Phi_\tau w_0\|_m^2. \tag{31.3}$$

Then $\mathcal{L}_m$ is well-defined and satisfies

$$\|w_0\|_m^2 \le \mathcal{L}_m(w_0) \le C_m^2 \|w_0\|_m^2. \tag{31.4}$$

Moreover, for every $s \ge 0$,

$$\mathcal{L}_m(\Phi_s w_0) \le e^{-2s} \mathcal{L}_m(w_0). \tag{31.5}$$

Proof. The Gallay–Wayne estimate with $\mu = 1$ gives

$$e^{2\tau} \|\Phi_\tau w_0\|_m^2 \le C_m^2 \|w_0\|_m^2,$$

which proves finiteness and the upper bound. The lower bound is the value at $\tau = 0$. By uniqueness,

$$\Phi_\tau(\Phi_s w_0) = \Phi_{\tau+s} w_0.$$

Hence

$$\mathcal{L}_m(\Phi_s w_0) = \sup_{\tau \ge 0} e^{2\tau} \|\Phi_{\tau+s} w_0\|_m^2 = e^{-2s} \sup_{\sigma \ge s} e^{2\sigma} \|\Phi_\sigma w_0\|_m^2 \le e^{-2s} \mathcal{L}_m(w_0).$$

□

31.4. **Ancient rigidity in the small weighted ball.**

Proof of Theorem 27.2. Let w be an ancient solution in the small ball and fix $\tau_0 \in \mathbb{R}$. For $s < \tau_0$, the shifted function

$$W_s(\theta) = w(s + \theta), \qquad \theta \ge 0,$$

is the Gallay–Wayne mild solution with initial datum $w(s)$. The small-ball hypothesis gives $\|w(s)\|_m \le \rho_m$, so the forward solution is in the domain of Φ_θ, and uniqueness gives

$$W_s(\theta) = \Phi_\theta w(s), \qquad \theta \ge 0.$$

Choosing $\theta = \tau_0 - s$ gives

$$w(\tau_0) = \Phi_{\tau_0 - s} w(s).$$

Since $\|w(\tau_0)\|_m \le \rho_m$, the Lyapunov functional is also defined at $w(\tau_0)$. Applying Proposition 31.3 with forward time $\tau_0 - s$,

$$\mathcal{L}_m(w(\tau_0)) \le e^{-2(\tau_0 - s)} \mathcal{L}_m(w(s)) \le C_m^2 \rho_m^2 e^{-2(\tau_0 - s)}.$$

Letting $s \to -\infty$ gives $\mathcal{L}_m(w(\tau_0)) = 0$. The coercive lower bound (31.4) gives $w(\tau_0) = 0$. Since τ_0 was arbitrary, $w \equiv 0$. □

32. Summary of PDE consequences

32.1. Zero conclusions.

Theorem 32.1 (Zero conclusions for Type I blow-up limits). *The following hypotheses cannot hold for a nonzero Type I blow-up limit:*

(i) *axisymmetry and $U_\theta \equiv 0$;*
(ii) *axisymmetry and the pointwise bound $|U(x,t)| \le C/r$;*
(iii) *axisymmetry and finite-L^p control of the swirl scalar,*

$$\Gamma \in L^\infty_t L^p_x, \qquad 1 \le p < \infty;$$

(iv) *axisymmetry, z-periodicity, and bounded Γ;*
(v) *the Gallay–Wayne small weighted-vorticity condition, whenever the velocity is recovered from the corresponding vorticity by the Biot–Savart law.*

Proof. Items (i)–(iv) are Theorem 27.1. Item (v) is Theorem 27.2; since the velocity in that case is recovered from w by the Biot–Savart law, $w \equiv 0$ gives the zero velocity. In each Type I case, the zero conclusion contradicts Lemma 28.3. □

32.2. Consequences for Type I blow-up arguments.

Proof of Theorem 27.3. The axisymmetric cases are ruled out by Theorem 27.1. The perturbative weighted-vorticity case is ruled out by Theorem 27.2 once the corresponding rescaled vorticity satisfies the small weighted condition; the Biot–Savart reconstruction then gives zero velocity. These zero conclusions contradict the nontriviality of a Type I blow-up limit. □

Theorem 32.2 (Combined zero conclusion). *Let U be a Type I ancient solution obtained as a nonzero Type I blow-up limit. If U satisfies one of the hypotheses in Theorem 32.1, then $U \equiv 0$, a contradiction.*

Proof. This is Theorem 32.1. □

32.3. Summary table.

Hypothesis	Conclusion	Analytic ingredient
no swirl	$U \equiv 0$	KNSS plus Type I constant removal
pointwise $1/r$	$U \equiv 0$	KNSS
$\Gamma \in L^\infty_t L^p_x$	$U \equiv 0$	LZZ plus Type I constant removal
periodic bounded Γ	$U \equiv 0$	LRZ plus Type I constant removal
small weighted vorticity	$w \equiv 0$	Gallay–Wayne Lyapunov functional

The no-swirl, pointwise scale-invariant, finite-L^p-swirl, and periodic bounded-swirl cases are covered by the cited axisymmetric Liouville theorems. The perturbative weighted-vorticity case is covered by a Lyapunov functional derived from Gallay–Wayne decay. Thus every zero conclusion used in the Type I application rests on one of the explicit PDE hypotheses in the table.

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